

United in Booms, Divided in Busts: Regional House Price Cycles and Monetary Policy

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Abstract

This paper shows that regional disparities in house price growth increase during house price busts. We link this to intensified house price belief extrapolation in busts and regional housing market heterogeneities. To explain this observation, we build a multi-region currency union model with a housing sector and extrapolative house price beliefs, introducing a novel method to handle such models efficiently. Analytically, we show that heterogeneities generate differences in house price growth, while extrapolation amplifies them. Quantitatively, the model matches the observed empirical patterns. Monetary policy targeting house prices stabilizes output and house prices, improving welfare relative to an inflation-targeting benchmark.

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I. INTRODUCTION

A substantial body of literature connects boom-bust cycles in house prices to over-optimism and over-pessimism regarding the future evolution of house prices.¹ The bust phases of these cycles are often associated with significant economic costs, making them a primary concern for policymakers, particularly central bankers.² However, housing markets are inherently regional, and consequently, housing cycles and their associated economic costs also exhibit pronounced regional variation. These regional divergences, especially during periods of economic downturn, therefore constitute an additional challenge for policymakers.³ While a substantial literature examines housing cycles and their implications for monetary policy at the federal level, there is comparatively little focus on regional heterogeneities in house price dynamics.

This paper makes several contributions to the literature. Empirically, we document a novel stylized fact: cross-regional dispersion in house price growth is systematically more pronounced during busts than during booms. We relate this asymmetry to amplified extrapolative house price expectation updating during busts and to heterogeneity on the housing supply side.

To rationalize these findings, we develop a multi-region currency union model incorporating regional heterogeneity in housing markets and extrapolative belief updating over house prices. We introduce a novel solution method that enables to efficiently solve structural models with extrapolative expectations. Within this framework, intensified extrapolation during busts amplifies house price responses and thereby increases the dispersion in house price growth that originates from supply-side heterogeneity. Calibrated to U.S. metropolitan areas, the model quantitatively matches both the aggregate house price response and the cross-regional dispersion in house price growth to monetary policy shocks, conditional on boom and bust regimes.

Finally, we illustrate that by leaning against aggregate house prices, the monetary authority can mitigate cross-regional disparities in house prices and economic activity, resulting in welfare improvements relative to standard inflation targeting.

We begin by presenting our central empirical finding that cross-regional dispersion in house price growth is more pronounced during busts, both unconditionally and conditionally on macroeconomic shocks. The cross-regional standard deviation of house price growth among major U.S.

¹See, e.g., [Case et al. \(2012\)](#); [Armona et al. \(2019\)](#); [Kuchler and Zafar \(2019\)](#); [Ma \(2020\)](#); [Kaplan et al. \(2020\)](#)

²During the global financial crisis of 2008, 20% of household wealth ([Detting et al., 2018](#)) and 8 million jobs were lost ([Mian and Sufi, 2015](#)). During this time, the Fed responded by drastically cutting interest rates.

³See, for instance, the Euro area crisis of 2009, which ultimately threatened the existence of the currency union.

cities is 50%-60% larger during busts than booms,⁴ a pattern also present at U.S. state level and for the Euro Area. Conditional on monetary policy shocks, we observe a significant increase in cross-city standard deviations during busts, while the response is muted during booms. This result is robust to other types of shocks and a variety of house price cycle definitions.

To understand the underlying dynamics, we examine two features: extrapolative house price expectations, which are known to amplify house price movements, and regional heterogeneity on the housing supply side. Estimating the pass-through from past house price growth to household expectations, we find that it roughly doubles during busts relative to booms, indicating that belief updating is significantly stronger in downturns. Turning to supply-side heterogeneity, we document that the differential house price response of supply-inelastic relative to supply-elastic regions increases significantly during busts, a finding that is robust to focusing on unconditional cross-regional standard deviations within each group.

Building on these empirical results, we develop a multi-region currency union New Keynesian model in the spirit of [Benigno \(2004\)](#) incorporating (i) regional heterogeneity in housing supply and (ii) extrapolative belief updating about house prices. Households form subjective expectations only about house prices, but their optimal choices—such as consumption—depend on expected future consumption, which in turn is shaped by future expected house prices. It therefore becomes necessary to characterize these expected endogenous choice variables as functions of prices that are perceived to be exogenous by households.

To address this, we propose a novel first-order solution method for general equilibrium models with time-consistent but non-rational expectations. Specifically, we show that a standard log-linearized Euler equation can be expressed in terms of current consumption, the expected future path of real interest rates, and the subjectively expected consumption at the limit. We refer to the latter as the *subjective expectations wealth effect*.⁵

We demonstrate that the *subjective expectations wealth effect* admits an explicit representation that depends solely on prices. Moreover, we show that knowledge of this term is sufficient to solve the model. Finally, the first-order approximation ensures compatibility with standard solution methods widely used in the DSGE literature, enabling the solution of large-scale models. In our quantitative analysis, we apply this approach to a currency union model comprising 20 regions.⁶

⁴Throughout the paper, a bust period is defined as one in which house prices declined on an aggregate or federal level in the previous period.

⁵Intuitively, when households extrapolate increases in house prices into the future, they perceive higher expected lifetime wealth, which raises expected consumption in the limit and, consequently, current consumption.

⁶Our solution method contrasts with that introduced by [Winkler \(2020\)](#), which, although based on first-order

Equipped with our model, we demonstrate analytically that regional differences in housing markets, combined with house price belief extrapolation, can account for the asymmetry in house price growth dispersion observed across the house price cycle.⁷ First, regional heterogeneity in housing markets creates regional differences in the responsiveness of house prices. Second, subjective expectations about house prices lead to amplification through extrapolation. Intuitively, when agents observe an increase in house prices, they anticipate further price increases in the future, leading to heightened demand for housing and, subsequently, driving prices even higher.

The interaction of these two forces generates the following dynamics: Regional heterogeneities in housing markets create differences in regional house price growth rates, which are amplified by extrapolation. This theoretical result directly aligns with the empirical findings.

In a rational expectations version of the model, the ability of cross-regional differences in housing markets to produce differences in house prices is much weaker. Additionally, this version lacks extrapolative belief updating and, thus, cannot account for amplified cross-regional heterogeneity in busts in the absence of any additional frictions.

We calibrate regional housing supply heterogeneities to the 20 largest U.S. metropolitan areas used in our empirical analysis and target the aggregate house price response to a monetary policy shock, independent of boom-bust asymmetries. When accounting for asymmetries in the belief updating process, the model accurately captures the magnitudes of aggregate house price changes in both booms and busts, even though these asymmetries were not targeted. The model also successfully replicates the regional house price growth dispersion observed during booms and busts in response to a monetary policy shock. In contrast, the rational expectations version of the model fails to match these dynamics, underestimating aggregate house price responses by a factor of three and generating no significant difference in regional house prices.

Finally, we investigate the extent to which leaning against house prices influences economic outcomes. Specifically, we assess the effects of including aggregate house prices in the Taylor rule compared to a baseline rule focused solely on inflation stabilization in response to a productivity shock. We find that house price targeting reduces volatility in both output and house prices, albeit at the cost of increased inflation volatility. Notably, it also reduces regional dispersion across all variables, synchronizing economic activity across regions. The underlying intuition is that by

approximation, does not confine subjective expectations to specified asset prices and results in belief spillovers to all other prices. Conversely, [Adam et al. \(2022\)](#) solve a similar model using non-linear solutions.

⁷We focus on monetary policy shocks, but the general proposition extends to all shocks that affect housing demand.

targeting house prices, the central bank limits the divergence in house price beliefs, mitigating the channel through which regional heterogeneities spill over to the real economy. We show that house price targeting is beneficial under subjective expectations, expressing the utilitarian welfare as in [Galí \(2015\)](#).⁸ Welfare improves by 9% in booms and by 13% in busts relative to the inflation-targeting baseline, with the larger bust gains reflecting stronger belief extrapolation in those periods.

Related literature. Our paper relates to a broad empirical literature emphasizing that house price beliefs deviate from rational expectations, highlighting momentum and revisions in belief formation ([Case et al., 2012](#); [Armona et al., 2019](#); [Kuchler and Zafar, 2019](#); [Ma, 2020](#)). On the theoretical side, it connects to the behavioral macro-finance literature on deviations from rational expectations in asset price formation, including models of diagnostic expectations ([Bordalo et al., 2018, 2021](#); [L’Huillier et al., 2023](#); [Bianchi et al., 2024a](#)), learning and extrapolation ([Adam et al., 2017](#); [Winkler, 2020](#); [Farhi and Werning, 2020](#); [Fontanier, 2022](#)), belief disagreement ([Caballero and Simsek, 2019, 2020](#)), and sentiments ([Krishnamurthy and Li, 2020](#); [Maxted, 2024](#)). Regarding housing specifically, [Glaeser and Nathanson \(2017\)](#) and [Schmitt and Westerhoff \(2019\)](#) model extrapolative expectations in partial equilibrium, while [Adam et al. \(2012\)](#), [Caines and Winkler \(2021\)](#), and [Adam et al. \(2022\)](#) study these mechanisms in general equilibrium. Other studies explain house price behavior through optimism and pessimism, concave demand curves, or exogenous belief shifts ([Burnside et al., 2016](#); [Guren, 2018](#); [Kaplan et al., 2020](#)). We contribute by emphasizing the asymmetric evolution of house price beliefs across booms and busts and by proposing a novel solution method for models with capital gains extrapolation. We also contribute to the literature on regional housing supply heterogeneity. [Mian et al. \(2013\)](#), [Mian and Sufi \(2014\)](#), and [Guren et al. \(2021\)](#) explore housing wealth effects through supply elasticities, while [Aastveit and Anundsen \(2022\)](#) and [Aastveit et al. \(2023\)](#) show that inelastic regions display stronger house price responses to monetary policy. We connect to [Glaeser et al. \(2008\)](#), who show that supply-inelastic regions face higher bubble probabilities, extending this work by linking supply-side heterogeneity to asymmetric belief formation in general equilibrium. Our analysis builds on the literature on cross-regional heterogeneities within currency unions, including optimal policy ([Benigno, 2004](#); [Galí and Monacelli, 2008](#); [Kekre, 2022](#)) and the role of heterogeneities in wage and price setting, hand-to-mouth shares, and mortgage markets ([Calza et al., 2013](#); [Slacalek et al., 2020](#); [Bletzinger and von Thadden, 2021](#); [Pica, 2021](#); [Almgren et al., 2022](#); [Corsetti et al., 2022](#)). For the United States, [Beraja et al. \(2019\)](#), [Chen \(2019\)](#), and [Gitti \(2024\)](#) document regional

⁸Specifically, the welfare loss is expressed in terms of the equivalent permanent decline in consumption, measured as a fraction of steady-state consumption.

variation in exposure to fiscal policy, housing, and migration. We contribute by linking supply-side housing constraints to asymmetric house price belief formation. Finally, we relate to the literature on leaning against asset prices. [Stein \(2012\)](#) argues for such policies to mitigate over-borrowing externalities, while [Svensson \(2017\)](#) and [Weidmann \(2018\)](#) emphasize their costs and the superior role of macroprudential tools, and [Gourio et al. \(2018\)](#) highlight the trade-off between crisis risk and cyclical volatility. In a rational expectations framework, [Bianchi et al. \(2024b\)](#) find that monetary policy should not target house prices—consistent with our rational expectations findings. Under deviations from rational expectations, however, leaning against the wind becomes beneficial ([Adam and Woodford, 2021](#); [Caines and Winkler, 2021](#); [Fontanier, 2022](#); [Caballero and Simsek, 2019](#)). We contribute by showing that house price targeting additionally mitigates regional dispersion, thereby helping harmonize an economically integrated area—a point we believe to be novel in the literature.

Outline. The remainder of the paper is structured as follows. Section (II) documents the empirical fact that regional house price growth is more dispersed in busts than in booms. Section (IV) presents the model and solution method, and Section (V) provides the analytical intuition. Section (VI) discusses evidence of stronger extrapolation in busts and the model calibration. Section (VII) reports the quantitative findings, Section (VIII) presents the policy analysis, and Section (IX) concludes.

II. REGIONAL HOUSE PRICES IN BOOMS AND BUSTS

In the following we document a novel empirical fact: regional disparities in house price growth are significantly larger during housing busts than during booms. This asymmetry in regional dynamics has, to our knowledge, not been previously emphasized in the literature.⁹ We establish this finding using both unconditional moments—capturing overall cross-regional dispersion—and conditional moments, when controlling for specific shocks. For our analysis, we classify periods with positive aggregate house price growth ($\Delta q_{t-1}^{agg} > 0$) as booms and those with negative growth ($\Delta q_{t-1}^{agg} < 0$) as busts. This definition follows [Glaeser et al. \(2008\)](#); our findings are robust with respect to other definitions.

⁹While regional heterogeneities and their effects on house prices are well studied (e.g., [Saiz, 2010](#); [Guren et al., 2021](#); [Aastveit and Anundsen, 2022](#)), asymmetries between booms and busts remain unexplored.

II.A Cross-regional house price variation: unconditional evidence

To investigate this pattern, we focus on regional house price growth data, specifically at the major city or state level, for the United States. For each point in time, we compute the cross-regional standard deviation of house price growth rates. We then split the sample into boom and bust episodes, where these episodes are defined based on the aggregate house price growth. Our preferred specification is based on U.S. major cities in the cross-section.¹⁰ The data is given in monthly frequencies, allowing for sufficient observations of booms and busts. For the U.S. state level, we use quarterly data. Our sample includes all states except for Alaska and Hawaii but includes the District of Columbia. Both U.S. data sets span from 1990 to 2024. Table (1) reports the mean and median cross-regional standard deviations for house price booms and busts, as well as their share in busts relative to booms.

Table 1: U.S. cross-regional house price growth standard deviation (σ_c) in booms and busts

	<i>cities</i>		<i>states</i>	
	<i>mean</i>	<i>median</i>	<i>mean</i>	<i>median</i>
<i>Bust</i>	1.60	1.64	1.23	1.26
<i>Boom</i>				
$\sigma_c : \text{Boom}$	0.47	0.41	0.87	0.80
$\sigma_c : \text{Bust}$	0.74	0.67	1.08	1.01
$p - \text{Val. } \text{Bust} > \text{Boom}$	0.000	0.000	0.007	0.007
<i>Number regions</i>	20		51	
<i>Obs. boom</i>	331		109	
<i>Obs. bust</i>	82		28	
<i>Sample</i>	1990M1 – 2024M5		1990Q1 – 2024Q1	

Notes: The Table reports the mean and median values, across time, of the estimates of cross-sectional standard deviations of house price growth rates within a given economic entity during house price booms and busts. $\frac{\text{Bust}}{\text{Boom}}$ denotes the regional standard deviation in busts relative to booms. The test for $\text{Bust} > \text{Boom}$ is based on a one-sided t-test.

Our findings indicate that the cross-regional standard deviations observed during busts are larger for both the mean and the median estimates than those observed during booms. Specifically, if one considers monthly data for U.S. cities, the cross-regional standard deviations during busts are 60% larger than during booms, judging by the relative shares. The ratio decreases to 23% – 30% for the U.S. states. We find that a t-test rejects the H_0 hypothesis of regional standard deviations being larger in booms for both regional specifications. Table (C.1) in the Appendix shows that the same

¹⁰We use the Case-Shiller U.S. National Home Price Index for the city level. The 20 major U.S. cities included are Atlanta, Boston, Charlotte, Chicago, Cleveland, Dallas, Denver, Detroit, Las Vegas, Los Angeles, Miami, Minneapolis, New York, Phoenix, Portland, San Diego, San Francisco, Seattle, Tampa, and Washington DC.

holds for the euro area at the country level.

II.B Cross-regional house price variation: conditional evidence

While the previous analysis focused on unconditional variations in house price growth during booms and busts, we now move to a conditional approach. Using local projections, we estimate the response of cross-regional standard deviations of house price growth to a monetary policy shock. We utilize city-level data due to its sufficiently large time series dimension. Equation (1) depicts the estimation equation. The left-hand-side variable (y_{t+h}) denotes the three-month moving average of the cross-regional standard deviation of house price growth at horizon h after the impact of the shock. α^h denotes a constant. The monetary policy shock, ϵ_t^{MP} , is the high frequency identified from [Degasperi and Ricco \(2021\)](#). The monetary policy shock further conditions on times of booms ($\Delta q_{t-1}^{agg} > 0$) and busts ($\Delta q_{t-1}^{agg} < 0$). We adopt this approach because monetary policy shocks are relatively small and do not influence the U.S. economy's transition between boom and bust phases.¹¹ We use the absolute values of the shocks as the cross-regional dispersion should be more responsive in busts independent of the sign of the shock. The controls, x_t , contain 6 lags of the left-hand side variable, the log of the house price index on a federal level, the log of industrial production, the log of the CPI, the Federal Funds Rate (FFR), and the shocks. The sample runs from 1991 to 2019.

$$y_{t+h} = \alpha^h + \mathbb{1}(\Delta q_{t-1}^{agg} > 0) \times \beta_1 |\epsilon_t^{MP}| + \mathbb{1}(\Delta q_{t-1}^{agg} < 0) \times \beta_2 |\epsilon_t^{MP}| + x_t + u_{t+h} \quad (1)$$

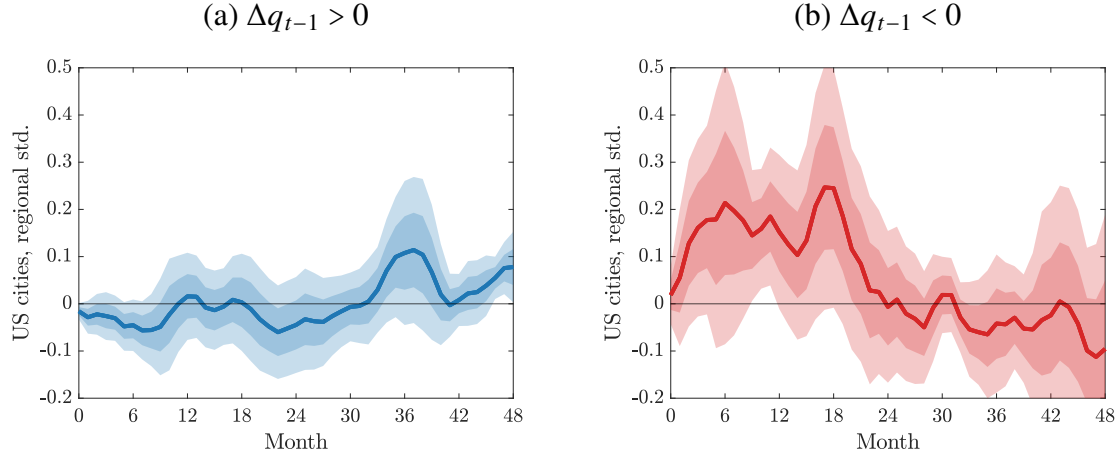
Figure (1) presents the results. We see no significant change in cross-regional standard deviations in booms; see panel (a). In times of busts, panel (b), we find that a monetary policy shock leads to a significant increase in the variation of cross-regional house price growth over the next two years. At the relative peaks, the bust response is approximately twice as high as the one in booms. Hence, conditional on monetary policy shocks, regional house price growth is more dispersed in busts relative to booms.

II.C Robustness

House price cycles and recessions. House price booms and busts do not simply mirror aggregate expansions and recessions. Figure (C.1), which plots the cross-city three-month moving average standard deviation of house price growth, illustrates this clearly. Cross-city dispersion was elevated

¹¹Regressing house price growth on the aggregate level on the monetary policy shock reveals that $R^2 = 0.0003$. This emphasizes that these shocks do not drive the cycle.

Figure 1: City-level std. house price growth response to a monetary policy shock, boom-bust



Notes: Responses to a monetary policy shock (25 bps on impact, absolute value) ; confidence intervals: 68% and 95% (Newey-West). The lhs variable is the three-month moving average of the cross-city standard deviation of house price growth.

both before and after the Global Financial Crisis, and rose again during 2022–2023 — a house price bust unaccompanied by a recession. Conversely, the recessions associated with the Dotcom bust (2001) and the COVID-19 crisis (2020–2023) were not accompanied by house price busts or elevated cross-regional dispersion. House price busts and recessions are therefore not perfectly correlated.

Alternative cycle definitions. The results in Table (1) are robust to alternative cycle classifications, including those of [Albuquerque et al. \(2025\)](#) and [Aastveit et al. \(2023\)](#), as well as the cycle-dating approach of [Harding and Pagan \(2002\)](#) (see Table (C.2)).

Regional cycles. [Ferreira and Gyourko \(2023\)](#) and [Sinai \(2012\)](#) note that local housing cycles may not be synchronized across regions. Since our cycle definition is based on aggregate house prices, we assess the degree to which regional cycles co-move with the national cycle by computing the frequency of alignment between city-level and aggregate house price cycles. Alignment occurs in approximately 85% of cases, rising to 92% once measurement noise in aggregate and regional house price growth is accounted for. Local and aggregate housing cycles are thus synchronized in the majority of our sample. Restricting the sample to observations aligned with the aggregate cycle reduces the relative dispersion in busts versus booms slightly, consistent with [Ferreira and Gyourko \(2023\)](#) and [Sinai \(2012\)](#), but bust dispersion continues to significantly exceed boom dispersion and our main findings remain intact (see Table (C.1)). Finally, we assess whether results are driven by any single city by recomputing cross-city standard deviations leaving out one city at a time. The

bust-to-boom standard deviation ratio ranges from 1.55 to 1.62 (mean) and 1.56 to 1.74 (median), confirming that no individual city is responsible for the finding.

Alternative conditional evidence. We also study the response of regional house price growth dispersion to a main business cycle shock constructed by [Angeletos et al. \(2020\)](#). The shock is given in quarterly frequencies and we, therefore, move to state-level data to extend our cross-section. The main idea is that the main business cycle shock should be strong enough to affect the house price cycle.¹² Consequently, we study the response of the cross-state standard deviation of house price growth to an expansionary main business cycle shock. We observe that this shock indeed reduces house price growth heterogeneities; see Figure (C.2). As a sanity check, we also report the response of house price growth to the shock and observe that it is increasing. Our results are, therefore, in line with the findings reported above.

Discussion and relation to existing literature. [Sinai \(2012\)](#) documents that regional dispersions in house price growth widen during bust periods. His analysis, conducted at annual frequency, ends in 2010, thereby omitting the period of peak regional dispersion during the 2006–2012 bust. Moreover, the subsequent recovery and boom from 2012 to 2021—characterized by markedly low regional dispersion, as shown in Figure (C.1)—falls entirely outside his sample. His reported peaks in regional standard deviations in 2006 and 2009, corresponding to the onset and trough of the bust respectively, are consistent with our findings.

III. HOUSE PRICE EXPECTATIONS AND REGIONAL HETEROGENEITY

This section provides an explanation for the dynamics documented above. We show that house price expectations violate rational expectations and are instead extrapolative, as agents project recent house price growth into the future. This mechanism amplifies volatility during busts. We further link these findings to heterogeneity on the housing supply side, showing that the differential response of house prices to monetary policy shocks between inelastic and elastic housing markets widens significantly during busts.

¹²Regressing house price growth on the shock gives an R^2 of 0.1, meaning that this shock can explain a non-trivial part of the variation in house price growth.

III.A House price beliefs and boom-bust dynamics

Violation of rational expectations. A substantial body of empirical literature documents deviations from full information rational expectations (FIRE) in housing markets.¹³ We extend the Coibion and Gorodnichenko (2015) regression to house prices, conditioning on boom and bust regimes. Adam et al. (2022) estimate this regression unconditionally and document deviations from FIRE. Specifically, using house price expectations from the Survey of Consumers and house price data from the Case-Shiller index, we estimate:

$$q_{t+12} - \mathbb{E}_t^{\mathcal{P}} q_{t+12} = a^{boom} + a^{bust} + b^{boom} \mathbb{1}(\Delta q_{t-1} > 0)(\mathbb{E}_t^{\mathcal{P}} q_{t+12} - \mathbb{E}_{t-1}^{\mathcal{P}} q_{t+12}) + b^{bust} \mathbb{1}(\Delta q_{t-1} < 0)(\mathbb{E}_t^{\mathcal{P}} q_{t+12} - \mathbb{E}_{t-1}^{\mathcal{P}} q_{t+12}) + u_t$$

q_{t+12} denotes house prices in levels at $t = 12$, and $\mathbb{E}_t^{\mathcal{P}} q_{t+12}$ one year ahead house price expectations formed at period t . The regression projects forecast errors on expectation revisions, separately for booms and busts. Under FIRE, b^{boom} and b^{bust} should both equal zero.¹⁴ Table (2) presents the

Table 2: Test for rational expectations

b^{boom}	b^{bust}	R^2	N
1.76*	3.26***	0.45	150
(1.00)	(1.17)		

Notes: Estimation of the Coibion and Gorodnichenko (2015) regression for house prices, conditioning on boom and bust regimes. Reported standard errors are Newey-West. Significant levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

results. Two findings stand out. First, both coefficients are statistically significant, providing clear evidence against FIRE. Second, the positive point estimates indicate that households update their beliefs too sluggishly relative to the information contained in forecast errors, consistent with Adam et al. (2022). In Appendix (C.3), we further validate our modeling approach by examining the response of forecast errors to monetary policy shocks. Under rational expectations, this response should be zero; instead, we find a substantial and statistically significant reaction, characterized by initial over-pessimism that transitions to over-optimism at longer horizons.

¹³See: Case et al. (2012); Armona et al. (2019); Kuchler and Zafar (2019); Ma (2020)

¹⁴The Survey of Consumers only reports expectations at one fixed horizon. We follow Adam et al. (2022) and use the previous month's expectations to proxy for $\mathbb{E}_{t-1}^{\mathcal{P}} q_{t+12}$. Under full-information rational expectations b^{boom} and b^{bust} remain zero under this proxy.

House price belief formation. The pattern of initial over-pessimism followed by over-optimism in the forecast error responses is consistent with extrapolative expectation updating (see [Adam et al. \(2022\)](#)). We therefore estimate two specifications: a linear model and a threshold model allowing for regime switching, with micro-foundations provided in Section (IV). The linear specification is:

$$\mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1} = c + \varrho(1 - g) \times \mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t + \varrho g \times \Delta q_{t-1}. \quad (2)$$

Expectations in house price growth depend on prior expectations, $\mathbb{E}_{t-1}^{\mathcal{P}}$, and an updating part based on observed price movements, Δq_{t-1} . c is a constant that can be linked to the long-run house price growth expectations. g denotes the Kalman gain and governs the degree of expectations updating. A higher g increases the path-through of observed house prices to expectations. ϱ governs the persistence of prior beliefs.

Equation (3) mirrors the linear setup but allows for heterogeneity in ϱ and g below and above a certain threshold:

$$\begin{aligned} \mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1} = & \mathbb{1}(\Delta q_{t-1} > \omega) (c^h + \varrho^h(1 - g^h) \times \mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t + \varrho^h g^h \times \Delta q_{t-1}) + \\ & \mathbb{1}(\Delta q_{t-1} < \omega) (c^l + \varrho^l(1 - g^l) \times \mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t + \varrho^l g^l \times \Delta q_{t-1}) \end{aligned} \quad (3)$$

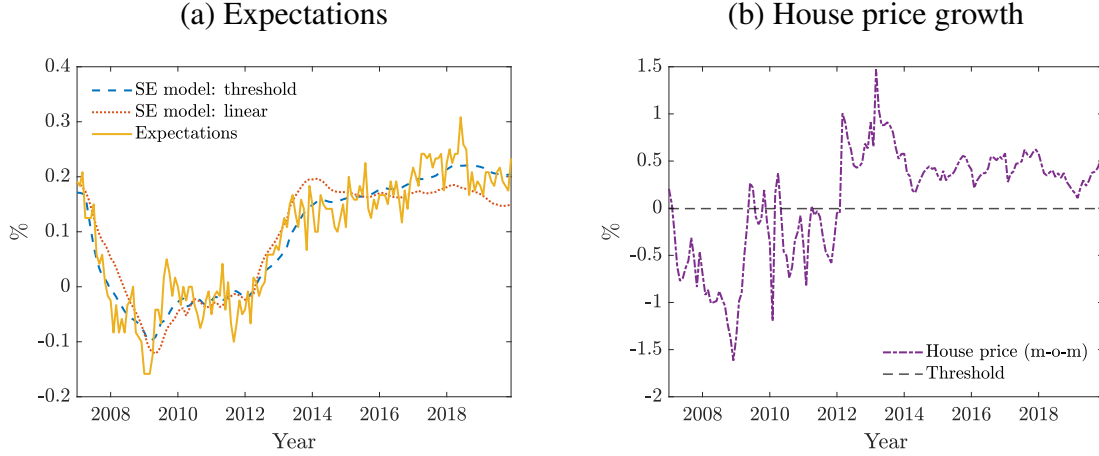
Both models are estimated at monthly frequency over 2007–2020. Month-on-month house price growth expectations, $\mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1}$, are obtained by dividing the year-on-year changes from the Michigan Survey by 12, and house price growth is taken from the Case-Shiller National House Price Index. The estimation replicates the recursive belief updating procedure employed by agents in our model. For a given parameter tuple (ϱ, g, c) , we feed in realized house price growth Δq_{t-1} and recursively update $\mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t$ to generate a sequence of expected house price growth rates. Parameters are chosen to minimize the mean squared error (MSE) between fitted and observed expectations, subject to $\varrho, g \in (0, 1)$.¹⁵ The threshold model is estimated analogously, with the parameter vector expanded to $(\varrho^h, \varrho^l, g^h, g^l, c^h, c^l, \omega)$; the threshold ω is estimated jointly with the remaining parameters.

The estimation results are presented in Figure (2). Panel (a) plots the expectations data from the Michigan Survey (solid yellow line), the threshold model fit (blue dashed line), and the linear model fit (dotted red line). Panel (b) shows month-on-month house price growth (dashed-dotted purple line) and the estimated threshold (dashed black line). Turning to the linear model first, we find that it does a reasonable job of explaining the expectation formation process. However, it lags

¹⁵The estimation follows [Adam et al. \(2017\)](#). A starting value for $\mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t$ is required; to minimize its influence on estimated parameters, we initialize the recursion using house price growth data from 1987, yielding starting values for 2007M1. Results are insensitive to this choice.

behind during the bust episode and underestimates the expectations for house price growth during the recovery following the global financial crisis. In comparison, the threshold model improves on these dimensions. It captures the bust without an obvious lag and performs better in recovery episodes. The estimated threshold for this model is close to zero, suggesting that the updating behavior differs during periods when house prices rise (booms) compared to when they fall (busts).

Figure 2: House price belief model, U.S.



Notes: SE model, threshold (blue dashed): fitted values of Equation (3); subjective expectations model, linear (red dotted): fitted values of Equation (2); expectations (yellow solid): mean expectations data from the Michigan survey; house price (m-o-m) (purple dashed-dotted): month-on-month percentage change in the house price. The black dashed line in panel (b) depicts the threshold in the threshold model.

The estimated parameters are reported in Table (3). The linear model yields $\varrho = 0.95$ and a Kalman gain of $g = 0.018$. The threshold model yields $\varrho^h = 0.99$ and $g^h = 0.013$ in booms, and $\varrho^l = 0.84$ and $g^l = 0.022$ in busts, all broadly consistent with the literature. The parameter differences across regimes explain why the threshold model outperforms the linear one. During busts, when house prices and beliefs fall sharply, a higher Kalman gain and lower persistence enable faster pass-through from realized to expected house price growth. During booms, when prices recover gradually, a higher persistence ϱ^h and lower updating g^h better match the slow-moving dynamics of expectations. The threshold specification reduces the MSE by approximately 45%. As a robustness check, we assess whether the parameter differences across regimes drive the performance gains of the threshold model. We do so by imposing the boom and bust parameters from the threshold model on the linear specification. In both cases, the MSE increases significantly, confirming that the regime-specific parameters are responsible for the threshold model's superior fit.

As a final robustness check, we follow [Case et al. \(2012\)](#), who document the pass-through from realized house prices to expectations by regressing expectations on lagged house price growth, and

Table 3: House price belief model: parameters

<i>Specification</i>	ϱ^{lin}	ϱ^h	ϱ^l	g^{lin}	g^h	g^l	<i>Threshold</i>	MSE^i/MSE^{lin}
<i>Baseline :</i>								
<i>linear</i>	0.95			0.018				1.00
<i>threshold</i>		0.99	0.84		0.013	0.022	-0.004	0.55
<i>Robustness :</i>								
<i>high regime beliefs</i>	0.99			0.013				5.9
<i>low regime beliefs</i>	0.84			0.022				5.5

Notes: Linear: estimated parameters from Equation (2); threshold: estimated parameters from (3); robustness exercises are explained in the main text.

extend their specification by conditioning on booms and busts:

$$\mathbb{E}_t^P \Delta q_{t+1} = a^{boom} + a^{bust} + b^{boom} \mathbb{1}(\Delta q_{t-1} > 0) \Delta q_{t-1} + b^{bust} \mathbb{1}(\Delta q_{t-1} < 0) \Delta q_{t-1} + u_t$$

We again use the monthly expectations data from the Michigan Survey and the Case-Shiller house price index for realized prices. Table (4) presents the results. The pass-through is statistically indistinguishable from zero in booms, while positive and significant in busts.¹⁶ Crucially, we reject the null hypothesis of equal pass-through across regimes ($H_0 : boom = bust$). These results are consistent with the findings documented above. Further, the large R^2 is in line with Case et al. (2012), confirming that lagged house price growth accounts for a substantial share of variation in expectations.

Table 4: Pass-through from lagged prices to expectations

b^{boom}	b^{bust}	R^2	N	$p - val. H_0 : b^{boom} = b^{bust}$
-0.05	0.06**	0.61	150	0.03
(0.04)	(0.03)			

Notes: Estimation of the Case et al. (2012) regression for house prices, conditioning on boom and bust regimes. Reported standard errors are Newey-West. Significant levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Column 5 shows the p-value for the null hypothesis that both coefficients are equal.

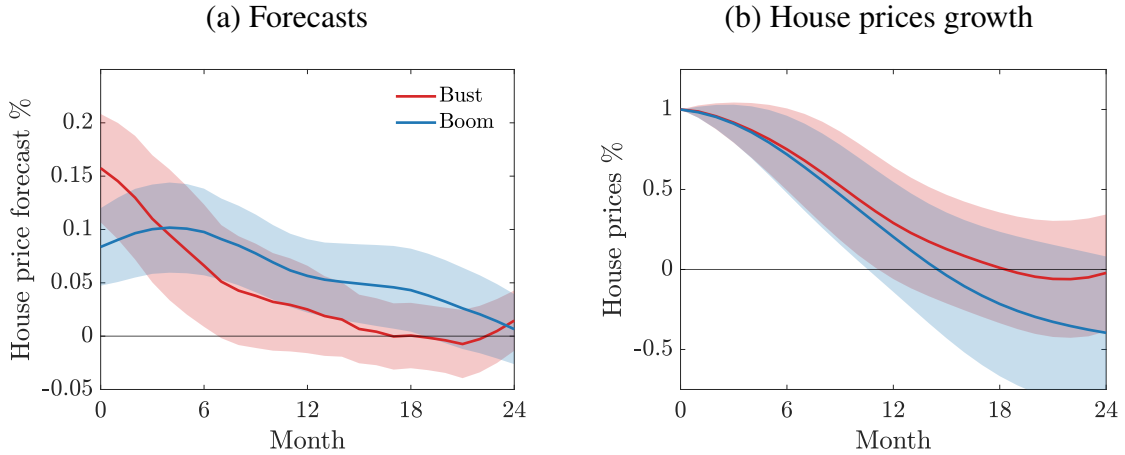
Sources of asymmetric updating. The results above raise the question of why house price expectations differ in booms and busts. We provide suggestive evidence for two complementary channels. The first focuses on fundamentals: house prices behave differently in booms and busts,

¹⁶Although the coefficient for the boom is insignificant, its sign is counterintuitive. We show in the Appendix that this anomaly is attributable to small sample limitations. For details see the discussion below and Appendix (C.6).

which spills over into expectations. To test this, we directly re-estimate equation (3) using actual house price realizations instead of expectations. We find that the realization gain is indeed larger in busts ($g^l = 0.99$) than in booms ($g^h = 0.66$), confirming that fundamentals alone justify stronger updating during downturns. House prices are more volatile and less persistent in busts, as visible in panel (b) of Figure (2), making it rational to pay more attention to recent price movements. Supporting this interpretation, Appendix (C.4) shows that mentions of "house price" in the New York Times increase substantially during busts when house price volatility is high, suggesting that public attention to housing market developments is significantly higher in busts than in booms, consistent with a rational inattention framework in which the degree of attention is endogenous to the informativeness of the signal.¹⁷

The second channel focuses on belief formation itself. The asymmetry in expectations exceeds what fundamentals alone would justify: while the realization gain is 50% larger in busts than in booms, the expectations gain is 69% larger — a gap that reflects a behavioral amplification over and above the fundamental asymmetry. This is further supported by the local projection evidence in Figure (3), which shows the forecast and house price growth responses to a 1% increase in house price growth in booms and busts. The realized price response is broadly similar across regimes. In contrast, the impact forecast response is 88% larger in busts than in booms — exceeding the 50% differential in realized dynamics and consistent with the larger asymmetry in the expectations gain.

Figure 3: Local projections: Forecast and house prices responses



Notes: Panel (a) depicts the response of one-year-ahead forecasts to a 1% increase in year-on-year house price growth. Panel (b) depicts the response of year-on-year house price growth to a 1% increase in year-on-year house price growth. The Responses are estimated using local projections and include a time trend as a control. Forecasts are taken from the Michigan Survey, and house prices from the Case-Shiller Index, confidence intervals: 68% (Newey-West).

¹⁷Pfäuti (2025) shows that our expectations formulation nests a rational inattention setup.

Consequences of asymmetric updating. Our findings indicate that house price expectations updating is more pronounced in busts. This, in turn, can affect house prices themselves: a larger reaction in expectations translates into economic decision-making and will eventually be reflected in prices. We test whether this is the case by studying the effect of a monetary policy shock on house prices during booms and busts. We find that house prices respond more strongly during times of a house price bust (see Appendix (C.5)).

Additional evidence and discussion. Aastveit and Anundsen (2022) and Glaeser and Nathanson (2017) argue that expectation updating is stronger in booms than in busts. The former relies on autoregressive regressions of realized house prices to test for momentum, while the latter — using annual data over 1980–2011 — notes that their model fit would improve with stronger extrapolation in booms. Crucially, neither paper directly estimates expectation updating across regimes. Our direct estimation of house price expectation formation yields the opposite conclusion.

A potential concern is that our sample omits the pre-crisis run-up. Since no house price expectations data at adequate frequency exist for this period, we rely on a proxy proposed by Jacobson (2022), combining backcasts until 1993 with the actual expectations data from 2007 onward. Re-estimating equation (3) leaves the results qualitatively unchanged: the threshold remains near zero, with lower persistence and a higher Kalman gain in busts. Distinguishing explicitly between booms, busts, and normal times yields $g^{bust} > g^{boom} > g^{normal}$ and $\varrho^{boom} > \varrho^{normal} > \varrho^{bust}$. Three findings are noteworthy. First, extrapolation remains strongest in busts, consistent with our baseline. Second, extrapolation is elevated in booms relative to normal times, in line with Aastveit and Anundsen (2022) and Glaeser and Nathanson (2017). Third, while belief updating is amplified in both regimes, persistence is markedly higher in booms than busts, reflecting the gradual and persistent nature of boom-phase dynamics relative to the sharp but short-lived dynamics of busts. These findings are robust to re-estimating the regressions of Case et al. (2012) using the extended series, as shown in Table (4). Full results are reported in Appendix (C.6).

III.B Regional heterogeneities in booms and busts

A large empirical literature has documented the link between regional house prices and housing supply side heterogeneities.¹⁸ More specifically, recent research by Aastveit and Anundsen (2022) has linked regional variations in housing supply elasticities across US metropolitan areas to differential responses in house prices to monetary policy shocks. Specifically, house prices are found to be more responsive in regions in which housing supply is more inelastic.

¹⁸See, for example, Glaeser et al. (2008); Saiz (2010); Mian et al. (2013); Mian and Sufi (2014); Guren et al. (2021).

Supply side heterogeneities and housing cycles. A potential explanation for the findings in Section (II) is that regional heterogeneity originates from supply-side differences across cities and is amplified through expectation dynamics during busts, as documented in Section (III.A). To test this, we rank the 20 Case-Shiller cities by the housing supply elasticities of Guren et al. (2021) and compute house price differences between cities at the 10th and 90th, and 25th and 75th percentiles of the elasticity distribution. We then estimate local projections of these differences on monetary policy shocks, conditioning on boom and bust regimes:

$$y_{t+h}^{p,1-p} = \alpha^h + \mathbb{1}(\Delta q_{t-1}^{agg} > 0) \times \beta_1 \epsilon_t^{MP} + \mathbb{1}(\Delta q_{t-1}^{agg} < 0) \times \beta_2 \epsilon_t^{MP} + x_t + u_{t+h}$$

The specification mirrors our baseline local projections setup, with identical shocks, regime indicators, and controls. $y_{t+h}^{p,1-p}$ denotes the house price differential between the least elastic cities (p th percentile) and the most elastic cities ($(1-p)$ th percentile). Peak responses are reported in Table (5). At the 10th-90th percentile split, inelastic cities respond slightly more strongly during booms, though insignificantly. During busts, the differential response is substantially larger and statistically significant. Results at the 25th-75th percentile split are qualitatively similar but attenuated in both regimes, as expected if supply-side heterogeneity is the underlying driver. These findings are consistent with the local projections results in Section (II).

As a complementary exercise, we recompute cross-city standard deviations separately for the most inelastic and most elastic cities (25th and 75th percentiles). For inelastic cities, the bust-to-boom standard deviation ratio is 1.58 (mean) and 1.77 (median), compared to only 1.30 and 1.36 for elastic cities, further corroborating the role of supply-side heterogeneity. Overall, these results suggest that supply-side heterogeneity is a primary driver of the regional disparities documented in Section (II).

Table 5: Peak differential house price response to monetary policy shock

	10th – 90th percentile		25th – 75th percentile	
	Boom	Bust	Boom	Bust
<i>Housing supply elasticities</i>	0.59	4.62	0.48	1.43
	(−0.51, 1.69)	(3.3, 5.94)	(−0.38, 1.34)	(0.70, 2.16)
<i>Employment cyclical</i>	−1.67	−5.53	0.97	0.76
	(−2.33, −1.01)	(−6.89, −4.17)	(0.48, 1.46)	(−0.01, +1.53)

Notes: Responses to an expansionary monetary policy shock (25 bp on impact). Numbers in brackets denote the 68% confidence interval (Newey-West) at the peak.

Alternative explanations. Supply-side heterogeneity need not be the only source of heterogeneity for our main empirical finding. As we show below, demand-side heterogeneities can theoretically generate the same boom-bust asymmetry. We test this by computing income cyclicality for the 20 cities, ranking them, and repeating the local projections exercise.¹⁹ At the 10th-90th percentile split, less income-cyclical cities exhibit a stronger house price response, particularly in busts (see Table (5)).²⁰ However, the signs reverse at the 25th-75th percentile split, suggesting that income cyclicality is unlikely to be a primary driver of the asymmetric house price response across booms and busts.

Further remarks. The presence of regional heterogeneities in housing markets and the observation that house price expectations updating depends on the underlying volatility of the fundamentals raises the question whether house price expectations themselves experience regional heterogeneities. Unfortunately, there are no city-level expectations at an appropriate frequency available to test this. However, our findings suggest that more supply-constrained regions plausibly experience stronger updating, which — as we show below — would reinforce our mechanism. By adopting a common expectation updating process across regions, we therefore take a conservative modelling stance.

Finally, Glaeser and Gyourko (2005) argues that housing supply is fixed on the downward margin, though their analysis operates at long-run rather than business cycle frequencies. Fernandes and Rigato (2023) model time-to-build, which correlates with the housing supply elasticities of Guren et al. (2021), as an endogenous decision, and show that firms decelerate projects during downturns. This implies that high time-to-build regions exhibit prolonged supply responses as construction remains ongoing, while low time-to-build regions reach the extensive margin earlier and halt construction altogether — consistent with our narrative. In our model, we rely on time-to-build as the margin of heterogeneity across regions, directly relating to Fernandes and Rigato (2023).

IV. THE MODEL

With these empirical findings in mind, we develop a multi-region currency union model inspired by Benigno (2004) to connect our empirical evidence and conduct policy experiments. Relative to

¹⁹We use MSA-level income per capita data from FRED, regressing MSA-level income on its national counterpart to obtain a cyclicality estimate used for ranking.

²⁰A possible explanation is that, within the 10th-90th split, cities with highly inelastic housing supply tend to have low income cyclicality.

Benigno, we depart from perfect consumption insurance by introducing incomplete bond markets across regions, and enrich the model with a housing sector along two key dimensions.

First, we incorporate subjective expectations in house price formation. Subjective expectations are applied exclusively to house prices, reflecting both the focus of the paper and the limited availability of forecast data across variables.

Second, we allow for regional heterogeneity in housing markets. Regions differ solely on the supply side, specifically in the time required to build new houses, referred to as time-to-build. This single parameter delivers tractable supply-side heterogeneity across regions and allows for a transparent calibration of the model. Below, we provide evidence that time-to-build relates closely to housing supply elasticities.

In what follows, we focus on a representative region i , with all remaining regions modeled symmetrically.

IV.A The Economy

Households. A representative domestic household in region i derives utility from consuming domestic and foreign varieties of goods from regions $j \neq i$, leisure, and housing. The preferences are as follows:

$$\mathbb{E}_0^{\mathcal{P}} \sum_{t=0}^{\infty} \beta^t \left(\frac{\xi_c c_{i,t}^{1-\sigma}}{1-\sigma} + \frac{\xi_{ih} h_{i,t}^{1-\nu}}{1-\nu} - \frac{\chi n_{i,t}^{1+\varphi}}{1+\varphi} \right)$$

$$c_{it} = [\lambda_{ii}^S c_{ii,t}^{1-S} + \sum_j^N \lambda_{ij}^S c_{ij,t}^{1-S}]^{\frac{1}{1-S}}, \quad c_{ii,t} = \gamma_i \left[\int c_{ii,t}(k)^{\frac{\epsilon-1}{\epsilon}} dk \right]^{\frac{\epsilon}{\epsilon-1}}$$

where $\mathbb{E}_0^{\mathcal{P}}$ denotes the subjective expectations operator discussed below. ξ_j and $j \in \{c, h, a, x\}$ denote model-exogenous shock terms, and $h_{i,t}$ and $n_{i,t}$ denote housing and hours worked, respectively. $\gamma_i \in (0, 1)$ denotes the measure of households in region i . Following Benigno (2004), γ_i is simultaneously the economic size of the region, that is, the mass of variety-producing firms. $c_{i,t}$ denotes the consumption goods basket of region i that is assembled from the varieties of domestically produced goods $c_{ii,t}(j)$ and the varieties of imported goods $c_{ij,t}(k)$, which in turn are aggregates of the constant elasticities of substitution (CES) of the two groups of goods and their varieties. Specifically, $c_{ij,t}(k)$ denotes the consumption of good k consumed by region i and produced by region j . A preference bias for the goods produced in the respective region of residence (“home bias”) arises if $\lambda_{ii} > \sum_{j \neq i}^N \lambda_{ij} \neq \gamma$.²¹ The slope of the housing demand curve is governed by the

²¹Throughout the paper we maintain the assumptions that (i) the degree of home bias is symmetric, and (ii) the bias is such that households favor domestically produced products ($\lambda_{ii} \geq \gamma_i$).

parameter ν .

The household budget constraint is then given by

$$c_{i,t} + q_{i,t}(h_{i,t} - (1 - \delta)h_{i,t-1}) + b_{i,t+1} + \frac{P_{ii,t}}{P_{i,t}}x_{i,t} = w_{i,t}n_{i,t} + (1 + r_{i,t})b_{i,t} + q_{i,t} \cdot H(x_{i,t-\tau_i}, \xi_x) - T_{i,t} + \Sigma_{i,t} + \mathbf{b}_{i,t}$$

The budget constraint is expressed in units of the region i 's final consumption basket, c_i . $\Sigma_{i,t}$ are profits from all domestic firms, which are owned evenly by all domestic households; $w_{i,t}$ is the real wage; $T_{i,t}$ represents government lump-sum taxes; and $b_{i,t}$ is a one-period nominal zero-coupon bond that is traded union-wide. $q_{i,t}$ denotes the real house price, and $x_{i,t}$ is the number of domestic consumption units dedicated to the production of new housing units.

To invest in housing production, households need to purchase domestically produced goods and transform them into housing investment units. Housing production is defined as $H(x_{i,t-\tau_i}, \xi_x) = \xi_x \frac{x_{i,t-\tau_i}^\eta}{\eta}$, $\eta \in (0, 1)$. τ_i denotes the number of quarters it takes to construct new housing units. Housing investment in period t will, therefore, lead to a return on this investment in period $t + \tau_i$. Regional heterogeneities on the housing supply side are modeled by allowing for cross-regional differences in time-to-build (τ_i).

In the steady state, we calibrate the steady-state values for ξ_{hj} , $\forall j$ s.t. $b_{j,ss} = 0$, $\forall j$. Hence, in the steady state, there are no net debtors or net creditor regions. This ensures that the only form of regional heterogeneity is situated on the housing side.

It is convenient to express bond holdings in units of region i 's final basket. Households take as given the real interest rate, $r_{i,t}$, which is determined in equilibrium by the following Fisher-type equation: The value of bond holdings in units of numéraire is $B_{i,t} = P_{i,t} \cdot b_{i,t}$ and the nominal bond pays $i_{t-1} - \psi b_{i,t}$ units of the currency as interest.²² The real interest rate is, thus, given by

$$1 + r_{i,t} = \frac{1 + i_{t-1} - \psi b_{i,t}}{1 + \pi_{i,t}}$$

where $\pi_{i,t} := P_{i,t}/P_{i,t-1} - 1$. Finally, $\mathbf{b}_{i,t} := (\beta^{-1} - 1)(\gamma_i + \sum_{j \neq i}^N \gamma_j \frac{P_{j,t-1}}{P_{i,t-1}})(1 + \pi_{i,t})^{-1} \bar{b}_i$, taken as exogenous by the household, captures the payment streams between i and j that guarantee that households are content with holding no bonds in the non-stochastic steady state with zero inflation and real exchange rate parity.²³

²²The nominal interest rate is elastic in domestic households' aggregate bond holdings. We follow [Schmitt-Grohé and Uribe \(2003\)](#) to ensure stationarity of the first-order dynamics. In Appendix A, we provide a simple micro-foundation for debt-elastic interest rates.

²³Given that bond holding entails a real cost in equilibrium, see footnote 22, introducing payments $\mathbf{b}_{i,t}$ as a way to

Subjective expectations for house price setup. We model house price expectations following the capital gains extrapolation literature (e.g. Adam and Marcet, 2011; Adam et al., 2017). Section (VI) provides empirical support for this choice; Adam et al. (2022) offers additional evidence. Households are endowed with a set of beliefs in the form of a probability measure over the full sequence of variables that they take as given, henceforth external variables $(\xi_t, r_{i,t}, w_{i,t}, \Sigma_{i,t}, T_{i,t}, \mathbf{b}_{i,t}, \pi_{i,t}, (P_{ii,t}/P_{ij,t}), q_{i,t})_{t \geq 0}$.²⁴ We denote this measure as $\mathcal{P}$. Rational expectations are a special case of this setup in the form where households' beliefs agree with the objective or, equivalently, with the "true" or "equilibrium-implied" distribution of the external variables, $\mathcal{P} = \mathbb{P}$. Although households may hold expectations that are generally inconsistent with the equilibrium-implied (conditional) distribution of the external variables, it is worth emphasizing that, first, households have a time-consistent set of beliefs and, second, they behave optimally, given their beliefs. That is, households are *internally rational* in the sense of Adam and Marcet (2011). Moreover, the fact that all households are identical in their beliefs and preferences is not common knowledge among agents, so households cannot discover the misspecification of their beliefs, $\mathcal{P} \neq \mathbb{P}$, by inductively reasoning through the structure of the economy. Given the observed path of the external variables up to period t , households then use this information and $\mathcal{P}$ to form a conditional expectation over the continuation sequence of the external variables, which we denote as $\mathbb{E}_t^{\mathcal{P}}$. As usual, we denote the conditional rational expectations operator by $\mathbb{E}_t$.

We assume that agents have rational expectations with respect to all external variables, except for house prices, $q_{i,t+s}$.²⁵ Households entertain the idea that house prices follow a simple state-space model:

$$\begin{aligned} \ln \frac{q_{i,t+1}}{q_{i,t}} &= \ln m_{i,t+1} + \ln e_{t+1} \\ \ln m_{i,t+1} &= \varrho \ln m_{i,t} + \ln v_{t+1}, \quad \varrho \in (0, 1) \\ (\ln e_t \quad \ln v_t)' &\sim \mathcal{N} \left(\begin{pmatrix} -\frac{\sigma_e^2}{2} & -\frac{\sigma_v^2}{2} \end{pmatrix}, \begin{pmatrix} \sigma_e^2 & 0 \\ 0 & \sigma_v^2 \end{pmatrix} \right) \end{aligned} \tag{4}$$

ensure that there are no bond holding costs in the non-stochastic steady state with zero inflation and real exchange rate parity (i.e., $1 + \pi_{i,t} = 1 + \pi_{j,t} = 1 + \pi_{ii,t} = 1 + \pi_{ij,t} = \frac{P_{ii,t}}{P_{ij,t}} = 1$). This ensures that this steady state is efficient, given that fiscal policy undoes the monopolistic competition distortion. $\mathbf{b}_{i,t}$ may be interpreted as the real interest rate paid by a non-marketable nominal consol that perpetually pays the nominal rate $(\beta^{-1} - 1)(\gamma_i + \sum_{j \neq i}^N \gamma_j P_{j,t-1}/P_{i,t-1})$ of which the household is endowed with $\bar{b}_i$ units. The endowments of these consols ensure that nominal payments balance; that is, $\gamma_i \bar{b}_i + \sum_{j \neq i}^N \gamma_j \bar{b}_j = 0$, see Appendix A. Since we will linearize the model around a steady-state with zero bond holding, $\mathbf{b}_{i,t}$ will be zero in equilibrium.

²⁴ ξ_t denotes the vector of exogenous shocks, which common across regions.

²⁵Formally, $\mathcal{P} := \mathbb{P}_{-q} \otimes \mathcal{P}_q$, where $\mathbb{P}_{-q}$ is the objective measure over the sequences of the external variables without house prices, $\mathcal{P}_q$ is the measure over the sequences of house prices implied by the described perceived model of house prices, and $\otimes$ is the product measure. Since we are interested in a first-order solution to the model, it does not matter what households perceive to be the dependence structure between house prices and the other external variables.

Hence, agents perceive house price growth rates as the sum of a transitory, and a persistent component. Crucially, $\ln e_t$ and $\ln v_t$ are not observable to the agents, rendering $\ln m_{i,t}$ unobservable. Agents apply the optimal Bayesian filter; that is, the Kalman filter, to arrive at the observable system.²⁶

Lemma 1 (House price belief updating). *Applying the Kalman filter to the state-space model and log-linearizing around the non-stochastic steady state gives*

$$\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+s} = \widehat{q}_{i,t} + \frac{1 - \varrho^s}{1 - \varrho} \varrho \widehat{m}_{i,t} \quad (5)$$

$$\widehat{m}_{i,t} = \varrho(1 - g)\widehat{m}_{i,t-1} + g\Delta\widehat{q}_{i,t-1}, \quad (6)$$

where $\ln \bar{m}_{i,t} := \mathbb{E}_t^{\mathcal{P}}(\ln m_{i,t})$ is the posterior mean, $g = \frac{\varrho^2 \sigma_v^2 + \sigma_e^2}{\varrho^2 \sigma_v^2 + \sigma_v^2 + \sigma_e^2}$ is the steady-state Kalman filter gain, $\sigma^2 = \frac{1}{2\varrho^2} [-(\sigma_v^2 + (1 - \varrho^2)\sigma_e^2) + \sqrt{(\sigma_v^2 + (1 - \varrho^2)\sigma_e^2)^2 + 4\varrho^2 \sigma_v^2 \sigma_e^2}]$ is the steady-state Kalman filter uncertainty, and $\ln \widehat{e}_t$ is perceived to be a white noise process.

Proof. See Appendix B.1 for the application of the Kalman filter. Log linearization around the steady state gives the result. ■

Variables denoted with a " $\widehat{\cdot}$ " express the respective variables in percent deviations from their steady-state value. Combining equation (5) and (6) yields equation (2).

Asymmetry in house price beliefs. Our empirical analysis documents that the degree of house price expectation updating varies across regimes. We capture this by allowing the belief persistence parameter ϱ and the Kalman gain g to differ above and below a threshold $\bar{\omega}$. Heterogeneity in belief persistence can be microfounded by assuming agents hold different perceptions of ϱ across regimes (ϱ^h, ϱ^l), while differential Kalman gains arise from allowing the variances of the transitory $((\sigma_e^h)^2, (\sigma_e^l)^2)$ and persistent $((\sigma_v^h)^2, (\sigma_v^l)^2)$ components to differ across regimes. Augmenting the house price belief model (4) with regime-dependent updating and proceeding analogously to Lemma (1) yields the threshold belief updating model:

$$\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} = \mathbb{1}(\Delta\widehat{q}_{i,t-1} > \bar{\omega}) \left[q_{i,t} + \varrho^h \widehat{m}_{i,t} \right] + \mathbb{1}(\Delta\widehat{q}_{i,t-1} < \bar{\omega}) \left[q_{i,t} + \varrho^l \widehat{m}_{i,t} \right] \quad (7)$$

$$\begin{aligned} \widehat{m}_{i,t} = & \mathbb{1}(\Delta\widehat{q}_{i,t-1} > \bar{\omega}) \left[\varrho^h (1 - g^h) \widehat{m}_{i,t-1} + g^h \Delta\widehat{q}_{i,t-1} \right] \\ & + \mathbb{1}(\Delta\widehat{q}_{i,t-1} < \bar{\omega}) \left[\varrho^l (1 - g^l) \widehat{m}_{i,t-1} + g^l \Delta\widehat{q}_{i,t-1} \right] \end{aligned} \quad (8)$$

Combining equation (7) and (8) yields equation (3).

²⁶We assume that the agents' prior variance equals the steady-state Kalman variance.

Firms and price setting. We assume a continuum of monopolistically competitive firms that produce varieties of intermediate goods and have the same beliefs as households. Firms' beliefs, however, concern only the variables over which households have rational expectations. Therefore, firms are rational. Firms buy labor from the representative labor packer and produce good varieties with a linear technology, with labor being the only production factor. The firm sets its retail price and maximizes the expected discounted stream of profits, subject to Rotemberg-type adjustment costs. The linearized Phillips curve reads:

$$\widehat{\pi}_{i,t} = \beta E_t \widehat{\pi}_{i,t+1} + \frac{\epsilon - 1}{\kappa} (\widehat{w}_{i,t} - \widehat{\xi}_{a,t} - \widehat{s}_{i,t}), \quad \widehat{s}_{i,t} = (1 - \lambda) \sum_{j \neq i}^N \widehat{s}_{ij,t}$$

The terms of trade between region i and j are given by $s_{ij,t} = \frac{P_{ii,t}}{P_{ji,t}}$.

Monetary authority. The monetary authority sets nominal interest rates according to a standard Taylor rule targeting currency union consumer price inflation:

$$i_t = \frac{1}{\beta} (\Pi_t^{cu})^{\phi_\pi} \xi_{r,t} \quad (9)$$

Currency union inflation is the average of region-level consumer price inflation ($\Pi_{i,t}$), weighted by the country size: $\Pi_t^{cu} = \prod_{j=1}^N (\Pi_{j,t})^{\gamma_j}$. $\xi_{r,t}$ denotes a monetary policy shock.

Market clearing. To achieve goods market clearing, each goods market for a variety j must clear. Additionally, labor and housing markets must clear within each region. Finally, the balance-of-payments equations must hold. Aggregation and the specific market clearing conditions are relegated to Appendix (B.3).

Equilibrium. We adopt the equilibrium concept of internally rational expectations equilibrium, as defined in [Adam and Marcet \(2011\)](#):

Definition 1 (Internally rational expectations equilibrium). *An IREE consists of three bounded stochastic processes: shocks $(\xi_t)_{t \geq 0}$, allocations $([c_{ij,t}(k)]_{k \in [0,1]}, b_{i,t}, h_{i,t}, x_{i,t}, n_{i,t})_{i \in [1,N], j \in [1,N], t \geq 0}$ and prices $(w_{i,t}, q_{i,t}, i_t, [P_{ij,t}(k)]_{k \in [0,1]})_{i \in [1,N], j \in [1,N], t \geq 0}$, such that in all t*

1. *households choose c, n, h, x, b optimally, given their beliefs $\mathcal{P}$,*
2. *firms set prices optimally, given their beliefs $\mathcal{P}$,*
3. *the monetary authority acts according to the Taylor rule (9),*

4. markets for varieties of consumption goods, hours, and housing clear, given the prices, and the household budget constraints hold.

IV.B The solution method

We solve our model to the first order around a non-stochastic and efficient steady state. This preserves the analytic tractability at many points in the model and allows us to derive results on household behavior under capital gains extrapolation in a closed form. The model presented in Section IV.A admits a unique non-stochastic steady state with zero net inflation and parity of the terms of trade:

Lemma 2 (Non-stochastic steady-state). *Consider the model economy presented in Section IV.A. As the variance of the shocks (actual or perceived) fades, $\text{Var}[\|\xi_t\|] \rightarrow 0$, $\xi_t = (\xi_{a,t}, \xi_{r,t}, e_t, v_t)^\top$, there exists one and only one steady state in which net inflation is zero, $\pi_{ij,ss} = \pi_{i,ss} = 0, \forall i, j \in [1, N]$, and in which the terms of trade are at parity, $s_{i,j,ss} = 1, \forall i, j \in [1, N]$.*

Proof. See Appendix B.4 ■

Linearizing models with capital gains extrapolation is not straightforward. In fact, to the best of our knowledge, we are the first to provide a first-order approximation of a model with capital gains extrapolation under the assumption that agents hold rational expectations outside of the asset pricing block.²⁷ The entire exposition focuses on the typical linearized household problem in region i with all derivations being analogous for the typical household in the remaining regions.²⁸ For expositional clarity, we omit the specification that housing investment occurs in domestically produced goods. This simplification does not affect the fundamental logic of our solution method. We also drop the regional subscript i .

External and internal variables. Standard first-order solution techniques for models with rational expectations rely on a recursive representation of the equilibrium conditions. For instance, the

²⁷Winkler (2020) proposes the “conditionally model-consistent expectations” (CMCE) concept as a starting point for linearizing models with capital gains extrapolation. Under CMCE, however, beliefs over all of the external variables are distorted relative to rational expectations. In our approach, linearized decision rules may be obtained under the assumption that belief distortions apply only to the asset prices, allowing us to confine the deviations from rational expectations to exactly those variables where the survey data allows us to discipline the expectations-modeling choices.

²⁸For any variable $\text{var}_t \notin \{b_{i,t}, \Sigma_{i,t}\}$, define $\widehat{\text{var}}_t := \frac{\text{var}_t - \text{var}_{ss}}{\text{var}_{ss}} \simeq \ln \text{var}_t - \ln \text{var}_{ss}$ to the first order. For $b_{i,t}, \Sigma_{i,t}$ define $\widehat{\text{var}}_t := \frac{\text{var}_t - \text{var}_{ss}}{y_{ss}}$, which allows for the case that $\text{var}_{ss} = 0$. (That is, we scale the deviations in the bond holdings and profits by the GDP.) Note, furthermore, that $\widehat{1 + r_{t+1}} \simeq \ln(1 + r_{t+1}) - \ln(1 + r_{ss}) \simeq r_{t+1} + \ln \beta$. We abuse the notation slightly and write $\widehat{r}$ instead of $\widehat{1 + r}$.

inter-temporal consumption decision is captured by the forward recursive Euler equation:

$$\widehat{c}_t = \mathbb{E}_t \widehat{c}_{t+1} - \frac{1}{\sigma} \mathbb{E}_t \widehat{r}_{t+1}.$$

Under subjective expectations, an equivalent formulation exists:

$$\widehat{c}_t = \mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+1} - \frac{1}{\sigma} \mathbb{E}_t^{\mathcal{P}} \widehat{r}_{t+1}.$$

At this point, it is important to note that, first, we can characterize the external variables for which households have distorted expectations. In our case, this is only the house price, $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+s}$, for which we derived a subjective expectation model. Second, for all other external variables for which households have rational expectations, we can formulate all of the equilibrium conditions recursively in the usual manner. In this case, the equilibrium-implied distribution measure applies and we can drop $\mathcal{P}$ from the expectations operator. The difficulty arises concerning the expectations for household choices or the internal variables, in our case $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+s}$. These variables depend on the external variables over which households hold subjective and rational expectations. In this case, we need to determine the subjective expectations distribution of $\widehat{c}_{t+1}$.²⁹ Our solution method allows us to determine these variables in closed form. We provide a detailed derivation in Appendix B.5 and concentrate here on conveying the intuition of our solution approach.³⁰

Characterizing subjective expectations over internal variables. In solving for the subjectively optimal consumption plan, we exploit two key insights into how households behave to first-order that are valid irrespective of which set of beliefs they hold.³¹ First, there is only one inter-temporal trade-off, namely in consumption, c . Given a path for consumption, the first-order conditions for housing, hours worked, and housing investment uniquely pin down a mapping from the external variables to the decisions for these internal variables. This insight allows us to concentrate on finding the optimal path for consumption.

The second insight is that to the first-order, the permanent income hypothesis holds and consumption depends only on the path of real interest rates, an external variable, and the subjectively

²⁹The reason is that households have distorted expectations over at least one price sequence and, therefore, will make distorted choices; in particular, they plan to make choices in the future that are inconsistent with what these choices will be in equilibrium. If we ignored this; that is, if we exchanged $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+s}$ for $\mathbb{E}_t \widehat{c}_{t+s}$ in the forward iteration above, then the computed equilibrium would be different from the IREE in Definition 1.

³⁰For tractability, we set habit formation, $\bar{h}$, to zero and we assume that housing investments are made on aggregate consumption of goods bundle c_t instead of only locally produced goods. Relaxing these assumptions does not affect our methodological approach.

³¹In particular, our solution method can be used to solve RE models.

expected lifetime income. By iterating forward the Euler Equation we obtain

$$\widehat{c}_t = -\frac{1}{\sigma} \sum_{s \geq 0} \mathbb{E}_t \widehat{r}_{t+s+1} + \mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty, \quad (10)$$

where we define $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty = \lim_{s \rightarrow \infty} \mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+s}$. This shows that the only subjective expectation variable to remain is $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$.³²

At this point, the question arises whether this term is merely a technical artifact originating from our methodological approach or whether it possesses intrinsic economic meaning. For expositional purposes it is informative to consider a transitory shock to the model. Suppose that the shock raises house prices, which leads households to believe that house prices will rise further. Consequently, they believe that they will be wealthier in the future, leading to an increase in their current and planned future consumption. This expectation is reflected in the limit by an increase in $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$, thereby capturing a subjective expectations wealth effect. In essence, due to extrapolation, households believe they will be richer than they actually will be. Under rational expectations, this term also exists. However, households recognize that the shock is purely transitory and will not influence their long-term wealth. Therefore, the wealth effect under rational expectations is always zero.

Characterizing the subjective expectations wealth effect. We now focus on deriving an analytical expression for the subjective expectations wealth effect, $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$. Using the insights from above, the household's first-order conditions are given by the equations in (11). (For expositional brevity, we dropped the exogenous shocks and set time-to-build to zero.)

Rational Expectations	Subjective Expectations	
$\varphi \widehat{n}_t + \sigma \widehat{c}_t = \widehat{w}_t$	$\varphi \widehat{n}_t + \sigma \widehat{c}_t = \widehat{w}_t$	
$\widehat{q}_t = (1 - \eta) \widehat{x}_t$	$\widehat{q}_t = (1 - \eta) \widehat{x}_t$	
$\widehat{c}_t = -\frac{1}{\sigma} \sum_{s \geq 0} \mathbb{E}_t \widehat{r}_{t+s+1} + \mathbb{E}_t \widehat{c}_\infty$	$\widehat{c}_t = -\frac{1}{\sigma} \sum_{s \geq 0} \mathbb{E}_t \widehat{r}_{t+s+1} + \mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$	
$\widehat{h}_t = \frac{\sigma}{\nu(1-\beta)} (\widehat{c}_t - \bar{\beta} \mathbb{E}_t \widehat{c}_{t+1}) - \frac{1}{\nu} \frac{\widehat{q}_t - \bar{\beta} \mathbb{E}_t \widehat{q}_{t+1}}{1-\bar{\beta}}$	$\widehat{h}_t = \frac{\sigma}{\nu(1-\beta)} (\widehat{c}_t - \bar{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+1}) - \frac{1}{\nu} \frac{\widehat{q}_t - \bar{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+1}}{1-\bar{\beta}},$	(11)

where $\bar{\beta} := \beta(1 - \delta)$ and under rational expectations $\mathbb{E}_t \widehat{c}_\infty = 0$, as explained above. From this representation, it becomes evident that the difference between subjective and rational expectations arises from the expectation operator $\mathcal{P}$ applied to next period's consumption and house prices,

³²We directly used the fact that households have rational expectations over external variables other than house prices, $\mathbb{E}_t^{\mathcal{P}} \text{var}_{t+1} = \mathbb{E}_t \text{var}_{t+1}$ for any variable $\text{var}_{t+1} \neq q_{t+1}$.

as well as from the presence of the subjective expectations wealth effect, $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$. In particular, while the labor supply and housing investment decisions remain identical under both expectation formations, the key differences manifest in the Euler equation and the housing demand condition. Subjective expectations with respect to house prices are easy to characterize, as these are pinned down by the subjective expectations model. Further note that subjective expectations for the next period's consumption can be characterized by leading the Euler Equation (10) one period, once we have a representation for the subjective expectations wealth effect. Therefore, to solve the subjective expectations model, it suffices to find a representation for the subjective expectations wealth effect. We find this characterization by combining the first-order conditions with the linearized household budget constraint, (BC) in Appendix-equation (B.6). After iterating over this equation, we can find a closed form expression for the subjective expectations wealth effect.

Proposition 1 (Subjective expectations wealth effect). *The first-order around the non-stochastic steady-state subjective expectations wealth effect, $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$, is given by*

$$\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty = \bar{Q} \cdot \underbrace{[\widehat{q}_t + \bar{M} \widehat{m}_t]}_{=\Lambda_{t,1}} + \bar{C} \underbrace{\left[\sum_{n=1}^{\infty} \beta^n \mathbb{E}_t \{ \bar{z}_{t+n} \} \right]}_{=\Lambda_{t,2}} + \widehat{b}_{t+1}, \quad (12)$$

where $\bar{Q}, \bar{C} \in (0, 1)$, $\bar{M} > 0$ and $\bar{z}_{t+n}$ is a function of external variables the household has rational expectations about, as is specifically stated in Appendix B.5.

Proof. See Appendix B.5 ■

The notation “ss” in the subscript of a variable denotes its’ steady-state values. Equation (12) shows that the subjective expectations wealth effect consists of three parts. First, there are variables for which the household has subjective expectations, $\Lambda_{t,1}$. In our case this is only the house price, $\widehat{q}_t$, and the posterior beliefs about house prices, $\widehat{m}_t$. Second, it depends on the expectations of all of the variables the household has rational expectations about, $\Lambda_{t,2}$, which are collected in z_{t+n}^* . Third, it depends on today's bond choices $\widehat{b}_{t+1}$. Intuitively, households evaluate the wealth effect under subjective expectations by applying the permanent income hypothesis. In doing so they account for their future decisions by using their first-order conditions resulting in a function that depends only on prices and today's saving choices. This formulation elucidates the previously discussed intuition regarding the role of subjective expectations in wealth effects. Specifically, an increase in house prices ($\widehat{q}_t$) or in beliefs about future house prices ($\widehat{m}_t$) lead to a rise in $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$, given that $\bar{Q}, \bar{M} > 0$. The agents, anticipating higher wealth as a result of these changes, adjust their expectations accordingly and expect an increase in consumption in the long run.

Discussion. Our approach offers two advantages over existing methods for solving asset price learning models. First, we employ a first-order approximation, making the solution fast, scalable, and amenable to Ramsey-optimal policy analysis. Prior work has relied on nonlinear (Adam et al., 2017) or hybrid (Adam et al., 2022) solution techniques, which are computationally demanding and limit model complexity. Second, subjective expectations are confined to house prices. The perturbation-based methods of Winkler (2020) and Caines and Winkler (2021) instead impose conditionally model-consistent expectations, under which subjective expectations about one variable spill over to all others, so that households form subjective expectations across the full set of model variables. In our approach, households hold subjective expectations only over house prices and remain rational with respect to all other variables. Our approach of explicitly characterizing choices in terms of lifetime income is general in the sense that it allows for solving household decisions under any time-consistent set of beliefs.

V. ANALYTIC INSIGHTS

We proceed by deriving the primary channel through which house price extrapolation and heterogeneities on the housing supply side can produce the patterns observed in Section (II). Furthermore, we establish that our findings do not hold under a rational expectations framework.

To do so we examine a simplified version of the model introduced above: we consider a one-region, zero liquidity endowment economy with fully sticky prices. We study the path of house prices within an exercise that, while keeping everything tractable, allows us to understand the effects of expansionary monetary policy: the consumption endowment process is selected such that the real rate drops by ε on impact of the shock and returns to 0 afterwards. This is arguably a parsimonious way to model expansionary monetary policy.

We assume that the housing supply is a function of endowment, and moves in response to the shock by $-\iota_i \varepsilon$. Hence, in response to an expansionary monetary policy shock housing supply increases. In the following regional differences in housing supply will be simply modeled as regional variations in ι_i . In Appendix (B.6.3) we also consider the case of demand-side heterogeneities and show that similar results can be attained as under supply-side heterogeneities.

Throughout this section we will consider parameter choices commonly used in the literature. Specifically, we consider parameter choices in line with Section (VI), which are stated in Table (6) and (C.6). Detailed derivations are relegated to Appendix B.6. Under the given assumptions the

house price and housing market clearing can be expressed as:

$$\widehat{q}_{i,t} = \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} - \widehat{r}_{t+1} - \nu(1 - \bar{\beta})\widehat{h}_{i,t} + \nu(1 - \bar{\beta})^2 \sum_{s=0}^{\infty} \bar{\beta}^s \mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1} \quad (13)$$

$$\widehat{h}_{i,t} - (1 - \delta)\widehat{h}_{i,t-1} = -\iota_i \varepsilon_t \quad (14)$$

House price dynamics under rational expectations. As we focus on housing supply side heterogeneities we simplify the housing demand side and assume that $\nu = 1$. The house price is pinned down jointly by the equations (13) and (14). The differences between rational and subjective expectations arise due to the expectation formation on future house prices, $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1}$, and future housing choices, $\mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1}$. Under rational expectations, we can simply use the housing market clearing condition (14) to pin down house prices. Doing so yields the following proposition.

Proposition 2 (House prices under Rational Expectations). *In the outlined one-region, zero liquidity endowment economy with exogenous housing supply, and considering a shock monetary policy shock such that $\varepsilon_1 < 0$ and $\varepsilon_t = 0$ for $t > 1$. House prices are given by:*

$$\widehat{q}_{i,t} = \mathbb{E} \widehat{q}_{i,t+1} - \widehat{r}_{t+1} - (1 - \bar{\beta})\widehat{h}_{i,t} + (1 - \bar{\beta})^2 (1 - \delta)^t \frac{1}{1 - \bar{\beta}(1 - \delta)} (-\iota_i \varepsilon_1)$$

And for the first two periods we have:

$$\widehat{q}_{i,1} = -\varepsilon_1 + \omega_h \iota_i \varepsilon_1$$

$$\widehat{q}_{i,2} = (1 - \delta) \omega_h \iota_i \varepsilon_1$$

with $\omega_h = \frac{1 - \bar{\beta}}{1 - \bar{\beta}(1 - \delta)} \in (0, 1)$.

Proof. See Appendix B.6.1 ■

Proposition (2) makes clear that an expansionary monetary policy shock raises house prices. The shock fades after the first period. If housing supply increases simultaneously ($\iota_i > 0$), the house price increase is less pronounced in the first period. Due to the increase in housing supply, house prices will be below the steady-state level after the first period, converging back to the steady-state going forward. The coefficient ω_h , which scales the effect through housing supply, captures the notion that greater housing supply exerts downward pressure on both current and future prices, owing to the durability of housing.³³

³³In the housing demand equation, $\mathbb{E} \widehat{q}_{i,t+1}$ appears. Iterating forward on this equation yields the closed-form expression for ω_h : since additional supply today depresses future prices, this feeds back into current prices. Proposition (2) derives this channel explicitly.

House price dynamics under subjective expectations. Moving on to the subjective expectation case, house price expectations, $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1}$, are pinned down by the subjective expectations model, equations (5) and (6). Future expected housing choices, $\mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1}$, can be characterized using the solution method described in Section (IV.B). We arrive at the following proposition:

Proposition 3 (House prices under Subjective Expectations). *In the outlined one-region, zero liquidity endowment economy with exogenous housing supply, and considering a shock monetary policy shock such that $\varepsilon_1 < 0$ and $\varepsilon_t = 0$ for $t > 1$. House prices are given by:*

$$\widehat{q}_{i,t} = -\widehat{h}_{i,t} - \omega_r \widehat{r}_{t+1} + \omega_m \widehat{m}_{i,t}$$

And for the first two periods we have:

$$\begin{aligned}\widehat{q}_{i,1} &= -\omega_r \varepsilon_1 + \iota_i \varepsilon_1 \\ \widehat{q}_{i,2} &= (1 - \delta) \iota_i \varepsilon_1 + \omega_m g \widehat{q}_{i,1}\end{aligned}$$

with $\omega_r = 1 + \frac{y_{ss}(1-\beta)}{y_{ss} + h_{ss} q_{ss} \delta} > 1$, $\omega_m > 1$, and $\frac{\partial \omega_m}{\partial \rho} < 0$.

Proof. See Appendix B.6.2 ■

Proposition (3) elucidates several key dynamics. First, an expansionary monetary policy shock raises house prices, and given that $\omega_r > 1$, this increase is more pronounced than under RE. The amplified impact response of house prices under SE follows directly from the dual role of prices: they signal not only current scarcity but also the expected future path of asset values. Under RE, agents anticipate mean-reversion in house prices; under SE, agents apply their perceived model and project elevated prices to persist. Formally, the heightened pass-through of the monetary shock to house prices, captured by $\omega_r > 1$ under SE relative to unity under RE, arises because expectations affect the house price not only directly through the future house price in the housing demand equation, but also indirectly through agents' choices.³⁴ Second, a positive housing supply response attenuates the monetary-policy-induced rise in house prices, and this mitigating effect is stronger than under RE, as reflected by $\omega_h < 1$. As noted above, under RE agents implicitly internalize the fact that higher housing supply depresses future prices, thereby restraining current prices as well; this forward-looking channel is absent under SE. Third, belief extrapolation introduces a backward-looking component that exerts upward pressure on house prices. An increase in the Kalman gain (g) amplifies this channel, as does a decrease in the persistence of belief formation (ρ). While both

³⁴Solving for $\sum_{s=0}^{\infty} \bar{\beta}^s \mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1}$ introduces a distinct interest rate pass-through, as future expected housing choices depend on all exogenous variables, including the path of interest rates.

mechanisms generate amplification in theory, the quantitative exercises below demonstrate that the Kalman gain is the quantitatively dominant channel.

Regional heterogeneities. Let us now turn towards understanding how these dynamics affect regional dispersions. Consider an expansionary monetary policy shock that is symmetric across regions and results in a heterogeneous increase in housing supply as ι_i differs across regions. Using the results from above, we can derive the regional standard deviation two periods after the impact of the shock:

Proposition 4 (Regional standard deviation). *The regional standard deviation of house price growth in period two under rational expectations is given by:*

$$\sigma_{\Delta q,2} = \delta \omega_h \sqrt{\frac{1}{N} \sum_i^N [(\bar{\iota} - \iota_i) \epsilon_1]^2}, \quad \bar{\iota} = \frac{1}{N} \sum_i^N \iota_i$$

The regional standard deviation of house price growth in period two under subjective expectations is given by:

$$\sigma_{\Delta q,2} = (\omega_m g - \delta) \sqrt{\frac{1}{N} \sum_i^N [(\iota_i - \bar{\iota}) \epsilon_1]^2}, \quad \bar{\iota} = \frac{1}{N} \sum_i^N \iota_i$$

where $\omega_m g \gg \delta$ generally holds. And a higher Kalman gain, g , or lower belief persistence, ϱ , amplifies regional differences:

$$\frac{\partial \sigma_{\Delta q,2}}{\partial g} = \omega_m \sqrt{\frac{1}{N} \sum_i^N [(\iota_i - \bar{\iota}) \epsilon_1]^2} > 0, \quad \frac{\partial \sigma_{\Delta q,2}}{\partial \varrho} = g \frac{\partial \omega_m}{\partial \varrho} \sqrt{\frac{1}{N} \sum_i^N [(\iota_i - \bar{\iota}) \epsilon_1]^2} < 0$$

Proof. The regional standard deviations are obtained by first computing house price growth from Propositions (2) and (3), respectively, and subsequently calculating the corresponding standard deviations across regions. ■

Proposition (4) summarizes the paper's main argument. Under both rational and subjective expectations, the presence of regional heterogeneity, $\sum_i^N (\iota_i - \bar{\iota}) \neq 0$, generates dispersion in house price growth. With rational expectations, this dispersion remains limited, as $\omega_h \in (0, 1)$ and δ is small and positive. Under subjective expectations, however, dispersion can become substantial since $(\omega_m g - \delta) > 0$ and $\omega_m g$ may be large.³⁵ Moreover, it becomes apparent that an increase in

³⁵Numerical simulations indicate that $\omega_m g$ can reach values up to 73, implying strong amplification. For realistic Kalman gains of $g \in [0.01, 0.025]$, we use $g = 0.02$ in our baseline example.

the Kalman gain g , or a decrease in the perceived persistence ρ , leads to an increase in house price growth dispersion. These findings therefore directly connect to the empirical evidence presented in Sections (II) and (III): Regional heterogeneities cause differences in house price growth rates which are amplified through expectations in busts.

VI. CALIBRATION

The model is calibrated at quarterly frequency. Regional supply-side heterogeneity is calibrated using 20 regions, each corresponding to a metropolitan area from the Case-Shiller U.S. National Home Price Index. The slope of the Phillips curve is chosen so that the model matches the peak response to a 25 basis point monetary policy shock, estimated unconditionally on boom and bust regimes. Remaining parameters follow standard values from the literature.

Calibrating expectations and regional heterogeneities. The house price expectation formation is calibrated according to our estimations in Table (3). To calibrate supply-side heterogeneity, we draw on time-to-build data from the U.S. Census, which measures the duration between building authorization and construction completion.³⁶ However, these data are only available at the Census Division level. To obtain corresponding measures at the metropolitan level, we employ the housing supply elasticities from [Guren et al. \(2021\)](#), aggregate them to the Census Division level, and regress them onto the Census time-to-build measures. As shown in Appendix (C.7), these variables exhibit a strong positive correlation. We then use the estimated slope coefficient from this regression to infer time-to-build measures at the city level. Table (C.5) shows the results. Since the final model is calibrated to quarterly frequency, we round the monthly time-to-build data to the nearest quarter.

Remaining model parameters. Table (6) lists the parameter values, all of which are symmetric across regions. First, focusing on the household sector, the labor disutility shifter, the inverse Frisch elasticity, and the intertemporal elasticity of substitution are set to standard values in line with the literature. The discount factor is set to achieve a 2.0% steady-state interest rate. We further assume the regions are symmetric in size. We set housing depreciation to 2% per quarter. Moving to the production sector, we set the home bias, the elasticity of substitution across regions and goods in accordance with [Bletzinger and von Thadden \(2021\)](#). We will choose the persistence parameter of house price beliefs and the Kalman gain as presented in Table (3). The Taylor rule weight on inflation is 1.27, in line with [Iacoviello \(2005\)](#). In terms of price adjustment, we choose the

³⁶We use annual averages from 1999 to 2017.

price adjustment costs such that the peak response of house prices to a monetary policy shock, unconditional on booms or busts, matches the response in the data. The slope of the Phillips Curve is given by $\frac{\epsilon-1}{\kappa} = 0.006$. This is in line with the recent literature.³⁷

Table 6: Model parameters (symmetric parameters)

Parameter		Value	Description	Source/ Target
Households	χ	1.000	labor disutility shifter	standard
	φ	1.250	inverse Frisch elasticity	standard
	σ	2.000	inverse of intertemporal EOS	standard
	ν	1.000	housing utility elasticity	Iacoviello (2005)
	δ	0.020	housing depreciation	2% quarterly depreciation
	β	0.995	discount factor	standard for quarterly frequency
	γ	0.500	relative region size	symmetric regions
Goods aggregation & production	λ	0.800	home bias	
	ς	1.000	EOS across regions	Bletzinger and von Thadden (2021)
	ϵ	6.000	EOS across varieties	
	κ	858.40	price adjustment costs	slope of 0.006 for the Phillips curve
	η	0.800	elasticity of housing production	Adam et al. (2022)
Policy	ϕ_π	1.270	Taylor coefficient	Iacoviello (2005)

Notes: All parameters depicted above are equal across countries. One period in the model is one quarter.

The steady-state allocation and prices are reported in Table (C.6). The housing preference shifter is chosen to ensure a symmetric steady state in allocation variables, so that all cross-regional differences arise solely from supply-side heterogeneity. The steady-state house price is the only variable that differs across regions.³⁸ Symmetric steady-state values for bond levels, which are zero for all regions, also imply that there is no Fisherian debt revaluation effect. We parameterize ξ_x to match the housing sector in the US economy and target a steady-state housing investment-to-consumption ratio, $\frac{x_{ss}}{c_{ss}}$, of 6.5% as in [Adam et al. \(2022\)](#).

³⁷[Adam and Billi \(2006\)](#) set the slope of the Phillips Curve to 0.057. However, it has been argued that the slope of the Phillips Curve has decreased, see i.e [Del Negro et al. \(2020\)](#) and [Hazell et al. \(2022\)](#). Our parameter choice is in line with their estimates.

³⁸Numerically, these differences are small: steady-state house prices range from 0.395 in the region with the shortest time-to-build to 0.405 in the region with the longest, with corresponding values of ξ_h ranging from 0.104 to 0.106.

VII. QUANTITATIVE RESULTS

In this section, we will present our findings. First, we show that the model is able to replicate aggregate house price responses to a monetary policy shock, conditional and unconditional on booms and busts. Second, the model captures the documented boom-bust-asymmetry in regional house price growth dispersion.

VII.A Aggregate house prices in booms and busts

We first assess model performance at the aggregate level. All aggregate variables are convex combinations of regional variables weighted by region size, i.e. $\widehat{q}_t^{agg} = \sum_{i=1}^N \gamma_i \widehat{q}_{i,t}$. We examine the aggregate house price response unconditionally and conditional on boom and bust regimes, targeting the unconditional peak response in calibration, and then compare the rational and subjective expectations models conditional on regime.

The unconditional model response uses the linear model with the baseline parameterization, while the unconditional empirical response is obtained by estimating Equation (C.3) without conditioning on regime. For the conditional responses, we adjust the belief updating parameters to match either boom or bust dynamics, using the estimates from Table (3). Specifically, the boom calibration sets $\varrho^h = 0.99$ and $g^h = 0.013$, while the bust calibration sets $\varrho^l = 0.84$ and $g^l = 0.022$. Since these parameters do not affect the steady state, both models are shocked from the same steady state but exhibit different dynamics due to regime-specific updating. We consider a 25 basis point expansionary monetary policy shock across all specifications.³⁹ As discussed above, shocks of this magnitude are empirically too small to trigger a regime transition, validating the quantitative exercise.

Figure (4) plots the IRFs of the models associated with table (7). In terms of dynamics, we typically observe a very sluggish response in the data (see for instance Appendix (C.5)).⁴⁰ The rational expectations model produces no hump-shaped dynamics, with house prices peaking on impact and returning immediately to steady state. In the subjective expectations model, we do observe some sluggishness in the house price dynamics. House prices respond on impact and increase for the first four quarters, after converging back to the steady state. The extrapolative belief structure drives this dynamic: following the initial shock, agents anticipate further price increases and invest in housing until their expectations are not met, at which point beliefs are

³⁹The empirical shock size is determined by estimating a local projection of the federal funds rate on the monetary policy shock and taking the trough of the impulse response in the first quarter as the reference point, both unconditionally and conditional on regime.

⁴⁰In the case of the local projections shown in Appendix (C.5), the responses are extremely sluggish.

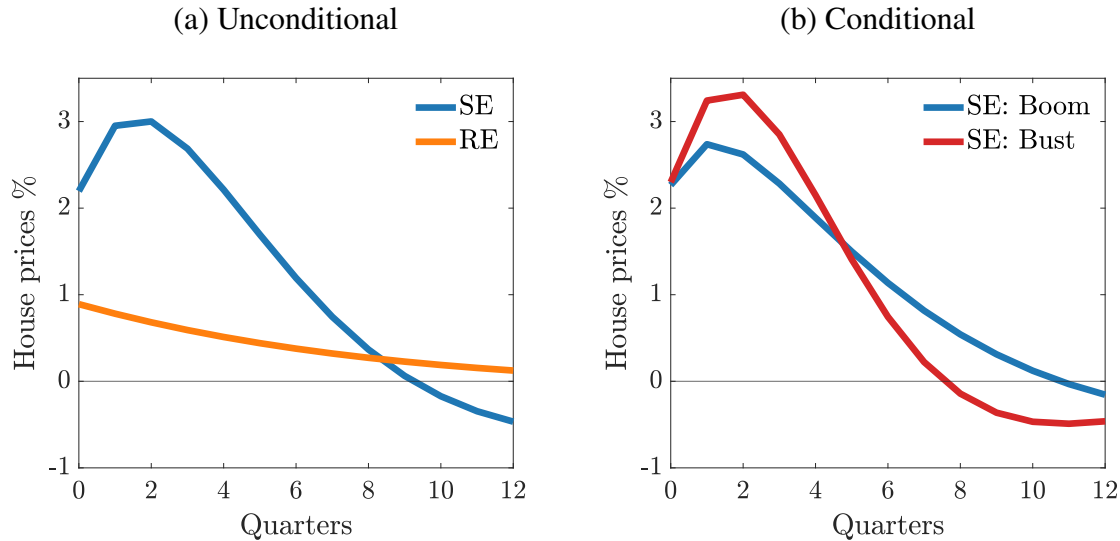
Table 7: Aggregate house price response to monetary policy shock at peak

	<i>Unconditional</i>	<i>Boom</i>	<i>Bust</i>
<i>Data</i>	3.00	2.21	4.28
	(1.43, 4.57)	(0.46, 3.97)	(1.40, 7.18)
<i>Subjective Expectations</i>	3.00	2.74	3.31
<i>Rational Expectations</i>	0.89	0.89	0.89

Notes: The unconditional model uses the calibration of the linear from Table (3), Boom and Bust the respective calibrations from the same table. Confidence intervals in the data are one standard deviation from the local projections exercise.

revised and the model returns to steady state. Quantitatively, however, this persistence falls short of what is observed in the data. This is a well-known limitation of DSGE models that typically requires additional backward-looking components, such as habit formation to fully match empirical persistence (Smets and Wouters, 2007).

Figure 4: House price response to monetary policy shock



Notes: Responses to expansionary monetary policy shock (25 bp on impact).

VII.B Cross-regional heterogeneity in booms and busts

Having demonstrated that our model can replicate aggregate house price responses during both boom and bust periods, we now turn our attention to analyzing cross-regional heterogeneities. We evaluate the model's performance in response to a monetary policy shock conditional on booms

and busts.

Regional house price growth dispersion. We begin by comparing the model-implied cross-regional standard deviation of house price growth and the inelastic-minus-elastic house price differential to their empirical counterparts from the local projections in Figure (1) and Table (5), using the 25th/75th percentile split for the latter. We compare peak impulse responses in booms and busts between the data and both a rational and a subjective expectations model. Results are presented in Table (8).

The rational expectations model generates virtually no regional differences: peak responses in both the standard deviation and the differential are essentially zero, with no distinction between booms and busts. The subjective expectations model performs considerably better. In booms, the model slightly overstates the peak standard deviation response and more substantially overstates the differential price response, though both remain within one standard deviation confidence interval. In busts, the model lies slightly above the point estimates for the standard deviation but again within the confidence intervals, while fitting the differential house price response well. Generally, our model does well in replicating the observed cross-regional variations in booms and busts, in particular in comparison to the rational expectations model.

These results are in line with Proposition (4): Housing supply-side heterogeneities create differences in house price growth, which in turn is amplified by extrapolation. Larger extrapolation in busts means stronger amplification and, therefore, larger cross-regional differences. Under rational expectations, the amplification mechanism is absent; therefore, the model is only able to create minor cross-regional differences.

Table 8: Peak Responses to a Monetary Policy Shock: Cross-Regional Dispersion and House Price Differentials, Model & Data

	<i>Standard deviations</i>		<i>Differential house prices</i>	
	<i>Boom</i>	<i>Bust</i>	<i>Boom</i>	<i>Bust</i>
<i>Data</i>	0.11 (0.03,0.19)	0.24 (0.12,0.37)	0.48 (−0.38, 1.34)	1.43 (0.70, 2.16)
<i>Subjective Expectations</i>	0.15	0.31	1.23	1.54
<i>Rational Expectations</i>	0.01	0.01	0.02	0.02

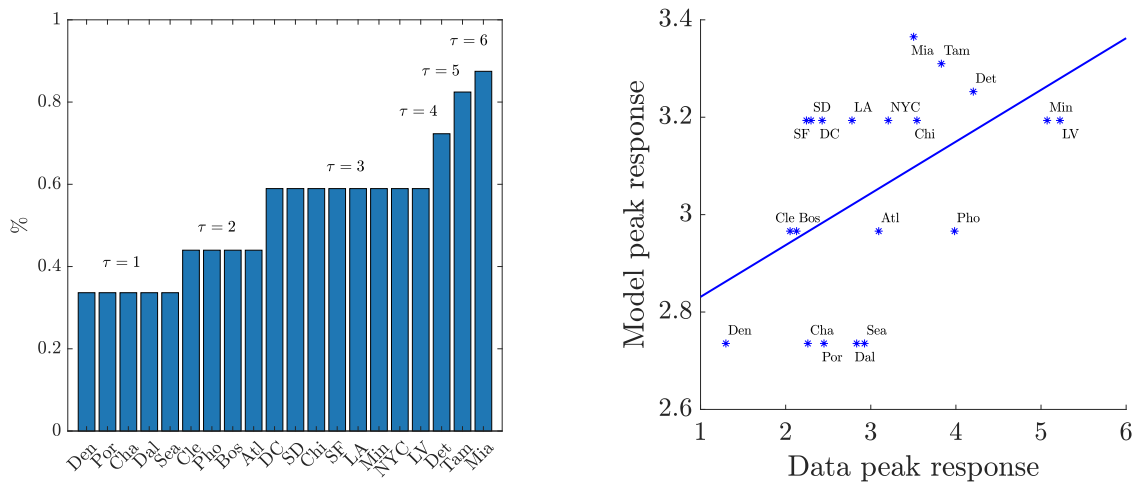
Notes: Response to a monetary policy shock (25 bp on impact); confidence intervals: 68% (Newey-West). The standard deviations and differential house prices in the data are taken from Figure (1) and Table (5). For the differential house prices we focus on the 25th/75th percentile split.

Figure (5) illustrates city-level house price responses. Panel (a) shows the cross-city standard

deviation of house prices in booms and busts: consistent with our theoretical results, cities with more inelastic housing supply exhibit stronger responses and larger boom-bust differentials. Panel (b) compares model-implied peak responses with their empirical counterparts from unconditional local projections, revealing a strong positive correlation. Notably, this close correspondence arises even though time-to-build constitutes the sole source of heterogeneity in the model, indicating that it captures a key dimension of regional variation in housing markets. These findings are consistent with [Aastveit and Anundsen \(2022\)](#).

Figure 5: City level responses

(a) Standard deviation across Boom & Bust (b) Unconditional peak responses: Model & data

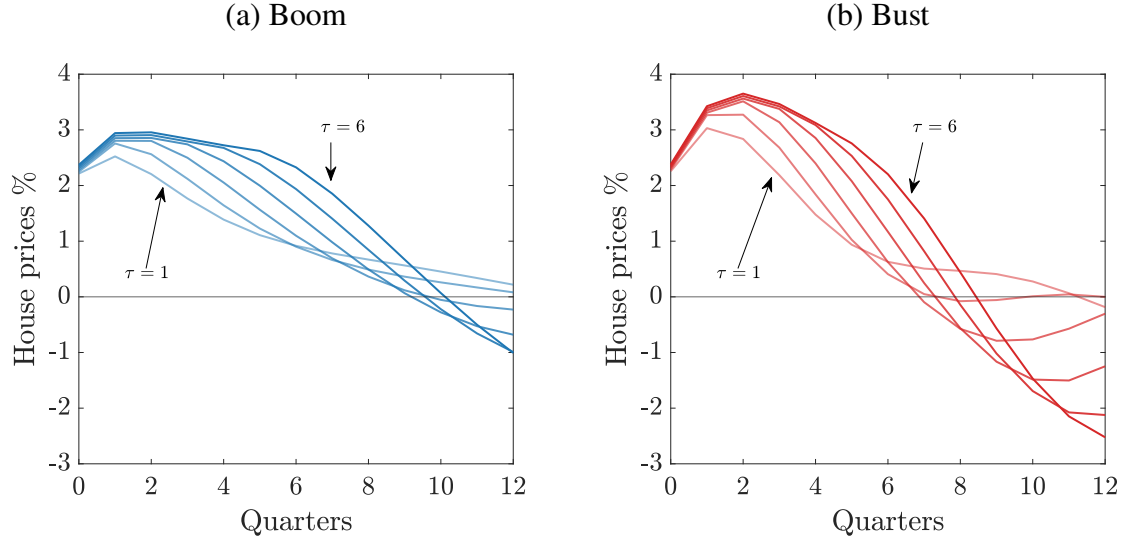


Notes: Panel(a): City level standard deviations of house prices across booms and busts to an expansionary monetary policy shock (25 bp on impact). The city name abbreviations are defined in Table (C.5). Panel (b): Peak response to the same shock in the data and model. The data response is generated by a local projections exercise. Specifically, we replace the aggregate house price in appendix (C.5) with the city counterpart unconditional on being in a boom or bust.

Figure (6) presents the corresponding impulse responses of house prices to a monetary policy shock, conditional on booms and busts. Because of the quarterly calibration, several cities share identical time-to-build durations, leading to overlapping IRFs. The dynamics confirm that regions with longer time-to-build exhibit stronger responses, and that regional dispersion is greater in busts—consistent with the analytical results in Section (V). For comparison, Appendix (D.1) shows that under rational expectations, the model generates no substantial regional dispersion.

House prices and economic activity. Having established the boom-bust asymmetry in regional house price responses, we now examine the implications for aggregate economic activity. These dynamics are of particular interest due to their implications for welfare and, consequently, their relevance for policy considerations. House prices directly affect output through housing investment:

Figure 6: Regional house price responses to a monetary policy shock in booms and busts



Notes: Responses to an expansionary monetary policy shock (25 bp on impact).

from the investment condition (11), setting time-to-build to zero for brevity, we have:

$$\hat{q}_{i,t} = (1 - \eta)\hat{x}_{i,t}$$

Higher house prices translate into increased housing investment which, through market clearing, raises aggregate output. Table (9) reports the peak responses of aggregate output and its cross-regional standard deviation. Both mirror the patterns observed in house price dynamics, reflecting the spillover from housing markets to output. As these regional disparities propagate through the model, they affect consumption, labor markets, and inflation, with direct implications for welfare and monetary policy, which we discuss in the next section.

Table 9: Peak output response to monetary policy shock

	Boom	Bust
<i>Aggregate Response</i>	1.47	1.71
<i>Cross – regional Variation</i>	0.13	0.21

Notes: Responses to an expansionary monetary policy shock (25 bp on impact). The Boom and Bust models use the calibrations from Table (3). The cross-regional variation is computed using the regional standard deviation.

VIII. POLICY ANALYSIS

Turning to policy implications, we study how targeting house prices in the Taylor rule changes aggregate variation and cross-regional variation. The linearized Taylor rule we consider is given by:

$$i_t = \phi^\pi \widehat{\pi}_t^{agg} + \phi^q \widehat{q}_t^{agg} \quad (15)$$

For our baseline case, we consider a rule with $\phi_\pi = 1.27$ and $\phi_q = 0$, as we did for all exercises above. Under house price targeting, we increase the weight on house prices such that $\phi_q > 0$. In the following, we will first focus on how targeting house prices changes the responses of key model variables. We then turn to the welfare implications of this policy. Going forward, we study the response of house prices to an expansionary productivity shock.

Consequences of house price targeting. We assess the effects of house price targeting by comparing aggregate and cross-regional volatility of inflation, output, and house prices in response to a productivity shock, computed over the first 24 quarters after the shock.

Under subjective expectations, house price targeting reduces aggregate volatility in both output and house prices, with the effect being particularly pronounced during busts. The reduction in output volatility follows from lower fluctuations in housing investment, which dampen aggregate output. This comes at the cost of higher inflation volatility, as the greater emphasis on house prices in the policy rule reduces the relative weight placed on inflation stabilization. Under rational expectations, qualitatively similar dynamics arise, but the changes in volatility are smaller and symmetric across booms and busts, reflecting the absence of any source of asymmetry in that framework.

Turning to regional dispersion, house price targeting significantly reduces cross-regional differences in output and house prices under subjective expectations, again with stronger effects in busts. The intuition is straightforward: extrapolation amplifies house price movements that spill over to the real economy; consequently, stabilizing house prices dampens the amplifying force and thereby also the differential regional responses they generate. The regional dispersion in inflation remains low throughout, owing to the flat Phillips curve. Under rational expectations, the model generates no meaningful regional variation to begin with, so targeting has little effect on dispersion. The results are reported in Tables (D.1) and (D.2) in the Appendix.

Welfare implications. The welfare-based loss function is derived as a second-order approximation of household utility around the efficient steady state, weighted by region size, and expresses welfare losses in terms of the equivalent permanent consumption decline as a fraction of steady-state consumption, following Galí (2015). We consider the paternalistic case. This loss function can be

expressed in terms of aggregate and cross-regional volatility (see Appendix (D.2) for details):

$$\frac{\mathbb{W}_{-1}}{c_{ss}} = -\frac{c_{ss}\sigma}{2} \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \left[\Omega_t^{agg} + \Omega_t^{var} + \Gamma^s \sum_{j \neq i}^N \hat{s}_{ijt}^2 + O(3) + \text{t.i.p.} \right] \quad (16)$$

Ω_t^{agg} collects the aggregate variation on a currency area level and Ω_t^{var} is the cross-regional variation:

$$\begin{aligned} \Omega_t^{agg} &= \Gamma^\pi (\hat{\pi}_t^{agg})^2 + \Gamma^c (\hat{c}_t^{agg})^2 + \Gamma^h (\hat{h}_t^{agg})^2 + \Gamma^n (\hat{n}_t^{agg})^2 + \Gamma^x (\hat{x}_t^{agg})^2 - \Gamma^{\xi_a} \hat{\xi}_{a,t} \hat{n}_t^{agg} \\ \Omega_t^{var} &= \Gamma^\pi \hat{\sigma}_{\pi,t}^2 + \Gamma^c \hat{\sigma}_{c,t}^2 + \Gamma^h \hat{\sigma}_{h,t}^2 + \Gamma^n \hat{\sigma}_{n,t}^2 + \Gamma^x \hat{\sigma}_{x,t}^2 + \Gamma^b \hat{\sigma}_{b,t}^2 \end{aligned}$$

Before we turn to the results, we briefly review the sources of inefficiency in this model. Three important inefficiencies arise. First, under subjective expectations, agents' choices are distorted due to their misspecified beliefs. In particular, households wishing to achieve capital gains in the future may be willing to shift resources from consumption units to housing demand and investment, eventually also distorting labor supply choices. In the welfare loss functions, this channel contributes to the quadratic deviations of $(\hat{c}_t^{agg})^2$, $(\hat{h}_t^{agg})^2$, $(\hat{n}_t^{agg})^2$, and $(\hat{x}_t^{agg})^2$ from their efficient benchmarks. While such allocation gaps may also arise from other frictions—such as nominal rigidities—even under rational expectations, misspecified beliefs represent an additional source of these inefficiencies.⁴¹ Second, welfare losses arise from regional heterogeneities through the convexity of the loss function: reducing dispersion by shifting volatility from more to less volatile regions enhances aggregate welfare. This is captured by the term Ω_t^{var} . Moreover, welfare losses directly occur through deviations in relative prices, $(\hat{s}_{ij,t})$. Finally, the model features the standard inefficiency associated with inflation, which arises from price adjustment costs.

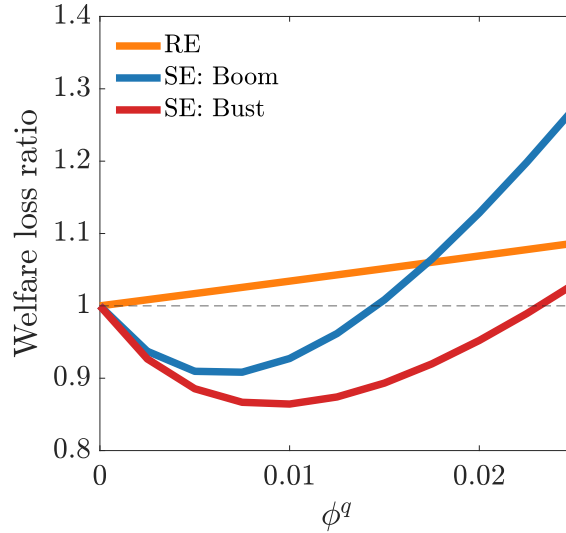
As discussed above, house price targeting reduces aggregate variance in house prices and output and compresses regional dispersion across all variables, but at the cost of higher inflation volatility. Under subjective expectations, this creates a trade-off: the policy corrects belief-distorted choices and mitigates regional heterogeneities, but raises inflation. Under rational expectations, both of the former two inefficiencies are either absent or negligible, leaving the policymaker with little incentive to lean against house prices.

We evaluate welfare losses under both the baseline and house price targeting regimes by computing the welfare loss ratio $\frac{\mathbb{W}(\phi^q=z)}{\mathbb{W}(\phi^q=0)}$, where z captures the degree of house price targeting. Values below unity indicate welfare improvements relative to inflation targeting. Figure (7) plots the results. Under subjective expectations, house price targeting enhances welfare, as both belief

⁴¹See Adam and Woodford (2021) for more details.

distortions and regional heterogeneities are present. The gains are particularly pronounced in busts, when stronger expectation updating amplifies both inefficiencies, and leaning against house prices dampens actual and expected price dynamics, thereby mitigating them. In contrast, under rational expectations, house price targeting does not yield welfare improvements, as the primary source of inefficiency stems solely from inflation. This finding aligns with the existing literature (Adam and Woodford, 2021; Bianchi et al., 2024b).

Figure 7: Welfare loss ratio across different degrees of house price targeting



Notes: The welfare loss ratio is defined as $\frac{\mathbb{W}(\phi^q=z)}{\mathbb{W}(\phi^q=0)}$. A welfare loss ratio smaller than unity indicates a policy that improves welfare relative to inflation targeting.

IX. CONCLUSION

This paper examines regional heterogeneity in housing cycles and its implications for monetary policy. We document that cross-regional dispersion in house price growth is systematically larger in busts than in booms, and attribute this asymmetry to stronger expectation extrapolation during busts and regional housing supply heterogeneity. A multi-region New Keynesian model incorporating both features jointly replicates the observed increase in dispersion during busts. Our findings suggest that a monetary authority placing greater weight on house prices can reduce volatility in output and house prices as well as cross-regional dispersion. This policy is welfare-improving under subjective beliefs.

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ONLINE APPENDIX

A. MICRO-FOUNDING THE DEBT-ELASTIC INTEREST RATE

Without loss of generality, we focus in this section on a two-region case, a home country H and a foreign country F , with all variables for the foreign region denoted by an asterisk. In the model, households in country H receive on their bond holdings the effective nominal interest rate $1 + i_{t-1} - \psi b_t$, with b_t being the real value of the aggregate bond holding in country H ; households in country F receive the effective nominal rate $1 + i_{t-1} - \psi b_t^*$. Moreover, the intermediation of bond positions entails a real cost $\gamma(1 + \pi_t)^{-1} \frac{\psi}{2} (b_t)^2 + (1 - \gamma)(1 + \pi_t)^{-1} \frac{\psi}{2} (b_t^*)^2$, of which $(1 + \pi_t)^{-1} \frac{\psi}{2} (b_t)^2$ is paid by each consumer in H and $(1 + \pi_t)^{-1} \frac{\psi}{2} (b_t^*)^2$ is paid by each consumer in F . In this appendix, we detail how these debt-elastic interest rates and associated intermediation costs can be parsimoniously micro-founded. We achieve this by introducing two competitive bond clearing houses, one in each country, that represent households' only access to financial markets and where they incur a real cost that is quadratic in the size of their balance sheet. The specific market arrangement is as follows: Households hold a consol and may hold liquid bonds.

Consol. Each household in H is endowed with $\bar{b} \in \mathbb{R}$ units of a non-marketable consol⁴² that pays as a coupon $(\beta^{-1} - 1)(\gamma P_{t-1} + (1 - \gamma)P_{t-1}^*)P_t^{-1}$ units of H 's consumption basket each period, per unit of consol. This implies that the nominal coupon rate, applied to the nominal coupon value $P_{t-1}\bar{b}$, is $(\beta^{-1} - 1)(\gamma P_{t-1} + (1 - \gamma)P_{t-1}^*)/P_{t-1}$; the real coupon rate applied to the real value $\bar{b}$, in turn, is $(\beta^{-1} - 1)\left(\gamma + (1 - \gamma)\frac{P_{t-1}^*}{P_{t-1}}\right)(1 + \pi_t)^{-1}$. The situation in country F is symmetric: Each household is endowed with $\bar{b}^*$ units of a consol that pays $(\beta^{-1} - 1)(\gamma P_{t-1} + (1 - \gamma)P_{t-1}^*)(P_t^*)^{-1}$ units of F 's consumption basket each period, per unit of consol. $\bar{b}$, $\bar{b}^*$ are model parameters selected such that (i) $\gamma\bar{b} + (1 - \gamma)\bar{b}^* = 0$ and (ii) all markets clear in the non-stochastic steady state with zero net inflation and terms-of-trade parity *without* households holding any liquid bonds. The latter fact ensures that there are no costs of financial intermediation in the steady state, shutting down this particular friction. The specific choice of the coupon payment scheme ensures two facts: (1) condition (i) implies that the nominal payments between H and F associated with the two consols exactly cancel out—whatever H/F receives as coupon payments on its consol endowment is paid for by F/H as a coupon service on its (endowed) short position of the consol; and (2) the real coupon rates paid by/ to the consol endowment only depend on the real exchange rate and the inflation rates and not on the price levels. Households cannot trade their consol holdings.

⁴²A consol is a type of bond that has infinite maturity and perpetually pays a constant or varying coupon rate.

Bonds. Households do have the possibility, however, to vary their position in the liquid bond. This liquid bond is a nominal, one-period, zero-coupon bond and the positions of the representative H -, respectively, the representative F -household, are denominated as b_t, b_t^* . If a household wants to hold a net balance of liquid bonds different from zero, they must go to one of the clearing houses in their country: In the H -country, there is a continuum of mass γ (respectively mass $1 - \gamma$ in F) of competitive clearing houses buying and selling bonds from and to the government and the respective country's citizens. Households themselves cannot directly buy/sell government bonds without having an account at the clearing house. The clearing house can costlessly buy/sell bonds but it incurs an operating cost that is quadratic in the size of its balance sheet, making this a model of costly financial intermediation. Thus, the interest rate that each citizen receives on their bond holdings is determined by the nominal rate paid on government bonds and the aggregate holding of liquid bonds. Each clearing house is owned equally by all citizens of the respective country so that it pays its profits to those citizens.⁴³ Consider an arbitrary clearing house in H (with symmetric arrangements in F). Denoting as $B_{c,t+1}$ the nominal value of the clearing house's net liabilities against H 's citizens and as $B_{g,t+1}$ the nominal value of the clearing house's position in the government bond, the profit maximization program is

$$\max_{B_{c,t+1}, B_{g,t+1} \in \mathbb{R}} -(1 + i_t^b)B_{c,t+1} + (1 + i_t)B_{g,t+1} - \frac{\psi}{2}P_t^{-1}(B_{c,t+1})^2, \text{ s.t. } B_{c,t+1} = B_{g,t+1},$$

where i_t^b is the nominal rate clearing of the market for household bond positions and i_t is the nominal rate the monetary authority sets on government bonds. $\frac{\psi}{2}P_t^{-1}(B_{c,t+1})^2 = \frac{\psi}{2}P_t(b_{c,t+1})^2$ is the nominal cost of intermediating—crucially, the real cost of intermediation does not directly depend on the price level. The first order conditions for this program are

$$\begin{aligned} 1 + i_t^b + \psi P_t^{-1}B_{c,t+1} &= \mu_t, \\ 1 + i_t &= \mu_t, \\ B_{c,t+1} &= B_{g,t+1}, \end{aligned}$$

⁴³In equilibrium, each clearing house makes a non-negative profit, and along the transition path back to the steady state after some shock, each clearing house makes a strictly positive profit. This fact is, in principle, incompatible with the notion of competitiveness (there is an incentive to open up more clearing houses or, equivalently, it is strictly profitable to split each clearing house). Therefore, it is better to interpret the clearing house's program as reflecting the capacity constraints: The program presented here can be thought of as the inner problem of a profit maximization program with an additional factor (say, managerial effort), that which (i) makes the intermediation service production function exhibit constant returns to scale (instead of decreasing RTS), (ii) is provided by households, and (iii) is in perfectly inelastic supply. When the clearing house is modeled this way, it behaves exactly as modeled here: In the current way of modelling, the clearing house always makes zero profits and the household's remuneration for providing the additional factor is the profit.

where μ_t is the Lagrange multiplier on the balance-sheet constraint $B_{c,t+1} = B_{g,t+1}$. Market clearing in the household's bond positions in H requires

$$\gamma B_{c,t+1} = \gamma P_t b_{t+1},$$

and market clearing in the government's bond positions requires

$$\gamma B_{g,t+1} + (1 - \gamma) B_{g,t+1}^* = 0,$$

so that by using the balance-sheet constraints $B_{c,t+1} = B_{g,t+1}$, $B_{c,t+1}^* = B_{g,t+1}^*$ and the clearing conditions for household bond positions in H and F , we recover the market clearing condition for government bonds in the main model:

$$\gamma P_t b_{t+1} + (1 - \gamma) P_t^* b_{t+1}^* = 0.$$

In sum, the aggregate conditions implied by this market arrangement are

$$\begin{aligned} 1 + i_t^b + \psi b_{t+1} &= 1 + i_t, \\ 1 + i_t^{b,*} + \psi b_{t+1}^* &= 1 + i_t, \\ \gamma P_t b_{t+1} + (1 - \gamma) P_t^* b_{t+1}^* &= 0. \end{aligned}$$

The nominal profits of the typical clearing house in H in equilibrium are

$$\begin{aligned} \text{Profit}_{t+1} &= (i_t - i_t^b) B_{c,t+1} - \frac{\psi}{2} P_t^{-1} (B_{c,t+1})^2 \quad \text{with optimal } B_{c,t+1} = \frac{i_t - i_t^b}{\psi} P_t \\ &= (i_t - i_t^b)^2 \psi^{-1} P_t - (i_t - i_t^b)^2 \psi^{-1} P_t \cdot \frac{1}{2} \\ &= P_t \frac{\psi}{2} (b_{t+1})^2, \quad \text{using market clearing in the household bond positions.} \end{aligned}$$

Of the $1 + i_t\%$ nominal interest collected on its (paid for) position of government bonds, each clearing house withholds $\psi b_{t+1}\%$ of the interest from its customers (respectively, charges $-\psi b_{t+1}\%$ of additional interest if $b_{t+1} < 0$). Half of these $\psi b_{t+1}\%$ are used for covering the operating cost (by buying this amount of H 's final basket and selling it in exchange for the numéraire), and the other half is paid as profit to the owners of the clearing house (which, in equilibrium, are its customers).

B. PROOFS AND DERIVATIONS

B.1 Proof of Lemma (1)

Without loss of generality, we drop the region-specific indexation in the following. The agents apply the optimal Bayesian filter, that is, the Kalman filter, to arrive at the observable system:⁴⁴

$$\begin{aligned}\ln \frac{q_{t+1}}{q_t} &= \varrho \ln \bar{m}_t + \ln \widehat{e}_{t+1} \\ \ln \bar{m}_t &= \varrho \ln \bar{m}_{t-1} - \frac{\sigma_v^2}{2} + g \cdot \left(\ln \widehat{e}_t + \frac{\sigma_e^2 + \sigma_v^2}{2} \right),\end{aligned}$$

where $\ln \bar{m}_t := \mathbb{E}_t^{\mathcal{P}}(\ln m_t)$ is the posterior mean, $g = \frac{\varrho^2 \sigma^2 + \sigma_v^2}{\varrho^2 \sigma^2 + \sigma_v^2 + \sigma_e^2}$ is the steady-state Kalman filter gain, $\sigma^2 = \frac{1}{2\varrho^2} [-(\sigma_v^2 + (1 - \varrho^2)\sigma_e^2) + \sqrt{(\sigma_v^2 + (1 - \varrho^2)\sigma_e^2)^2 + 4\varrho^2 \sigma_v^2 \sigma_e^2}]$ is the steady-state Kalman filter uncertainty, and $\ln \widehat{e}_t$ is perceived to be a white noise process.

To avoid simultaneity in the house price, we modify the belief setup following [Adam et al. \(2017\)](#).⁴⁵ We obtain the same observable system but with lagged information being used in the posterior mean updating equation:

$$\ln \bar{m}_t = \varrho(1 - g) \left(\ln \bar{m}_{t-1} - \frac{\sigma_v^2}{2} \right) + g \left(\ln \frac{q_{t-1}}{q_{t-2}} + \frac{\sigma_e^2}{2} \right). \quad (\text{B.1})$$

Under this formulation, the posterior mean is predetermined. We may now derive the posterior mean on the $s > 0$ periods-ahead of price:

$$\mathbb{E}_t^{\mathcal{P}} q_{t+s} = q_t \cdot \exp \left(\ln \bar{m}_t \cdot \varrho \frac{1 - \varrho^s}{1 - \varrho} + \frac{1}{2} \sigma^2 \left(\varrho \frac{1 - \varrho^s}{1 - \varrho} \right)^2 \right) \cdot \exp(V), \quad V \propto \sigma_v^2 \quad (\text{B.2})$$

For the derivation of equation(B.2) see Appendix (B.2). This completes the proof.

⁴⁴We assume the the agents' prior variance equals the steady-state Kalman variance.

⁴⁵ q_t appears twice: in the forecast equation, and in the Kalman-updating Equation through $\ln \widehat{e}_t$. Since q_t depends on $\bar{m}_t$, but the latter also depends on the former, it is not assured that at any point an equilibrium asset price exists and whether it is unique. See [Adam et al. \(2017\)](#) for the details. The idea of the modification is to alter the agents' perceived information setup in that they observe each period one component of the lagged transitory price growth.

B.2 Derivation of Equation (B.2)

Without loss of generality, we drop the region-specific indexation in the following. Equation (B.2) is the result of the following calculations:

$$\begin{aligned}
\mathbb{E}_t^{\mathcal{P}} q_{t+s} &= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \frac{q_{t+s}}{q_t} = q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp \left(\ln q_{t+s} - \ln q_t \right) = q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp \left(\sum_{n=1}^s \Delta \ln q_{t+n} \right) \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp \left(\sum_{n=1}^s \ln m_{t+n} \right) \cdot \underbrace{\mathbb{E}_t^{\mathcal{P}} \left[\prod_{n=1}^s e_{t+n} \right]}_{= \prod_{n=1}^s \mathbb{E}_t^{\mathcal{P}} e_{t+n} = 1} \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp \left(\sum_{n=1}^s \left[\sum_{j=0}^{n-1} \varrho^j \ln v_{t+n-j} + \varrho^n \ln m_t \right] \right) \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp \left(\ln m_t \cdot \sum_{n=1}^s \varrho^n \right) \cdot \underbrace{\mathbb{E}_t^{\mathcal{P}} \exp \left(\sum_{n=1}^s \sum_{j=0}^{n-1} \varrho^j \ln v_{t+n-j} \right)}_{\sim \mathcal{N}} \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp \left(\varrho \frac{1 - \varrho^s}{1 - \varrho} \ln m_t \right) \cdot \exp(V), \quad V \propto \sigma_v^2 \\
\iff \mathbb{E}_t^{\mathcal{P}} q_{t+s} &= q_t \cdot \exp \left(\ln \bar{m}_t \cdot \varrho \frac{1 - \varrho^s}{1 - \varrho} + \frac{1}{2} \sigma^2 \left(\varrho \frac{1 - \varrho^s}{1 - \varrho} \right)^2 \right) \cdot \exp(V), \quad V \propto \sigma_v^2
\end{aligned}$$

B.3 Aggregation and market clearing

To achieve goods market clearing, each goods market for a variety k must clear. $\Psi_{i,t} := (1 + \pi_t)^{-1} \frac{\psi}{2} (b_{i,t})^2$ is the real cost of intermediating the position of an i -citizen in the union-wide bond. This cost, just like consumption and housing investment, gets passed along down to the varieties $\Psi_{i,t} := \left(\frac{P_{ii,t}}{P_{i,t}} \right)^{-\frac{1}{\varsigma}} \lambda \Psi_t$. Goods market clearing across all goods markets requires

$$\int y_{it}(k) dk = \gamma_i \left(\int [c_{iit}(k) + x_{iit}(k) + \Psi_{iit}(k)] dk + \sum_{j \neq i}^N \gamma_j \int [c_{jit}(k) + \Psi_{jit}(k)] dk \right) + \int \Phi_t(k) dk$$

where $\Phi_t(k) = \left(\frac{P_{ii,t}(j)}{P_{i,t}} \right)^{-\epsilon} \Phi_t$ and $\Phi_t = \frac{\kappa}{2} (\Pi_{ii,t} - 1)^2 y_{ii,t}$ account for the price adjustment costs from the firm side. Since housing investment requires domestically produced goods, we have that $x_i = x_{ii,t}$. Aggregation and successive substitution eventually yield the domestic and foreign

aggregate goods market clearing conditions:

$$\gamma_i \left(1 - \frac{\kappa}{2} (\Pi_{ii,t} - 1)^2 \right) y_{i,t} = \gamma_i \left(\frac{P_{ii,t}}{P_{i,t}} \right)^{(-1/\varsigma)} \lambda_{ii} (c_{i,t} + \psi_{i,t}) + \gamma_i x_{i,t} + \sum_{j \neq i} \gamma_j \left(\frac{P_{ii,t}}{P_{j,t}} \right)^{(-1/\varsigma)} \lambda_{ji} (c_{j,t} + \psi_{j,t})$$

Further, the bond market clearing condition is given by

$$\sum_j^N \gamma_j P_{j,t} b_{j,t+1} = 0.$$

Market clearing in the housing sectors for any region i is given by

$$H(x_{i,t-\tau}) = (h_{i,t} - (1 - \delta)h_{i,t-1})$$

Finally, the household budget constraints are satisfied.

B.4 Proof of Lemma 2

Without loss of generality, we focus in this section on a two-region case, a home country H and a foreign country F , with all variables for the foreign region denoted by an asterisk. For simplicity we also assume that time-to-build is zero in both regions. We prove that there exists a unique non-stochastic steady state with zero net inflation and parity in the terms of trade $P_{H,t}/P_{F,t}$ (which implies parity in the real exchange rate P_t/P_t^* , see the definition of $\mathfrak{p}$).⁴⁶ We prove this first for arbitrary parameters $\eta, \eta^*, \xi_{h,ss}, \xi_{x,ss}, \xi_{h,ss}^*, \xi_{x,ss}^*$ and then show that it is possible to select parameters such that the steady-state allocation is symmetric. The non-stochastic steady state with zero net inflation and real exchange rate parity (“SS” for short) is obtained by setting $\text{Var}[\|\xi_t\|] = 0$, where the shock vector contains both actual shocks and shocks that are only perceived (and never observed) by the household within their perceived house price model: $\xi_t = (\xi_{a,t}, \xi_{c,t}, \xi_{h,t}, \xi_{x,t}, \xi_{i,t}, e_t, v_t)^\top$. Thus, the non-stochastic steady state represents the time-invariant equilibrium that obtains if the agents do not expect any shock to ever materialize and indeed no shock ever does materialize, and we have $\xi_t = (1, 1, \xi_{h,ss}, \xi_{x,ss}, 1, 1, 1)^\top$ almost surely where $\xi_{h,ss}, \xi_{x,ss}$ are the model parameters. We are interested in, and prove the existence and uniqueness of a non-stochastic steady state in which the net rates of inflation are zero and the terms of trade are at parity. The latter assumption implies $\mathfrak{p}_H(1) = \mathfrak{p}_H^*(1) = \mathfrak{p}_F(1) = \mathfrak{p}_F^*(1) = \mathfrak{p}(1) = 1$.

⁴⁶With H being region i and F being region j . Define $P_{H,t} = P_{ii,t}, P_{F,t} = P_{jj,t}, P_t = P_{i,t}, P_t^* = P_{j,t}$. The relative prices are denoted by $\mathfrak{p}_{H,t} = \frac{P_{H,t}}{P_t}, \mathfrak{p}_{F,t} = \frac{P_{F,t}}{P_t}, \mathfrak{p}_{H,t}^* = \frac{P_{H,t}}{P_t^*}, \mathfrak{p}_{F,t}^* = \frac{P_{F,t}}{P_t^*}, \mathfrak{p}_t = \frac{P_t}{P_t^*}$.

Beliefs are irrelevant in the non-stochastic steady state. The first relevant insight is that in the non-stochastic steady state, defined as above, the presence and precise parameterization of subjective beliefs over house prices is irrelevant. To see this, recall that the subjective house price expectation dynamics are fully characterized by

$$\begin{aligned} \forall s > 0, \mathbb{E}_t^{\mathcal{P}} q_{t+s} &= q_t \cdot \exp \left(\ln \bar{m}_t \cdot \varrho \frac{1-\varrho^s}{1-\varrho} + \frac{1}{2} \sigma^2 \left(\varrho \frac{1-\varrho^s}{1-\varrho} \right)^2 \right) \cdot \exp(V), \quad V \propto \sigma_v^2 \\ \ln \bar{m}_t &= (1-g) \left(\ln \bar{m}_{t-1} - \frac{\sigma_v^2}{2} \right) + g \left(\ln \frac{q_{t-1}}{q_{t-2}} + \frac{\sigma_e^2}{2} \right), \end{aligned} \quad (\text{B.3})$$

+ equations determining the equilibrium evolution of the price level, q_t .

Now recall that the non-stochastic steady state represents the unique equilibrium of the economy when the variances of actual and perceived external shocks tend to zero and the initial conditions are selected such that they give rise to a constant path of equilibrium variables. Formally, the SS arises by replacing $\text{Var}[\|\xi_t\|]$ with $\varphi^2 \cdot \text{Var}[\|\xi_t\|]$ in the model, taking the limit $\varphi \rightarrow 0$ and solving for the fixed point of the equilibrium equations. Applying this logic, $\sigma_e, \sigma_v, \sigma \rightarrow 0$, to the equations (B.3) delivers

$$\begin{aligned} q_{ss} &= q_{ss} \cdot \exp \left(\ln \bar{m}_{ss} \cdot \varrho \frac{1-\varrho^s}{1-\varrho} + 0 \right) \cdot \exp(0) \\ \ln \bar{m}_{ss} &= (1-g) \ln \bar{m}_{ss} + g \cdot 0, \end{aligned}$$

from which we see that there exists a unique fixed point, which is $\bar{m}_{ss} = 1$, regardless of the level of q_{ss} . This means that the presence of subjective beliefs has no consequences for the non-stochastic steady state provided the agents have the prior that house prices do not change, $\bar{m}_{ss} = 1$, and that the absence of perceived and actual shocks means that house prices indeed never change, which in turn means that the prior of constant house prices is never changed.

In summary, if a non-stochastic steady state with zero net inflation and terms of trade parity exists and is unique, then it solves the following system of equations.

Household

$$\begin{aligned} (h) \quad q_{ss} &= \xi_{h,ss} h_{ss}^{-\nu} c_{ss}^{\sigma} + \beta(1-\delta)q_{ss}, \\ (n) \quad \chi n_{ss}^{\varphi} c_{ss}^{\sigma} &= w_{ss}, \\ (b) \quad 1 &= \beta(1+r_{ss}), \\ (x) \quad 1 &= q_{ss} \cdot \xi_{x,ss} x_{ss}^{\eta-1}, \end{aligned} \quad (\text{B.4})$$

Household*

$$(h^*) \quad q_{ss} = \xi_{h,ss}^* (h_{ss}^*)^{-\nu} (c_{ss}^*)^\sigma + \beta(1 - \delta)q_{ss}^*,$$

$$(n^*) \quad \chi(n_{ss}^*)^\varphi (c_{ss}^*)^\sigma = w_{ss}^*,$$

$$(b^*) \quad 1 = \beta(1 + r_{ss}^*),$$

$$(x^*) \quad 1 = q_{ss}^* \cdot \xi_{x,ss}^* (x_{ss}^*)^{\eta-1},$$

Interest rates

$$(r) \quad 1 + r_{ss} = 1 + i_{ss} - \psi b_{ss},$$

$$(r^*) \quad 1 + r_{ss}^* = 1 + i_{ss} - \psi b_{ss}^*,$$

Firm

$$(PC) \quad w_{ss} = 1 \quad \text{using that } \tau^\ell = 1/\epsilon,$$

Firm*

$$(PC^*) \quad w_{ss}^* = 1 \quad \text{using that } \tau^\ell = 1/\epsilon,$$

Bond market clearing

$$(B) \quad \gamma b_{ss} + (1 - \gamma)b_{ss}^* = 0,$$

Goods market clearing

$$(GMC) \quad n_{ss}\gamma = \gamma\lambda y_{ss} + (1 - \gamma)(1 - \lambda^*)y_{ss}^*,$$

$$(GMC^*) \quad n_{ss}^*(1 - \gamma) = \gamma(1 - \lambda)y_{ss} + (1 - \gamma)\lambda^*y_{ss}^*,$$

$$(BOP) \quad \gamma(1 - \lambda)y_{ss} - (1 - \gamma)(1 - \lambda^*)y_{ss}^* - \gamma(i_{ss}b_{ss} + (\beta^{-1} - 1)\bar{b}) = 0,$$

Housing market clearing

$$(HMC) \quad \eta^{-1}\xi_{x,ss}(x_{ss})^\eta = \delta h_{ss},$$

$$(HMC^*) \quad (\eta^*)^{-1}\xi_{x,ss}^*(x_{ss}^*)^{\eta^*} = \delta h_{ss}^*,$$

where $\bar{b}, \bar{b}^*$ are the model parameters (see Appendix A for an interpretation) chosen so as to (i) ensure equilibrium existence, and (ii) ensure that (§1) $\gamma\bar{b} + (1 - \gamma)\bar{b}^* = 0$. We solve for the SS in 4 steps:

1. First, we explicitly solve a number of equations, thus, substituting out a number of variables:

(a) (b) and (b^*) together with (r) and (r^*) and (B) imply $b_{ss} = b_{ss}^* = 0$ and $i_{ss} = \beta^{-1} - 1$, hence it follows that $y_{ss} = c_{ss} + x_{ss}$ and analogously * . (§1) then implies $\bar{b} = -\bar{b}^* \cdot (1 - \gamma)/\gamma$, where $\bar{b}^*$ is not pinned down yet. We solve for this in the very last step.

(b) (PC) and (PC^*) imply $w_{ss} = w_{ss}^* = 1$;

- (c) together with (n) and (n^*) , this implies $n_{ss} = (\chi^{-1})^{1/\varphi} \cdot c_{ss}^{-\sigma/\varphi} =: \phi(c_{ss})$ with $\phi' < 0$ and analogously for n_{ss}^* , with the *same* function ϕ ;
- (d) (HMC) and (HMC^*) imply $h_{ss} = (\delta)^{-1} \xi_{x,ss} \eta^{-1} x_{ss}^\eta$ and analogously for h_{ss}^* ;
- (e) (h) and (h^*) imply (with $\bar{\beta} := \beta(1 - \delta)$)

$$q_{ss} = (1 - \bar{\beta})^{-1} \xi_{h,ss} \left[\delta^{-1} \xi_{x,ss} \eta^{-1} x_{ss}^\eta \right]^{-\nu} c_{ss}^\sigma$$

$$q_{ss}^* = (1 - \bar{\beta})^{-1} \xi_{h,ss}^* \left[\delta^{-1} \xi_{x,ss}^* (\eta^*)^{-1} (x_{ss}^*)^\eta \right]^{-\nu} (c_{ss}^*)^\sigma.$$

- (f) (BOP) now reads $\gamma(1 - \lambda)y_{ss} = (1 - \gamma)(1 - \lambda^*)y_{ss}^* - (1 - \gamma)(\beta^{-1} - 1)\bar{b}^*$, and the symmetric (BOP^*) , which is redundant by Walras's law, reads $(1 - \gamma)(1 - \lambda^*)y_{ss}^* = \gamma(1 - \lambda)y_{ss} + (1 - \gamma)(\beta^{-1} - 1)\bar{b}^*$; using this in (GMC) , (GMC^*) produces

$$(GMC) \quad \phi(c_{ss}) = y_{ss} + (1 - \gamma)/\gamma \cdot (\beta^{-1} - 1)\bar{b}^*,$$

$$(GMC^*) \quad \phi(c_{ss}^*) = y_{ss}^* - (\beta^{-1} - 1)\bar{b}^*.$$

2. The remaining equations are (x) , (x^*) and (GMC) , (GMC^*) , (BOP) with unknowns $x_{ss}, x_{ss}^*, c_{ss}, c_{ss}^*, \bar{b}^*$. In this step, we show there are strictly increasing functions that yield x_{ss}, x_{ss}^* , given c_{ss}, c_{ss}^* , respectively. We start by plugging q_{ss} into (x) :

$$(1 - \bar{\beta})^{-1} \xi_{h,ss} \left[\delta^{-1} \xi_{x,ss} \eta^{-1} x_{ss}^\eta \right]^{-\nu} c_{ss}^\sigma \xi_{x,ss} x_{ss}^{\eta-1} = 1$$

(the equation for $*$ is symmetric.) Now since $\eta \in (0, 1)$ and $\nu > 0$, the expression on the left-hand side is a strictly decreasing function of x_{ss} for any c_{ss} . Moreover, for $x_{ss} \rightarrow 0$, the $LHS \rightarrow +\infty$ and for $x_{ss} \rightarrow \infty$, the $LHS \rightarrow 0$, hence Bolzano's intermediate value theorem (and continuity) ensures that for each c_{ss} there exists a unique x_{ss} . We call this implicitly defined mapping $x_{ss} = \psi(c_{ss})$. As the implicit function theorem shows, $\eta, \nu, \sigma > 0$ imply $\psi' > 0$. Analogous arguments hold for $*$.

3. We now insert our previous findings into the only remaining equations:

$$(GMC) \quad \phi(c_{ss}) - c_{ss} - \psi(c_{ss}) - (1 - \gamma)/\gamma \cdot (\beta^{-1} - 1)\bar{b}^* =: \zeta(c_{ss}, \bar{b}^*) = 0,$$

$$(GMC^*) \quad \phi(c_{ss}^*) - c_{ss}^* - \psi^*(c_{ss}^*) + (\beta^{-1} - 1)\bar{b}^* =: \zeta^*(c_{ss}^*, \bar{b}^*) = 0,$$

Observe, now, that $c_{ss} \mapsto \zeta$ is continuous and strictly decreasing with $\lim_{c \rightarrow 0} \zeta = +\infty$ (by $\lim_{c \rightarrow 0} \phi = +\infty$ and $\lim_{c \rightarrow 0} \psi < +\infty$) and $\lim_{c \rightarrow +\infty} \zeta = -\infty$ (by $\lim_{c \rightarrow +\infty} c, \psi = +\infty$ and $\lim_{c \rightarrow \infty} \phi = 0$). Therefore, Bolzano's intermediate value theorem ensures that there exists a

unique c_{ss} for each $\bar{b}^*$. The exactly analogous argument ensures the existence and uniqueness of c_{ss}^* . We call these mappings $\varpi : \bar{b}^* \mapsto c_{ss}$, $\varpi^* : \bar{b}^* \mapsto c_{ss}^*$. The implicit function theorem now yields

$$\partial \varpi / \partial \bar{b}^* < 0 \text{ and } \partial \varpi^* / \partial \bar{b}^* > 0.$$

4. Finally, only one equation remains, (*BOP*), with only one variable, $\bar{b}^*$:

$$\begin{aligned} & \gamma(1-\lambda)[\varpi(\bar{b}^*) + \psi(\varpi(\bar{b}^*))] - (1-\gamma)(1-\lambda^*)[\varpi^*(\bar{b}^*) + \psi^*(\varpi^*(\bar{b}^*))] \\ & + (1-\gamma)(\beta^{-1}-1)\bar{b}^* =: \mathcal{H}(\bar{b}^*) = 0 \end{aligned}$$

with $\bar{b}^* \mapsto \mathcal{H}$ continuous. It also holds that $\lim_{\bar{b}^* \rightarrow -\infty} \mathcal{H}(\bar{b}^*) = -\infty$.⁴⁷ On the other hand, as $\bar{b}^* \rightarrow +\infty$, $\mathcal{H} \rightarrow +\infty$.⁴⁸ Thus, Bolzano's intermediate value theorem ensures the **existence** of a $\bar{b}^* \in \mathbb{R}$, which satisfies the BOP-equation and, thus, the existence of a non-stochastic steady state.

Uniqueness of the steady-state can be shown by establishing the strict positive monotonicity of $\mathcal{H}$: (we suppress our arguments, for brevity)

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial \bar{b}^*} &= \gamma(1-\lambda)(1+\psi')\varpi' - (1-\gamma)(1-\lambda^*)(1+\psi'^*)\varpi^{*'} \\ &\quad + (1-\gamma)(\beta^{-1}-1) \\ \text{step 2, cf. notes below:} &> -(1-\gamma)(1-\lambda)(\beta^{-1}-1) - (1-\gamma)(1-\lambda^*)(\beta^{-1}-1) \\ &\quad + (1-\gamma)(\beta^{-1}-1) \\ &= (1-\gamma)(\beta^{-1}-1)[1-1+\lambda-1+\lambda^*] \\ &\stackrel{\text{sign}}{=} \lambda + \lambda^* - 1 \\ \text{step 3, cf. notes below:} &= \lambda + 1 - \frac{\gamma}{1-\gamma}(1-\lambda) - 1 \\ &\stackrel{\text{sign}}{=} (1-\gamma)\lambda - \gamma(1-\lambda) \\ &= \lambda - \gamma \end{aligned}$$

⁴⁷Proof: (1) $\gamma(1-\lambda)[\varpi^*(\bar{b}^*) + \psi^*(\varpi^*(\bar{b}^*))] \geq 0$ by non-negativity of consumption & housing investment; (2) $\gamma(1-\lambda)[\varpi + \psi \circ \varpi] + (1-\gamma)(\beta^{-1}-1)\bar{b}^* = \gamma(1-\lambda)[\phi - \frac{1-\gamma}{\gamma}(\beta^{-1}-1)\bar{b}^*] + (1-\gamma)(\beta^{-1}-1)\bar{b}^* = \gamma(1-\lambda)\phi + \lambda(1-\gamma)(\beta^{-1}-1)\bar{b}^*$, where the second equality is a consequence of (*GMC*); (3) $\lim_{\bar{b}^* \rightarrow -\infty} \varpi = +\infty$ (assuming the contrary will produce a contradiction with (*GMC*)); (4) $\lim_{c \rightarrow \infty} \phi = 0$; (5) steps 1–4 now imply $\mathcal{H}(\bar{b}^*) \leq \gamma(1-\lambda)\phi + \lambda(1-\gamma)(\beta^{-1}-1)\bar{b}^* \rightarrow -\infty$ as $\bar{b}^* \rightarrow -\infty$. $\square$

⁴⁸Proof: (1) $\gamma(1-\lambda)[\varpi(\bar{b}^*) + \psi(\varpi(\bar{b}^*))] \geq 0$ by non-negativity of consumption & housing investment; (2) $-(1-\lambda^*)[\varpi^*(\bar{b}^*) + \psi^*(\varpi^*(\bar{b}^*))] + (\beta^{-1}-1)\bar{b}^* = -(1-\lambda^*)[\phi(c_{ss}^*) + (\beta^{-1}-1)\bar{b}^*] + (\beta^{-1}-1)\bar{b}^* = -(1-\lambda^*)\phi(c_{ss}^*) + \lambda^*(\beta^{-1}-1)\bar{b}^*$ where the substitution is made using the definition of ϖ^* ; (3) $\lim_{\bar{b}^* \rightarrow +\infty} \varpi^* = +\infty$ (assuming the contrary will produce a contradiction with (*GMC*)); (4) Fact 3 and $\lim_{c \rightarrow +\infty} \phi(c) = 0$ imply $\mathcal{H}(\bar{b}^*) \geq -(1-\lambda^*)\phi(c_{ss}^*) + \lambda^*(\beta^{-1}-1)\bar{b}^* \rightarrow +\infty$ as $\bar{b}^* \rightarrow +\infty$. $\square$

≥ 0 by assumption.

Step 3 follows by symmetric home bias, that is, $\gamma(1 - \lambda) = (1 - \gamma)(1 - \lambda^*) \iff \lambda^* = 1 - \frac{\gamma}{1-\gamma}(1 - \lambda)$; Step 2 requires slightly more work: First, we use the implicit function theorem on (GMC) and (GMC^*) , respectively, to obtain

$$\begin{aligned}\varpi' &= -\frac{\partial \zeta / \partial \bar{b}^*}{\partial \zeta / \partial c_{ss}} = \frac{\frac{1-\gamma}{\gamma}(\beta^{-1} - 1)}{\phi' - (1 + \psi')} < 0, \\ \varpi^{*'} &= -\frac{\partial \zeta^* / \partial \bar{b}^*}{\partial \zeta^* / \partial c_{ss}^*} = \frac{-(\beta^{-1} - 1)}{\phi' - (1 + \psi^{*'})} > 0;\end{aligned}$$

Second, we recognize that since $\phi' < 0$ then

$$\frac{1 + \psi'}{1 + \psi' - \phi'} < 1 \iff \frac{1 + \psi'}{-(1 + \psi') + \phi'} > -1$$

and symmetrically for * . This shows that $\frac{\partial \mathcal{H}}{\partial \bar{b}^*} > 0$ and the SS is unique.

Finally, notice that since existence and uniqueness follow for arbitrary parameters $\eta \in (0, 1)$, $\eta^* \in (0, 1)$, $\xi_{h,ss}, \xi_{x,ss}, \xi_{h,ss}^*, \xi_{x,ss}^* > 0$, it is possible to set $\eta^* > \eta$ and then choose $\xi_{x,ss}, \xi_{x,ss}^*$ to ensure that both (HMC) and $(HMC)^*$ hold if $h_{ss} = h_{ss}^*$ and $x_{ss} = x_{ss}^*$. Given this choice of $\xi_{x,ss}, \xi_{x,ss}^*$, and the symmetry in the housing stock and housing investment, it is then possible to select $\xi_{h,ss}, \xi_{h,ss}^*$ such that equations (x) and (x^*) hold. It then follows that $c_{ss} = c_{ss}^*, n_{ss} = n_{ss}^*, \bar{b} = \bar{b}^* = 0$, and the allocation is symmetric. This completes the proof.

B.5 Derivation of the household's subjectively optimal plans

In this appendix, we provide a formal derivation of the linearized subjectively optimal household decision rules presented in equations (11) and (12). Throughout the derivation, we concentrate on the representative household in i , with the understanding that the situation in all other regions is symmetric. We therefore omit the regional subscript i for the sake of conciseness. For simplicity, we omit domestically produced housing investment goods. This does not affect the layout of the proof below. We further allow for potential external innovations to ξ_c, ξ_h, ξ_x .

Consider the household program presented in Section IV.A, which is restated here for convenience.⁴⁹ We first clarify the shape of the underlying probability space to set the appropriate frame for the following derivations: denote $\Omega \ni \omega_t := (\xi_t, r_t, w_t, \Sigma_t, \pi_t, (P_{ij,t})_{i,j \in [1,N]}, q_t)^\top$ as

⁴⁹We ignore the terms T_{t+s} and b_{t+s} since they will be zero in the equilibrium we analyze. All derivations go through if these terms are non-zero.

the vector of the external decision-relevant variables that the household takes as given; denote $\Omega^t \ni \omega^t := (\omega_{t-s})_{s \geq 0}$ as the one-sided infinite history of past external variables; and denote $\Omega^\infty \ni \omega := (\omega_t)_{t \in \mathbb{Z}}$ as the typical elements from the set of possible realizations of the full sequences of the external variables;⁵⁰ and denote $\mathcal{B}^\infty$ as the Borel-sigma-algebra over Ω^∞ . Each household is now endowed with a probability measure $\mathcal{P}$ over $(\Omega^\infty, \mathcal{B}^\infty)$ which encodes their subjective beliefs over the realizations of the external variables ω . Rational expectations, denoted $\mathcal{P} = \mathbb{P}$, are a special case of this setup, where $\mathbb{P}$ is the (unique) measure generated by the distribution of $\xi := (\xi_t)_{t \in \mathbb{Z}}$ and the equilibrium conditions that allow us to compute $\omega \backslash \xi$ as a deterministic function of ξ .⁵¹ As explained in Section IV.A, we assume $\mathcal{P}$ to be of a particular form: that is, we assume that $\mathcal{P} = \mathcal{P}_q \otimes \mathbb{P}_{-q}$, where $\otimes$ is the product measure (statistical independence), $\mathcal{P}_q$ is the distribution over $(q_t)_{t \in \mathbb{Z}}$ generated by the unobserved components model (4), and $\mathbb{P}_{-q}$ is the rational expectations measure without house prices—that is, the measure over $\omega \backslash q = (\xi_t, r_t, w_t, \Sigma_t, \pi_t, (P_{ij,t})_{i,j \in [1,N]})_{t \in \mathbb{Z}}$ that which is consistent with the equilibrium-implied joint probability distribution of $\omega \backslash q$. Consequently, for any two measurable functions $f : \mathbb{R} \rightarrow \mathbb{R}, g : \mathbb{R}^{\dim \Omega - 1} \rightarrow \mathbb{R}$, we have

$$\mathbb{E}^{\mathcal{P}}[f(q_t) \cdot g(\omega_t \backslash q_t)] = \int f \mathcal{P}_q(dq) \cdot \int g \mathbb{P}_{-q}(d\omega \backslash q).$$

Finally, $\mathbb{E}_t^{\mathcal{P}}$ is the expectation implied by $\mathcal{P}$ conditional on the sigma-algebra generated by ω^t .

Now, at each calendar date $t \in \mathbb{Z}$, the household takes as given ω^t and chooses today's consumption, labor, housing, housing investment, and bond levels, $(c_t, n_t, h_t, x_t, b_{t+1})$, as well as contingent plans for the future, $\{(c_{t+s}, n_{t+s}, h_{t+s}, x_{t+s}, b_{t+s+1}) : \Omega^{t+s} \rightarrow \mathbb{R}_+^4 \times \mathbb{R}\}_{s \in \mathbb{N}_+}$, to maximize

$$\begin{aligned} \mathbb{E}_t^{\mathcal{P}} \sum_{s=0}^{\infty} \beta^s \left(\frac{\xi_{c,t+s} c_{t+s}^{1-\sigma}}{1-\sigma} + \frac{\xi_{h,t+s} h_{t+s}^{1-\nu}}{1-\nu} - \chi \frac{n_{t+s}^{1+\varphi}}{1+\varphi} \right) \text{ subject to} \\ c_{t+s} + q_{t+s}(h_{t+s} - (1-\delta)h_{t+s-1}) + b_{t+s+1} + x_{t+s} = w_{t+s}n_{t+s} \\ + (1+r_{t+s})b_{t+s} + q_{t+s} \cdot \frac{\xi_{x,t+s}}{\eta} x_{t+s-\tau}^\eta + \Sigma_{t+s}, \quad \forall s \geq 0, \mathcal{P}\text{-almost surely,} \end{aligned} \tag{B.5}$$

as well as being subject to a standard no-Ponzi-game condition on b . The first-order conditions for

⁵⁰We restrict Ω to be the set of absolutely summable sequences.

⁵¹For any three vectors x, y, z where $x = (y, z)$, we define $x \backslash y := z$.

program (B.5) (suppressing the transversality conditions) are

$$\begin{aligned}
\forall s \geq 0, \quad (h) \quad & \xi_{c,t+s} c_{t+s}^{-\sigma} q_{t+s} = \xi_{h,t+s} h_{t+s}^{-\gamma} + \beta(1-\delta) \mathbb{E}_{t+s}^{\mathcal{P}} \{ \xi_{c,t+s+1} c_{t+s+1}^{-\sigma} q_{t+s+1} \}, \\
(n) \quad & \chi n_{t+s}^{\varphi} \xi_{c,t+s}^{-1} c_{t+s}^{\sigma} = w_{t+s}, \\
(b) \quad & \xi_{c,t+s} c_{t+s}^{-\sigma} = \beta \mathbb{E}_{t+s}^{\mathcal{P}} \{ (1+r_{t+s+1}) \xi_{c,t+s+1} c_{t+s+1}^{-\sigma} \}, \\
(x) \quad & \beta^{\tau} \mathbb{E}_{t+s}^{\mathcal{P}} \{ q_{t+s+\tau} \xi_{c,t+s+\tau} c_{t+s+\tau}^{-\sigma} \cdot \xi_{x,t+s+\tau} x_{t+s+\tau}^{\eta-1} \} = \xi_{c,t+s} c_{t+s}^{-\sigma}, \\
(BC) \quad & c_{t+s} + q_{t+s} (h_{t+s} - (1-\delta) h_{t+s-1}) + b_{t+s+1} + x_{t+s} = w_{t+s} n_{t+s} \\
& + (1+r_{t+s}) b_{t+s} + q_{t+s} \cdot \frac{\xi_{x,t+s}}{\eta} x_{t+s-\tau}^{\eta} + \Sigma_{t+s}.
\end{aligned} \tag{B.6}$$

In close analogy to the standard procedure in a model with fully rational expectations, $\mathcal{P} = \mathbb{P}$, we now derive a linear approximation to (B.6) that—together with the other linearized equilibrium conditions—allows us to solve the model to the first order in the amplitude of the shocks. The specific challenge here, with $\mathcal{P} = \mathcal{P}_q \otimes \mathbb{P}_{-q}$, will be to compute all of the expectations explicitly that depend on house prices, q . This includes the house prices, themselves, $(q_{t+s})_{s>0}$, as well as the expectations over future contingent choices, $(c_{t+s}, n_{t+s}, h_{t+s}, x_{t+s}, b_{t+s+1})_{s>0}$.

To the first order around the steady state⁵² from Lemma 2, in which we have $1+r_{ss} = \beta^{-1}$, $\chi n_{ss}^{\varphi} c_{ss}^{\sigma} = w_{ss} (=1)$, $q_{ss} = \xi_{h,ss} h_{ss}^{-\gamma} c_{ss}^{\sigma} / (1-\bar{\beta})$, $\beta^{\tau} q_{ss} \xi_{x,ss} x_{ss}^{\eta-1} = 1$, it holds that, $\forall s \geq 0$:

$$\begin{aligned}
(h) \quad & \widehat{h}_{t+s} = \frac{1}{\gamma} \xi_{h,t+s} + \frac{\sigma}{\gamma} \frac{\widehat{c}_{t+s} - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} - \bar{\beta} \mathbb{E}_{t+s}^{\mathcal{P}} \{ \widehat{c}_{t+s+1} - \frac{1}{\sigma} \widehat{\xi}_{c,t+s+1} \}}{1-\bar{\beta}} - \frac{1}{\gamma} \frac{\widehat{q}_{t+s} - \bar{\beta} \mathbb{E}_{t+s}^{\mathcal{P}} \{ \widehat{q}_{t+s+1} \}}{1-\bar{\beta}}, \\
(n) \quad & \varphi \widehat{n}_{t+s} + \sigma \widehat{c}_{t+s} - \widehat{\xi}_{c,t+s} = \widehat{w}_{t+s}, \\
(b) \quad & \widehat{c}_{t+s} - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} = -\frac{1}{\sigma} \mathbb{E}_{t+s} \{ \widehat{r}_{t+s+1} \} + \mathbb{E}_{t+s}^{\mathcal{P}} \{ \widehat{c}_{t+s+1} - \frac{1}{\sigma} \widehat{\xi}_{c,t+s+1} \}, \\
(x) \quad & \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{q}_{t+s+\tau} - \mathbb{E}_{t+s}^{\mathcal{P}} \{ \sigma \widehat{c}_{t+s+\tau} - \widehat{\xi}_{c,t+s+\tau} \} \\
& + (\sigma \widehat{c}_{t+s} - \widehat{\xi}_{c,t+s}) + \mathbb{E}_{t+s} \widehat{\xi}_{x,t+s+\tau} = (1-\eta) \widehat{x}_{t+s}, \\
(BC) \quad & \widehat{b}_{t+s+1} = \frac{1}{\bar{\beta}} \widehat{b}_{t+s} + \frac{w_{ss} n_{ss}}{y_{ss}} (\widehat{w}_{t+s} + \widehat{n}_{t+s}) + \widehat{\Sigma}_{t+s} - \frac{c_{ss}}{y_{ss}} \widehat{c}_{t+s} \\
& - \frac{q_{ss} h_{ss}}{y_{ss}} [\widehat{h}_{t+s} - (1-\delta) \widehat{h}_{t+s-1}] + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} (\widehat{x}_{t+s-\tau} - \beta^{\tau} \widehat{x}_{t+s} + \widehat{\xi}_{x,t+s}).
\end{aligned} \tag{B.7}$$

To characterize the household's expectations over their own choice variables, we start with the optimality condition for liquid bonds. Iterations over future instances of the condition reveal

$$(b) \quad \widehat{c}_{t+s} - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} = -\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{ \widehat{r}_{t+s+n} \} + \lim_{n \rightarrow \infty} \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{c}_{t+s+n}.$$

To keep the notation concise, we define, for any random process, $(k_t)_t$: $\mathbb{E}_{t+s}^{\mathcal{P}} k_{\infty} := \lim_{n \rightarrow \infty} \mathbb{E}_{t+s}^{\mathcal{P}} k_{t+s+n}$.

⁵²Technically, we are scaling $\text{Var}^{\mathcal{P}}[\|\omega\|] \rightarrow 0$. This is consistent, however, with the definition of $\mathcal{P}$ and $\text{Var}[\|\xi\|] \rightarrow 0$.

Here, we implicitly assume that the law of iterated expectations holds for $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+s}$ —at the end of this section, we verify that the $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{\infty}$ we recover does indeed satisfy a law of iterated expectations. Careful inspection of equations (B.7) reveals that $\mathbb{E}_{t+s}^{\mathcal{P}} \widehat{c}_{\infty}$ is the only subjective expectation of a choice variable left unknown. This is because the law of iterated expectations for $\mathcal{P}$ allows substituting (b) into

$$\begin{aligned}
(h) \quad & \widehat{h}_{t+s} = \frac{1}{\nu} \xi_{h,t+s} - \frac{\sigma}{\nu} \frac{\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n}\} - \bar{\beta} \frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n+1}\}}{1-\bar{\beta}} - \frac{1}{\nu} \frac{\widehat{q}_{t+s} - \bar{\beta} \mathbb{E}_{t+s}^{\mathcal{P}} \{\widehat{q}_{t+s+1}\}}{1-\bar{\beta}} + \frac{\sigma}{\nu} \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{c}_{\infty}, \\
(n) \quad & \varphi \widehat{n}_{t+s} + \sigma \widehat{c}_{t+s} - \widehat{\xi}_{c,t+s} = \widehat{w}_{t+s}, \\
(b) \quad & \widehat{c}_{t+s} - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} = -\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n}\} + \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{c}_{\infty}, \\
(x) \quad & \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{q}_{t+s+\tau} - \sum_{n=1}^{\tau} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n}\} + \mathbb{E}_{t+s} \widehat{\xi}_{x,t+s+\tau} = (1-\eta) \widehat{x}_{t+s}, \\
(BC) \quad & \widehat{b}_{t+s+1} = \frac{1}{\beta} \widehat{b}_{t+s} + \frac{w_{ss} n_{ss}}{y_{ss}} (\widehat{w}_{t+s} + \widehat{n}_{t+s}) + \widehat{\Sigma}_{t+s} - \frac{c_{ss}}{y_{ss}} \widehat{c}_{t+s} \\
& \quad - \frac{q_{ss} h_{ss}}{y_{ss}} [\widehat{h}_{t+s} - (1-\delta) \widehat{h}_{t+s-1}] + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} (\widehat{x}_{t+s-\tau} - \beta^{\tau} \widehat{x}_{t+s} + \widehat{\xi}_{x,t+s}).
\end{aligned} \tag{B.8}$$

Now to find $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{\infty}$ (and characterize the choices at calendar date t), we first use the budget constraint for some $s > \tau$. We start by plugging in the following optimality conditions:

$$\begin{aligned}
(BC) \quad & \widehat{b}_{t+s+1} = \frac{1}{\beta} \widehat{b}_{t+s} + \frac{w_{ss} n_{ss}}{y_{ss}} (1 + 1/\varphi) \widehat{w}_{t+s} + \widehat{\Sigma}_{t+s} - \frac{c_{ss} + n_{ss} w_{ss} \sigma / \varphi}{y_{ss}} (\widehat{c}_{t+s}^* - \frac{1}{\sigma} \widehat{\xi}_{c,t+s}) - \frac{c_{ss}}{y_{ss} \sigma} \widehat{\xi}_{c,t+s} \\
& \quad - \frac{q_{ss} h_{ss}}{y_{ss}} [\widehat{h}_{t+s}^* - (1-\delta) \widehat{h}_{t+s-1}^*] + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} (\widehat{x}_{t+s-\tau}^* - \beta^{\tau} \widehat{x}_{t+s}^* + \widehat{\xi}_{x,t+s}) \\
& \quad - \frac{c_{ss} + n_{ss} w_{ss} \sigma / \varphi + \delta q_{ss} h_{ss} \sigma / \nu}{y_{ss}} \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{c}_{\infty} \\
& \quad + \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \frac{1}{1-\bar{\beta}} (\widehat{q}_{t+s} - \bar{\beta} \mathbb{E}_{t+s}^{\mathcal{P}} \{\widehat{q}_{t+s+1}\} - (1-\delta) (\widehat{q}_{t+s-1} - \bar{\beta} \mathbb{E}_{t+s-1}^{\mathcal{P}} \{\widehat{q}_{t+s}\})) \\
& \quad + \frac{x_{ss}}{y_{ss}} \frac{1}{\beta^{\tau} (1-\eta)} (\mathbb{E}_{t+s-\tau}^{\mathcal{P}} \{\widehat{q}_{t+s}\} - \beta^{\tau} \mathbb{E}_{t+s}^{\mathcal{P}} \{\widehat{q}_{t+s+\tau}\}),
\end{aligned}$$

where we defined the auxiliary variables

$$\begin{aligned}
\widehat{h}_{t+s}^* &:= \frac{1}{\nu} \xi_{h,t+s} - \frac{\sigma}{\nu} \frac{\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n}\} - \bar{\beta} \frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n+1}\}}{1-\bar{\beta}}, \\
\widehat{c}_{t+s}^* - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} &:= -\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n}\}, \\
(1-\eta) \widehat{x}_{t+s}^* &:= -\sum_{n=1}^{\tau} \mathbb{E}_{t+s} \{\widehat{r}_{t+s+n}\} + \mathbb{E}_{t+s} \widehat{\xi}_{x,t+s+\tau}.
\end{aligned}$$

Notice that the auxiliary variables are composed only of terms over which the household has rational expectations; thus, the terms that fade to zero as $s \rightarrow \infty$.

Next, we apply the operator $\mathbb{E}_t^{\mathcal{P}}$ to both sides of (BC) and consider the limit as $s \rightarrow \infty$. This delivers

$$\begin{aligned}
 (BC) \quad \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty &= \frac{1}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty - \frac{c_{ss} + n_{ss} w_{ss} \sigma / \varphi + \delta q_{ss} h_{ss} \sigma / \nu}{y_{ss}} \mathbb{E}_{t+s}^{\mathcal{P}} \widehat{c}_\infty \\
 &\quad + \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1 - \eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty \\
 \iff \quad \mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty &= \frac{y_{ss}}{c_{ss} + n_{ss} w_{ss} \sigma / \varphi + \delta q_{ss} h_{ss} \sigma / \nu} \left[\frac{1 - \beta}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty + \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1 - \eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty \right].
 \end{aligned}$$

Now since $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty$ can be computed from the subjectively perceived house price model, the only unknown left is $\mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty$. To find this, we plug $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$ into (BC). At some $s > 0$, we have

$$\begin{aligned}
 (BC) \quad \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s+1} &= \frac{1}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s} + \frac{w_{ss} n_{ss}}{y_{ss}} (1 + 1/\varphi) \mathbb{E}_t \widehat{w}_{t+s} + \mathbb{E}_t \widehat{\Sigma}_{t+s} - \frac{c_{ss} + n_{ss} w_{ss} \sigma / \varphi}{y_{ss}} \mathbb{E}_t \left(\widehat{c}_{t+s}^* - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} \right) - \frac{c_{ss}}{y_{ss} \sigma} \mathbb{E}_t \widehat{\xi}_{c,t+s} \\
 &\quad - \frac{q_{ss} h_{ss}}{y_{ss}} \mathbb{E}_t \left[\widehat{h}_{t+s}^* - (1 - \delta) \widehat{h}_{t+s-1}^* \right] + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \mathbb{E}_t^{\mathcal{P}} \left(\widehat{x}_{t+s-\tau} - \beta^\tau \widehat{x}_{t+s} + \widehat{\xi}_{x,t+s} \right) \\
 &\quad - \frac{1 - \beta}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty - \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1 - \eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty \\
 &\quad + \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \frac{1}{1 - \beta} \mathbb{E}_t^{\mathcal{P}} \left(\widehat{q}_{t+s} - \bar{\beta} \mathbb{E}_{t+s}^{\mathcal{P}} \{ \widehat{q}_{t+s+1} \} - (1 - \delta) \left(\widehat{q}_{t+s-1} - \bar{\beta} \mathbb{E}_{t+s-1}^{\mathcal{P}} \{ \widehat{q}_{t+s} \} \right) \right).
 \end{aligned}$$

After defining the auxiliary variables

$$\begin{aligned}
 z_{t+s}^* &:= \frac{w_{ss} n_{ss}}{y_{ss}} (1 + 1/\varphi) \widehat{w}_{t+s} + \widehat{\Sigma}_{t+s} - \frac{c_{ss} + n_{ss} w_{ss} \sigma / \varphi}{y_{ss}} \left(\widehat{c}_{t+s}^* - \frac{1}{\sigma} \widehat{\xi}_{c,t+s} \right) - \frac{c_{ss}}{y_{ss} \sigma} \widehat{\xi}_{c,t+s} \\
 &\quad - \frac{q_{ss} h_{ss}}{y_{ss}} \left[\widehat{h}_{t+s}^* - (1 - \delta) \widehat{h}_{t+s-1}^* \right] + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \widehat{\xi}_{x,t+s}, \\
 Q_{t+s} &:= \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \frac{1}{1 - \beta} \left(\widehat{q}_{t+s} - \bar{\beta} \mathbb{E}_{t+s}^{\mathcal{P}} \{ \widehat{q}_{t+s+1} \} - (1 - \delta) \left(\widehat{q}_{t+s-1} - \bar{\beta} \mathbb{E}_{t+s-1}^{\mathcal{P}} \{ \widehat{q}_{t+s} \} \right) \right),
 \end{aligned} \tag{B.9}$$

we can rewrite (BC) into

$$\begin{aligned}
 (BC) \quad \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s+1} &= \frac{1}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s} + \mathbb{E}_t z_{t+s}^* + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \mathbb{E}_t^{\mathcal{P}} \left(\widehat{x}_{t+s-\tau} - \beta^\tau \widehat{x}_{t+s} \right) + \mathbb{E}_t^{\mathcal{P}} Q_{t+s} \\
 &\quad - \frac{1 - \beta}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty - \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1 - \eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty.
 \end{aligned}$$

Performing backward substitution until $s = 1$ delivers

$$\begin{aligned}
 (BC) \quad \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s+1} &= \beta^{-s} \widehat{b}_{t+1} - \beta^{-s} (1 - \beta^s) \frac{\beta}{1 - \beta} \left[\frac{1 - \beta}{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty + \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1 - \eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty \right] \\
 &\quad + \beta^{-s} \sum_{n=1}^s \beta^n (\mathbb{E}_t z_{t+n}^* + \mathbb{E}_t^{\mathcal{P}} Q_{t+n}) + \beta^{-s} \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \sum_{n=1}^s \beta^n \mathbb{E}_t^{\mathcal{P}} \left(\widehat{x}_{t+n-\tau} - \beta^\tau \widehat{x}_{t+n} \right) \\
 \iff \quad \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s+1} - \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty &= \beta^{-s} \cdot \left[\widehat{b}_{t+1} + \sum_{n=1}^s \beta^n (\mathbb{E}_t z_{t+n}^* + \mathbb{E}_t^{\mathcal{P}} Q_{t+n}) + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \sum_{n=1}^s \beta^n \mathbb{E}_t^{\mathcal{P}} \left(\widehat{x}_{t+n-\tau} - \beta^\tau \widehat{x}_{t+n} \right) \right. \\
 &\quad \left. - \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty - (1 - \beta^s) \frac{\beta}{1 - \beta} \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1 - \eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_\infty \right].
 \end{aligned}$$

Now since we know that $\lim_{s \rightarrow \infty} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{t+s}$ exists, it must be that the left-hand side of this equation tends to zero as $s \rightarrow \infty$. But since $\beta^{-s} \rightarrow +\infty$ as $s \rightarrow \infty$, this implies that

$$\begin{aligned} \mathbb{E}_t^{\mathcal{P}} \widehat{b}_{\infty} &= \widehat{b}_{t+1} + \sum_{n=1}^{\infty} \beta^n (\mathbb{E}_t z_{t+n}^* + \mathbb{E}_t^{\mathcal{P}} Q_{t+n}) + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \sum_{n=1}^{\infty} \beta^n \mathbb{E}_t^{\mathcal{P}} (\widehat{x}_{t+n-\tau} - \beta^{\tau} \widehat{x}_{t+n}) - \frac{\beta}{1-\beta} \frac{\delta q_{ss} h_{ss} / \nu + x_{ss} (\beta^{-\tau} - 1) / (1-\eta)}{y_{ss}} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{\infty}, \\ &= \beta^{-\tau} \sum_{n=1}^{\tau} \beta^n \widehat{x}_{t+n-\tau} \text{ by rearranging a convergent sum} \end{aligned}$$

so that we can characterize $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{\infty}$ entirely in terms of the variables that are either predetermined, current choice variables (i.e., known under $\mathbb{E}_t^{\mathcal{P}}$) or variables of which we can compute expectations in closed form (be they rational or not):

$$\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{\infty} = \frac{y_{ss}}{c_{ss} + n_{ss} w_{ss} \sigma / \varphi + \delta q_{ss} h_{ss} \sigma / \nu} \frac{1-\beta}{\beta} \left[\widehat{b}_{t+1} + \sum_{n=1}^{\infty} \beta^n (\mathbb{E}_t z_{t+n}^* + \mathbb{E}_t^{\mathcal{P}} Q_{t+n}) + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \sum_{n=1}^{\tau} \beta^n \widehat{x}_{t+n-\tau} \right].$$

In the last step, we explicitly characterize the subjective expectation $\mathbb{E}_t^{\mathcal{P}} \sum_{n=1}^{\infty} \beta^n Q_{t+n}$ in terms of the processes governing subjective house price expectations, namely $\widehat{q}_t, \widehat{m}_t$.

Recall that for any $s > 0$, we have $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+s} = \widehat{q}_t + (1 - \varrho^s) \frac{\varrho}{1-\varrho} \widehat{m}_t$, so that we have

$$\begin{aligned} \mathbb{E}_t^{\mathcal{P}} Q_{t+s} &= \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \frac{1}{1-\beta} \left(\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+s} - \bar{\beta} \mathbb{E}_t^{\mathcal{P}} \{\widehat{q}_{t+s+1}\} - (1-\delta) (\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+s-1} - \bar{\beta} \mathbb{E}_t^{\mathcal{P}} \{\widehat{q}_{t+s}\}) \right) \\ &= \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \frac{1}{1-\beta} \left(\delta (1-\bar{\beta}) \widehat{q}_t + (1-\varrho^s) \frac{\varrho}{1-\varrho} \widehat{m}_t - \bar{\beta} (1-\varrho^{s+1}) \frac{\varrho}{1-\varrho} \widehat{m}_t \right. \\ &\quad \left. - (1-\delta) (1-\varrho^{s-1}) \frac{\varrho}{1-\varrho} \widehat{m}_t + (1-\delta) \bar{\beta} (1-\varrho^s) \frac{\varrho}{1-\varrho} \widehat{m}_t \right) \\ &= \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \frac{1}{1-\beta} \left(\delta (1-\bar{\beta}) \widehat{q}_t + \widehat{m}_t \frac{\varrho}{1-\varrho} \left[\delta (1-\bar{\beta}) + (1-\bar{\beta}\varrho)(1-\varrho-\delta)\varrho^{s-1} \right] \right), \end{aligned}$$

which in turn implies

$$\sum_{s \geq 1} \beta^s \mathbb{E}_t^{\mathcal{P}} Q_{t+s} = \beta \frac{q_{ss} h_{ss}}{y_{ss}} \frac{1}{\nu} \left(\frac{\delta}{1-\beta} \widehat{q}_t + \widehat{m}_t \frac{\varrho}{1-\varrho} \left[\frac{\delta}{1-\beta} + \frac{1-\bar{\beta}\varrho}{1-\beta} \frac{1-\varrho-\delta}{1-\beta\varrho} \right] \right). \quad (\text{B.10})$$

Notice that, as claimed in the main text, that

$$\frac{\delta}{1-\beta} + \frac{1-\bar{\beta}\varrho}{1-\beta} \frac{1-\varrho-\delta}{1-\beta\varrho} > 0, \quad \forall \beta, \delta, \varrho \in (0, 1), \quad ^{53}$$

so that the coefficient of the posterior expected house price growth rate onto the series is positive.

We have now characterized the household's decisions at an arbitrary calendar date t up to first

⁵³*Proof.* This follows from the claim being equivalent to the statement $f(\varrho) := \frac{\delta}{1-\beta} (1-\beta\varrho) + \frac{1-\bar{\beta}\varrho}{1-\beta} (1-\varrho-\delta) > 0$, $\forall \beta, \delta, \varrho \in (0, 1)$, from f being an upward-open parabola with $f(1) = 0$, and $f'(1) < 0$. $\square$

order around the deterministic steady state:

$$\begin{aligned}
(h) \quad & \widehat{h}_t = \frac{1}{\nu} \xi_{h,t} - \frac{\sigma}{\nu} \frac{\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_t \{\widehat{r}_{t+n}\} - \bar{\beta} \frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_t \{\widehat{r}_{t+n+1}\}}{1 - \bar{\beta}} - \frac{1}{\nu} \widehat{q}_t + \frac{1}{\nu} \frac{\bar{\beta}}{1 - \bar{\beta}} \varrho \widehat{m}_t + \frac{\sigma}{\nu} \mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty, \\
(n) \quad & \varphi \widehat{n}_t + \sigma \widehat{c}_t - \widehat{\xi}_{c,t} = \widehat{w}_t, \\
(b) \quad & \widehat{c}_t - \frac{1}{\sigma} \widehat{\xi}_{c,t} = -\frac{1}{\sigma} \sum_{n \geq 1} \mathbb{E}_t \{\widehat{r}_{t+n}\} + \mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty, \\
(x) \quad & \widehat{q}_t + (1 - \varrho^\tau) \frac{\varrho}{1 - \varrho} \widehat{m}_t - \sum_{n=1}^{\tau} \mathbb{E}_t \{\widehat{r}_{t+n}\} + \mathbb{E}_t \widehat{\xi}_{x,t+\tau} = (1 - \eta) \widehat{x}_t, \\
(BC) \quad & \widehat{b}_{t+1} = \frac{1}{\beta} \widehat{b}_t + \frac{w_{ss} n_{ss}}{y_{ss}} (\widehat{w}_t + \widehat{n}_t) + \widehat{\Sigma}_t - \frac{c_{ss}}{y_{ss}} \widehat{c}_t \\
& \quad - \frac{q_{ss} h_{ss}}{y_{ss}} [\widehat{h}_t - (1 - \delta) \widehat{h}_{t-1}] + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} (\widehat{x}_{t-\tau} - \beta^\tau \widehat{x}_t + \widehat{\xi}_{x,t}),
\end{aligned} \tag{B.11}$$

where, after a few rearrangements,

$$\begin{aligned}
\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty = & \frac{\delta q_{ss} h_{ss} / \nu}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu} \cdot \left[\widehat{q}_t + \widehat{m}_t \cdot \frac{\varrho}{1 - \varrho} \underbrace{\left(1 + \frac{1 - \bar{\beta} \varrho}{1 - \bar{\beta}} \frac{1 - \varrho - \delta}{1 - \beta \varrho} \frac{1 - \beta}{\delta} \right)}_{> 0 \ \forall \beta, \delta, \varrho \in (0, 1)} \right] \\
& + \frac{y_{ss}}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu} \frac{1 - \beta}{\beta} \cdot \left[\sum_{n=1}^{\infty} \beta^n \mathbb{E}_t \{z_{t+n}^*\} + \frac{x_{ss}}{y_{ss}} \sum_{n=0}^{\tau-1} \beta^{-n} \widehat{x}_{t-n} + \widehat{b}_{t+1} \right].
\end{aligned}$$

Define $\bar{Q} = \frac{\delta q_{ss} h_{ss} / \nu}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu}$, $\bar{M} = \frac{\varrho}{1 - \varrho} \left(1 + \frac{1 - \bar{\beta} \varrho}{1 - \bar{\beta}} \frac{1 - \varrho - \delta}{1 - \beta \varrho} \frac{1 - \beta}{\delta} \right)$, and $\bar{C} = \frac{y_{ss}}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu} \frac{1 - \beta}{\beta}$.

Further, $\sum_{n=1}^{\infty} \beta^n \bar{z}_{t+n} = \sum_{n=1}^{\infty} \beta^n \mathbb{E}_t \{z_{t+n}^*\} + \frac{x_{ss}}{y_{ss}} \sum_{n=0}^{\tau-1} \beta^{-n} \widehat{x}_{t-n}$. Lastly, we verify that the law of iterated expectations holds for the explicit formula given for $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty$, above. We define

$$A_t := \widehat{b}_{t+1} + \sum_{n=1}^{\infty} \beta^n (\mathbb{E}_t z_{t+n}^* + \mathbb{E}_t^{\mathcal{P}} Q_{t+n}) + \frac{x_{ss}}{y_{ss}} \beta^{-\tau} \sum_{n=1}^{\tau} \beta^n \widehat{x}_{t+n-\tau}$$

and note that the law of iterated expectations holds if and only if $\mathbb{E}_t^{\mathcal{P}} A_{t+1} = A_t$. Using equation (BC) to substitute $\widehat{b}_{t+2}$ in $\mathbb{E}_t^{\mathcal{P}} A_{t+1}$ we arrive at

$$\begin{aligned}
\mathbb{E}_t^{\mathcal{P}} A_{t+1} = & A_t / \beta - \frac{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss}}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu} \frac{1 - \beta}{\beta} \mathbb{E}_t^{\mathcal{P}} A_{t+1} - \frac{q_{ss} h_{ss} \sigma / \nu}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu} \frac{1 - \beta}{\beta} \mathbb{E}_t^{\mathcal{P}} A_{t+1} \\
& + \frac{(1 - \delta) q_{ss} h_{ss} \sigma / \nu}{c_{ss} + \sigma / \varphi \cdot n_{ss} w_{ss} + \delta q_{ss} h_{ss} \sigma / \nu} \frac{1 - \beta}{\beta} A_t,
\end{aligned}$$

which can be rearranged into $\mathbb{E}_t^{\mathcal{P}} A_{t+1} = A_t$.

B.6 Derivations for Section V

In this appendix we derive analytical results on the behavior of house prices in a one-region, zero-liquidity endowment economy with instantaneous housing production.

The equations describing the economy are

$$\begin{aligned}
(h) \quad & \widehat{q}_{i,t} - \bar{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} = \sigma(\widehat{c}_{i,t} - \bar{\beta} \mathbb{E}_t^{\mathcal{P}} \widehat{c}_{i,t+1}) - \nu(1 - \bar{\beta}) \widehat{h}_{i,t}, \\
(b) \quad & \widehat{c}_{i,t} = -\sigma^{-1} \mathbb{E}_t \widehat{r}_{t+1} + \mathbb{E}_t^{\mathcal{P}} \widehat{c}_{i,t+1}, \\
(BC) \quad & \widehat{b}_{i,t+1} = \beta^{-1} \widehat{b}_{i,t} + \widehat{y}_t - \frac{\varepsilon}{y} \widehat{c}_{i,t} - \frac{qh}{y} (\widehat{h}_{i,t} - (1 - \delta) \widehat{h}_{i,t-1}) + \frac{qh\delta}{y} \widehat{q}_{i,t}, \\
(B) \quad & \widehat{b}_{i,t+1} = 0, \\
(HMC) \quad & \widehat{h}_{i,t} - (1 - \delta) \widehat{h}_{i,t-1} = -\iota_i \varepsilon_t,
\end{aligned} \tag{B.12}$$

with the endowment process $\widehat{y}_t$ being the only exogenous disturbance to the economy. Iterating on (h) and plugging the results into the Euler equation gives our key equation:

$$\widehat{q}_{i,t} = \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} - \widehat{r}_{t+1} - \nu(1 - \bar{\beta}) \widehat{h}_{i,t} + \nu(1 - \bar{\beta})^2 \sum_{s=0}^{\infty} \bar{\beta}^s \mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1}.$$

B.6.1 Proof of Proposition (2)

Under rational expectations, and using the fact that the shocks only hit in period one, we can use housing market clearing to rewrite the house price equation as

$$\widehat{q}_{i,t} = \mathbb{E} \widehat{q}_{i,t+1} - \widehat{r}_{t+1} - (1 - \bar{\beta}) \widehat{h}_{i,t} + (1 - \bar{\beta})^2 (1 - \delta)^t \frac{1}{1 - \bar{\beta}(1 - \delta)} (-\iota_i \varepsilon_1)$$

Iteration on this equation and using the fact that the economy returns to the steady-state in expectations gives

$$\begin{aligned}
\widehat{q}_{i,1} &= -\varepsilon_1 - \omega_h (-\iota_i \varepsilon_1) \\
\widehat{q}_{i,2} &= -(1 - \delta) \omega_h (-\iota_i \varepsilon_1),
\end{aligned}$$

with $\omega_h = \frac{1 - \bar{\beta}}{1 - \bar{\beta}(1 - \delta)} \in (0, 1)$. This completes the proof.

B.6.2 Proof of Proposition (3)

Under subjective beliefs, we need to specifically characterize $\mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1}$. We can make use of the method described in section (IV.B). Importantly, we can make use of the fact that the shocks on the endowment and, hence, the interest rate only hit in the first period. As households have rational expectations about these processes, they possess knowledge of this fact. Applying our solution

method and substituting the subjective beliefs housing model, we get

$$\begin{aligned}\widehat{q}_{i,t} = \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} - (1 - \bar{\beta})\widehat{h}_{i,t} - \widehat{r}_{t+1} + (1 - \bar{\beta}) \left[-\widehat{q}_{i,t} - \frac{y_{ss}}{y_{ss} + \delta q_{ss} h_{ss}} \frac{1 - \beta}{1 - \bar{\beta}} \widehat{r}_{t+1} \right. \\ \left. + \left(\beta \frac{q_{ss} h_{ss}}{y_{ss}} \frac{\varrho}{1 - \varrho} \left(\frac{1 - \bar{\beta} \varrho}{1 - \bar{\beta}} \frac{1 - \varrho - \delta}{1 - \beta \varrho} + \frac{1}{1 - \beta \varrho} \right) + \frac{1 - \varrho}{1 - \varrho \bar{\beta}} - \frac{1}{1 - \varrho} \right) \widehat{m}_{i,t} \right]\end{aligned}$$

Rearranging yields

$$\widehat{q}_{i,t} = -\widehat{h}_{i,t} - \omega_r \widehat{r}_{t+1} + \omega_m \widehat{m}_{i,t},$$

where $\omega_r = 1 + \frac{y(1-\beta)}{y_{ss} + h_{ss} q_{ss} \delta}$ and $\omega_m = (1 - \bar{\beta})^{-1} \left[\varrho + (1 - \bar{\beta}) \left(\beta \frac{q_{ss} h_{ss}}{y_{ss}} \frac{\varrho}{1 - \varrho} \left(\frac{1 - \bar{\beta} \varrho}{1 - \bar{\beta}} \frac{1 - \varrho - \delta}{1 - \beta \varrho} + \frac{1}{1 - \beta \varrho} \right) + \frac{1 - \varrho}{1 - \varrho \bar{\beta}} - \frac{1}{1 - \varrho} \right) \right]$. Further, we have that

$$\begin{aligned}\frac{\partial \omega_m}{\partial \varrho} = (1 - \bar{\beta})^{-1} \left[1 + (1 - \bar{\beta}) \beta \frac{q_{ss} h_{ss}}{y_{ss}} \left(\frac{\varrho}{1 - \varrho} \left(\frac{(1 - \bar{\beta})(\beta \bar{\beta} \varrho (\varrho + \delta) - 1)}{((1 - \bar{\beta})(1 - \beta \varrho))^2} + \frac{\beta}{(1 - \beta \varrho)^2} \right) \right. \right. \\ \left. \left. \frac{1 - 2\varrho}{(1 - \varrho)^2} \left(\frac{1 - \bar{\beta} \varrho}{1 - \bar{\beta}} \frac{1 - \varrho - \delta}{1 - \beta \varrho} + \frac{1}{1 - \beta \varrho} \right) - \frac{1 - \bar{\beta}}{(1 - \bar{\beta} \varrho)^2} + \frac{1}{(1 - \varrho)^2} \right) \right],\end{aligned}$$

which is smaller than zero for standard parameter choices. The first two periods of house prices are given by

$$\begin{aligned}\widehat{q}_{i,1} &= -\omega_r \varepsilon_1 - (-\iota_i \varepsilon_1) \\ \widehat{q}_{i,2} &= -(1 - \delta)(-\iota_i \varepsilon_t) + \omega_m g \widehat{q}_1\end{aligned}$$

This completes the proof.

B.6.3 Demand-side heterogeneity

Turning to regional differences on the housing demand side, we focus on the differences in the housing demand elasticity (ν). Following the same logic as in propositions (2) and (3), we derive closed-form characterizations of the house price with respect to a monetary policy shock.

Under rational expectations, the house price is given by

$$\widehat{q}_{i,t} = \mathbb{E} \widehat{q}_{i,t+1} - \widehat{r}_{t+1} - \nu_i (1 - \bar{\beta}) \widehat{h}_{i,t} + \nu_i (1 - \bar{\beta})^2 (1 - \delta)^t \frac{1}{1 - \bar{\beta}(1 - \delta)} (-\iota \varepsilon_1).$$

Using the same steps as described in the Proof of Proposition (B.6.1) we get

$$\begin{aligned}\widehat{q}_{i,1} &= -\varepsilon_1 - \omega_h \nu_i (-\iota \varepsilon_t) \\ \widehat{q}_{i,2} &= -(1 - \delta) \omega_h \nu_i (-\iota \varepsilon_t)\end{aligned}$$

Given that $\iota > 0$, a higher elasticity of housing demand, hence a lower (ν_i), leads to a stronger response in house prices to a monetary policy shock.

Moving to the subjective beliefs model, we can use the same approach as in the proof of Proposition (B.6.2) and arrive at

$$\widehat{q}_{i,t} = -\nu_i \widehat{h}_{i,t} - \tilde{\omega}_{r,i} \widehat{r}_{t+1} + \tilde{\omega}_{i,m} \widehat{m}_{i,t},$$

where $\tilde{\omega}_{i,r} = 1 + \frac{y_{ss}(1-\beta)}{y_{ss} + h_{ss} q_{ss} \delta \nu_i^{-1}}$ and $\tilde{\omega}_{i,m} = (1 - \bar{\beta})^{-1} \left[\varrho + (1 - \bar{\beta}) \left(\beta \frac{q_{ss} h_{ss} \nu_i^{-1}}{y_{ss}} \frac{\varrho}{1-\varrho} \left(\frac{1-\bar{\beta}\varrho}{1-\bar{\beta}} \frac{1-\varrho-\delta}{1-\beta\varrho} + \frac{1}{1-\beta\varrho} \right) + \frac{1-\varrho}{1-\varrho\beta} - \frac{1}{1-\varrho} \right) \right]$. Assuming that $\iota = 0$, the house price response to the shock in the first two periods is given by

$$\begin{aligned}\widehat{q}_{i,1} &= -\tilde{\omega}_{i,r} \varepsilon_1 \\ \widehat{q}_{i,2} &= \tilde{\omega}_{i,m} g \widehat{q}_{i,1}\end{aligned}$$

Under subjective beliefs, we arrive at a similar result without conditioning on a positive supply response ($\iota \geq 0$). As under the supply-side differences, the subjective beliefs model is more responsive to a monetary policy shock. Furthermore, regional heterogeneities enter the model through $\tilde{\omega}_{i,r}$ and $\tilde{\omega}_{i,m}$, and are scaled by the factor g . This yields a structure analogous to that in Propositions (2) and (3), whereby regional heterogeneities give rise to regional variations, which are subsequently amplified by extrapolation.

C. EMPIRICAL PART AND MODEL CALIBRATION

C.1 Data sources

For the monthly house price data on the federal level, we use the SP CoreLogic Case-Shiller U.S. National Home Price Index, seasonally adjusted from the FRED database (CSUSHPISA). The FFR (FEDFUNDS), industrial production (INDPRO), and CPI (CPIAUCSL) on a monthly

frequency are also taken from the FRED. The house price expectations data are the mean of the expected change in home values during the next year, taken from Michigan University's Survey of Consumers. For the quarterly state-level data, we use all transactions house price indices ("State code"STHPI) from the FRED and seasonally adjust them. The remaining state-level data on building permits ("State code"BPPRIVSA), employment in construction ("State code"CONS), unemployment ("State code"UR), and employment in retail are all taken from the FRED and are already seasonally adjusted. We omit the code for employment in retail, as these vary across states. GDP (GDP) and the GDP deflator (GDPDEF) are also obtained from the FRED. As already mentioned in the main text, the house price sensitivity indicator is taken from [Guren et al. \(2021\)](#), and the monetary policy shock from [Degasperis and Ricco \(2021\)](#). The time to build measures was taken from the Census.

C.2 Robustness: Cross-regional heterogeneity in booms and busts

For the euro area, the sample covers the period 2000-2019, also on a quarterly basis. The countries included in the euro area sample are Austria, Germany, Spain, Finland, France, Ireland, Italy, the Netherlands, and Portugal. For the non-aligned US cities, we check whether the sign of house price growth in a certain city at a certain time aligns with the US aggregate sign. If they do not align, we ignore them in our exercise.

Table C.1: Cross-regional house price growth standard deviation (σ_c) in booms and busts, euro area, U.S: cities ignoring non-aligned regional cycles

	<i>E.A. countries standard measure</i>		<i>U.S. cities : ignore non – aligned regions</i>	
	<i>mean</i>	<i>median</i>	<i>mean</i>	<i>median</i>
<i>Bust Boom</i>	1.30	1.27	1.47	1.40
<i>σ_c : Boom</i>	1.57	1.59	0.40	0.33
<i>σ_c : Bust</i>	2.04	2.02	0.59	0.45
<i>p – Val. Bust > Boom</i>	0.002	0.004	0.000	0.000
<i>Number regions</i>	8		20	
<i>Obs. boom</i>	55		331	
<i>Obs. bust</i>	24		82	
<i>Sample</i>	2000Q1 – 2019Q3		1990M1 – 2023M12	

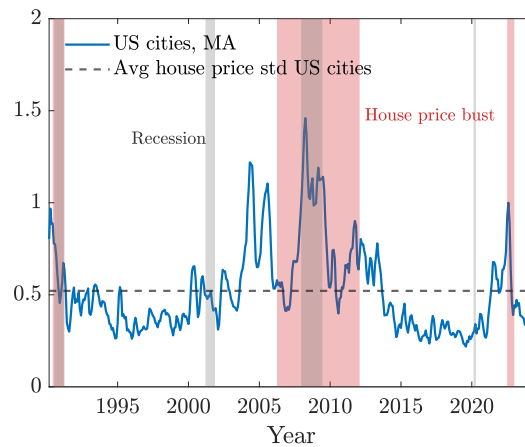
Notes: The table reports the means and medians, across time, of the estimates of cross-sectional standard deviations of house price growth rates for the euro area countries and U.S. cities, ignoring observations when city house price growth is non-aligned with the aggregate cycle. The test for *Bust > Boom* is based on a one-sided t-test.

Table C.2: Cross-regional house price growth standard deviation (σ_c) in booms and busts, US cities: alternative measures

	Aastveit et al. (2023)		Albuquerque et al. (2025)		Harding and Pagan (2002)	
	<i>mean</i>	<i>median</i>	<i>mean</i>	<i>median</i>	<i>mean</i>	<i>median</i>
<i>Bust</i>	1.36	1.47	1.17	1.16	1.70	1.73
<i>Boom</i>						
$\sigma_c : Boom$	0.53	0.45	0.49	0.41	0.46	0.41
$\sigma_c : Bust$	0.73	0.66	0.57	0.48	0.78	0.71
$p - Val. Bust > Boom$	0.000	0.000	0.000	0.006	0.000	0.000

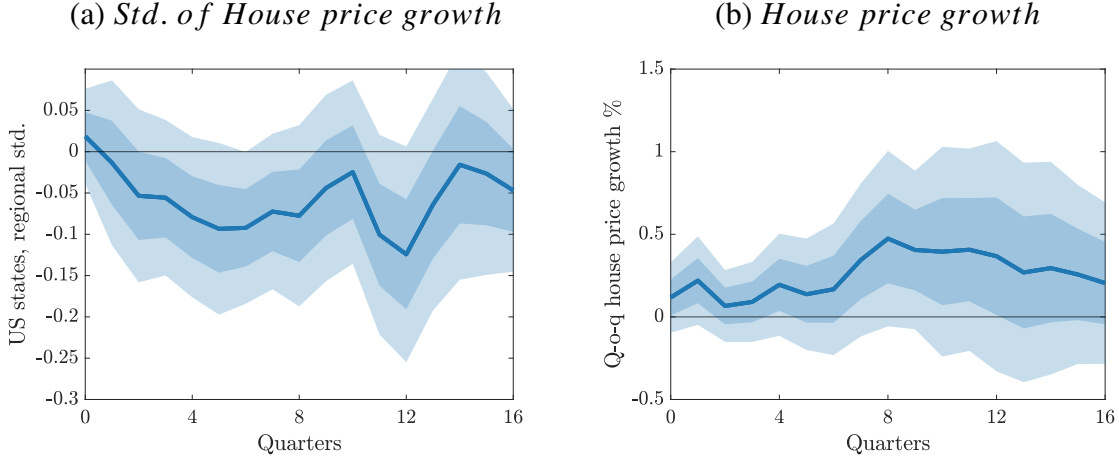
Notes: The table reports the means and medians, across time, of the estimates of cross-sectional standard deviations of house price growth rates for U.S. cities using the boom/bust several measures. The test for *Bust* > *Boom* is based on a one-sided t-test. Sample: 1990M1-2023M12.

Figure C.1: Regional house price variations in booms and busts



Notes: Three- month moving average of cross-city std.

Figure C.2: State-level std. house price growth response to main business cycle shock



Notes: Responses to a main business cycle shock (1 std) ; confidence intervals: 68% and 95% (Newey-West); three-month moving average of std. across states of quarter-on-quarter house price growth (left); q-o-q house price growth at the federal level (right.)

C.3 Forecast error response to monetary policy shock

We are first interested in whether house price expectations are formed according to rational expectations. To answer this question, we study the response of forecast errors to a monetary policy shock. Forecast errors are interesting for two reasons. First, they show whether house price expectations follow rational expectations. Under rational expectations, forecast errors should not respond to the shock. This becomes obvious in our local projections setup. We define forecast errors as follows:

$$fe_t = \Delta q_{t+12}^y - \mathbb{E}_t^{\mathcal{P}} \Delta q_{t+12}^y$$

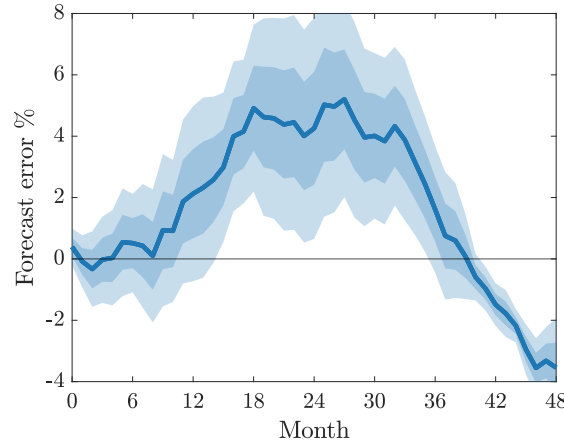
, where Δq_{t+12}^y is the year-on-year percentage change in house prices and $\mathbb{E}_t^{\mathcal{P}} \Delta q_{t+12}^y$ is the expected year-on-year percentage change in 12 month formed today. The local projections we estimate are given by Equation (C.1). At the time the shock hits, in period t , the forecast errors seen on the left-hand side already contain the forecast formed today for 12 months ahead. In other words, the shock is contained in the agent's information set and should be taken into account. Under rational expectations, agents will perfectly use this information and forecast errors should not respond. The second insight we may gain from this exercise is on how house price expectations are formed. Given the response of the forecast errors is non-zero, the dynamics reveal how households' expectations evolve over time. This will enable us to zoom in on a specific belief updating process that explains our observations. Due to data availability, we focus on the U.S. in this exercise. The monetary policy shock is taken from [Degasperi and Ricco \(2021\)](#). We use the expectations data from the Michigan Survey on Consumer Sentiment. The frequency is monthly and the sample runs from

2007 to 2019. As controls, we include six lags of the forecast error and the monetary policy shock. We estimate the following equation:

$$fe_{t+h} = \alpha^h + \beta^h \epsilon_t^{MP} + x_t + u_{t+h} \quad h = 0, 1, \dots, H \quad (C.1)$$

Figure (C.3) shows the results. The impulse response reveals a non-zero and highly significant response in forecast errors. We can therefore reject rational expectations. On the dynamic behavior, we find that for the first 30 months the forecast error is positive, indicating over-pessimism on the side of the agents. After 30 months, the forecast error turns negative, indicating over-optimism. For both cases, over-pessimism and over-optimism, the IRF is significant at the 95% confidence level. Further, the hump-shaped dynamics in the forecast errors indicate sluggish updating in the expectations formation process. These findings are in line with [Adam et al. \(2022\)](#). [Angeletos et al. \(2021\)](#) document similar dynamics in an SVAR setup for unemployment and inflation in response to an unemployment and inflation shock.

Figure C.3: Forecast error response to a monetary policy shock, U.S.



Notes: Response to an expansionary monetary shock (25 bp); confidence Intervals: 68% and 95% Newey-West).

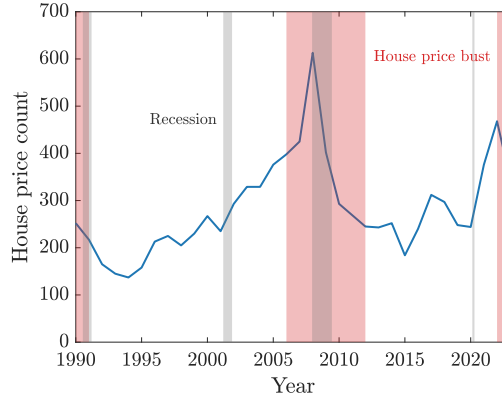
C.4 Potential sources of heterogeneity in the expectation formation

We estimate equation (3) using actual house price realizations instead of expectations. Hence, we estimate the following equation using the same procedure as before:

$$\Delta q_{t+1} = \mathbb{1}(q_{t-1} > \bar{\omega}) \left(\bar{c}^h + \bar{\varrho}^h(1 - \bar{g}^h) \times \Delta q_t + \bar{\varrho}^h \bar{g}^h \times \Delta q_{t-1} \right) + \mathbb{1}(\Delta q_{t-1} < \bar{\omega}) \left(\bar{c}^l + \bar{\varrho}^l(1 - \bar{g}^l) \times \Delta q_t + \bar{\varrho}^l \bar{g}^l \times \Delta q_{t-1} \right) \quad (C.2)$$

We estimate the following parameter values: $\bar{\varrho}^h = 1.00$, $\bar{g}^h = 0.66$, $\bar{\varrho}^l = 0.54$, $\bar{g}^l = 0.99$, and $\bar{\omega} = 0.10$. Consistent with the results from the expectations formation exercise, we find a higher degree of persistence and a lower degree of updating in the boom regime. Moreover, the estimated threshold is close to zero. We therefore conclude that the perceived law of motion mirrors the properties of the actual law of motion.

Figure C.4: Occurrence of "House Price" in New York Times



Number of times the New York Times mentioned the word "house price" in a given year.

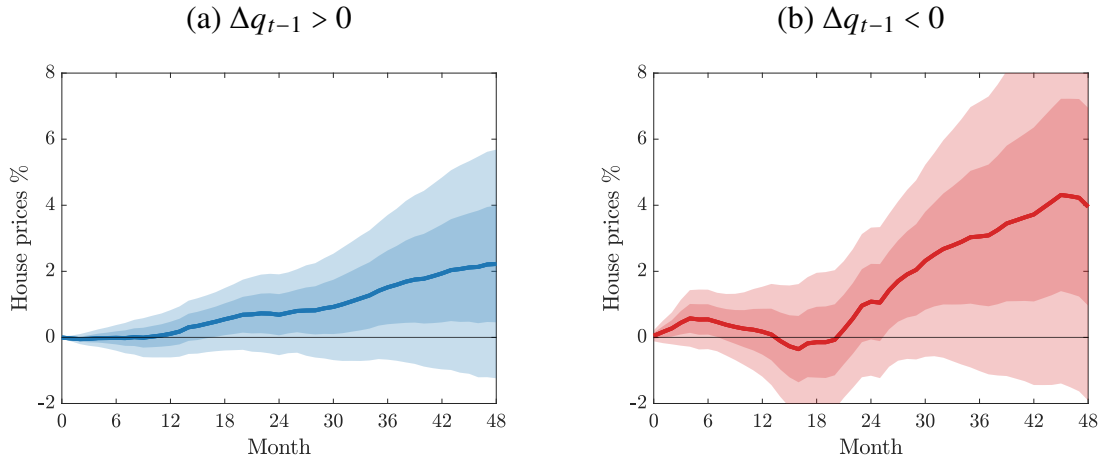
C.5 House price response to monetary policy shocks in booms and busts

To test whether house prices are more responsive in busts, we run local projections on house prices responding to a monetary policy shock, conditioning on booms and busts. The estimation equation is given by:

$$q_{t+h} = \alpha^h + \mathbb{1}(\Delta q_{t-1} > 0) \times \beta_1 \epsilon_t^{MP} + \mathbb{1}(\Delta q_{t-1} < 0) \times \beta_2 \epsilon_t^{MP} + x_t + u_{t+h} \quad (\text{C.3})$$

The monetary policy shock, ϵ_t^{MP} , is the high frequency identified and orthogonalized shock from [Degasperi and Ricco \(2021\)](#). The left-hand side variable, q_{t+h} , is the log house price. The sample runs from 1991-2019 and is in monthly frequency. The controls, x_t , contain 6 lags of the left-hand side variable, log of industrial production, the log of CPI, the FFR, and the shocks. We will focus on expansionary monetary policy shocks throughout the whole empirical analysis. Figure (C.5) plots the results. The blue line shows the response if house prices were increasing in the past, and the red line if they were decreasing. We find that house prices are notably more responsive to monetary policy shocks in times when they have been decreasing. The peak response of the point estimates doubles. Quantitatively, the response seems sizable in both cases; however, asset prices, including house prices, generally exhibit strong responsiveness to variations in interest rates. Further, the estimates exhibit large estimation uncertainty.

Figure C.5: House price response to monetary policy shock, boom-bust

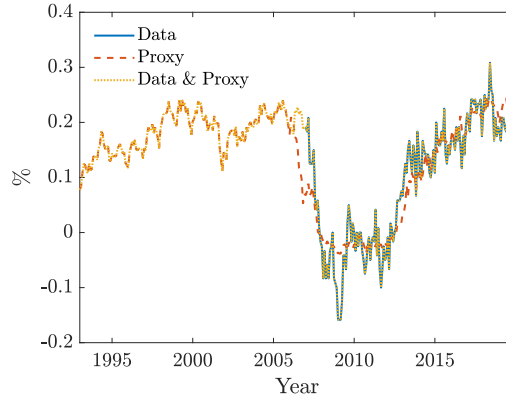


Notes: Responses to expansionary MP shock (25 bps); Confidence Intervals: 68% and 95% (Newey-West).

C.6 House price expectations: extended dataset

Jacobson (2022) proposes to backcast by regression expected house price changes on perceived buying conditions taken from the Michigan Survey. As the survey question about buying condition is goes back to 1993, this relation allows a backcast. We follow her proposal and combine the backcast with the actual house price expectations. However, the backcast leads the actual house price expectations by roughly one year at the beginning of the housing bust in 2007. We therefore use the proxy up until December 2005 and the actual data starting in January 2007. To bridge the gap, we regress house price expectations on three lags of past house price growth and compute the gap using the estimated coefficients. Figure (C.6) shows the results: one can clearly see that the proxy leads at the beginning of the crisis. Our approach corrects for this. For the remainder of the sample, the proxy does rather well.

Figure C.6: House price expectations proxy



Data is the actual house price expectations series from the Michigan Survey. Proxy is the approximation as suggested by [Jacobson \(2022\)](#). Data & proxy combine the two after accounting for the adjustment described in the text.

Using this series, we re-estimate the equation (3). We also directly control for booms, bust and normal times. We define a bust in the same manner as before, while a boom is a time when house price growth is twice as high as the long-run mean. Table (C.3) shows the results. First, for the baseline estimation, we find that qualitatively, our results remain valid. Even the estimated threshold is close to our value presented in the main text. Second, our results also remain robust if we specifically control for boom periods.

Table C.3: House price belief model: parameters for proxy

<i>Specification</i>	ϱ^{normal}	ϱ^h	ϱ^l	g^{normal}	g^h	g^l	<i>Threshold</i>	MSE^i/MSE^{lin}
<i>Baseline</i>		0.95	0.91		0.003	0.015	0.002	0.48
<i>Boom/Bust/Normal</i>	0.95	0.96	0.92	0.012	0.013	0.016		0.45

Notes: Baseline: estimated parameters from (3) using the proxy data series; Boom/Bust/Normal: Busts are defined by negative house price growth in the previous month, Booms are defined by house price growth being twice as high as the long-run mean. The share in the MSE is computed relative to using a linear model for the same sample.

Finally, we re-run the regressions proposed by [Case et al. \(2012\)](#) using the extent expectations series. Table (C.4). As before, the pass-through from realized house prices to expectations is larger in a bust. Furthermore, the previously insignificant negative coefficient for the boom becomes positive and significant. Finally, we can still reject the null of both coefficients being equal.

C.7 Time-to-build in the U.S.

The time it takes to obtain a building permit or, generally, to build a house is not available for the U.S. on a state level. However, the Census Bureau documents construction times for homes on

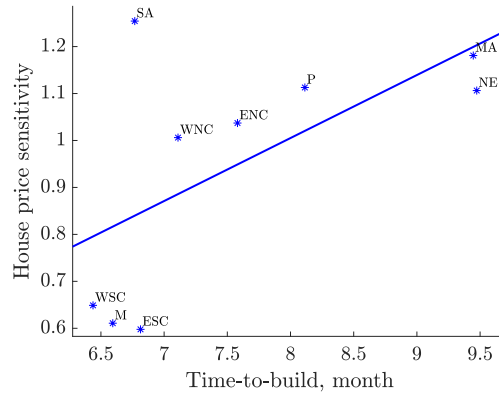
Table C.4: Pass-through from lagged prices to expectations, proxy

b^{boom}	b^{bust}	R^2	N	$p - val. H_0 : b^{boom} = b^{bust}$
0.06*** (0.02)	0.13*** (0.03)	0.59	323	0.06

Notes: Estimation of the [Case et al. \(2012\)](#) regression for house prices, conditioning on boom and bust regimes. Reported standard errors are Newey-West. Significant levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Column 5 shows the p-value for the null hypothesis that both coefficients are equal.

a census division level. Below, we aggregate the housing sensitivity indicator from [Guren et al. \(2021\)](#) to a census division level and correlate them with the housing construction time measured from the date of authorization to completion. One can see in figure (C.7) that these are clearly positively correlated.

Figure C.7: Correlation of house price sensitivity indicator with time-to-build



Correlation of house price sensitivity indicator from [Guren et al. \(2021\)](#) with time-to-build a house across census divisions. Time-to-build is measured from the date of the authorization until the completion of the building activities.

Table C.5: Regional housing supply side heterogeneities

City	Elasticity	Time-to-build	City	Elasticity	Time-to-build
Atlanta (Atl)	1.05	7.2	Minneapolis (Min)	1.3	8.5
Charlotte (Cha)	0.65	4.0	Miami (Mia)	2.4	17.1
Dallas (Dal)	0.65	4.2	New York (NYC)	1.4	9.7
Denver (Den)	0.41	2.4	Phoenix (Pho)	0.95	6.4
Detroit (Det)	1.50	10.5	Portland (Por)	0.61	3.9
Boston (Bos)	0.98	6.7	San Diego (SD)	1.1	7.7
Chicago (Chi)	1.13	7.8	San Francisco (SF)	1.2	8.1
Cleveland (Cle)	0.94	6.4	Seattle (Sea)	0.68	4.4
Las Vegas (LV)	1.43	10.0	Tampa (Tam)	2.0	14.6
Los Angeles (LA)	1.2	8.2	Washington D.C. (DC)	1.1	7.6

Notes: The elasticity data is taken from [Guren et al. \(2021\)](#). The time-to-build is in monthly frequency. Since the model is in quarterly frequency, we will round the time-to-build to the nearest quarter.

C.8 Model: steady-state values

Table C.6: Steady-state values

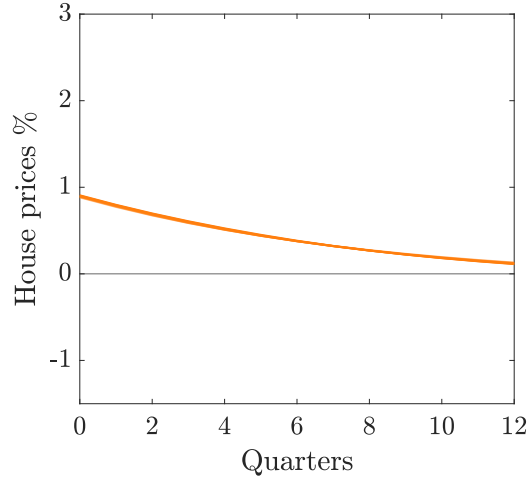
Variable	Value	
$\xi_{x,ss}$	1.46	housing productivity shifter
c_{ss}	0.98	consumption
x_{ss}	0.06	housing investment
h_{ss}	10.09	housing
y_{ss}	1.04	output
b_{ss}	0.00	bond holdings
w_{ss}	1.00	wage

Notes: All steady states except the house price are symmetric across countries. Time-to-build and the steady-state housing preference shifter are not symmetric across countries. All elements of ξ_{ss} not explicitly mentioned assume the value 1.

D. QUANTITATIVE MODEL

D.1 Further results

Figure D.1: Regional house price and output response under rational expectations



Notes: Responses to an expansionary monetary policy shock (25 bp).

Table D.1: Aggregate standard deviations across policy rules

<i>Model</i>	ϕ^π	ϕ^q	<i>Boom</i>			<i>Bust</i>		
			π	y	q	π	y	q
<i>Rational Expectations</i>	1.5	0.00	0.05	0.14	0.17	0.05	0.14	0.17
	1.5	0.125	0.07	0.09	0.11	0.07	0.09	0.11
<i>Subjective Expectations</i>	1.5	0.00	0.01	0.34	0.70	0.02	0.37	0.78
	1.5	0.125	0.06	0.10	0.20	0.06	0.10	0.20

Notes: The table shows standard deviations of several variables in response to an expansionary productivity shock (100 bp); The standard deviations are computed over the first 24 quarters.

D.2 Loss function

We assume that the policy-maker's objective is the utilitarian average welfare in the economy, evaluated under rational expectations. This definition leaves open the question of what information the policy-maker possesses. Conditional welfare, $\mathbb{W}_{-1}$, is the average expected welfare in the union, conditional on (i) the economy being in the steady state in the current and all past periods $t \leq 0$,

Table D.2: Cross-region standard deviations across policy rules, Subjective expectations

	ϕ^π	ϕ^q	<i>Boom</i>			<i>Bust</i>		
			π	y	q	π	y	q
<i>Subjective Expectations</i>	1.5	0.00	0.00	0.05	0.17	0.00	0.09	0.28
	1.5	0.125	0.00	0.02	0.05	0.00	0.02	0.06

Notes: The table shows standard deviations of several variables in response to an expansionary productivity shock (100 bp) under subjective expectations; the standard deviations are computed over the first 24 quarters.

and on (ii) the policy-maker knowing that in $t = 0$ a set of shocks $(\xi_t)_{t \geq 0} \in (\mathbb{R}^n)^\infty$ will realize with a Gaussian probability distribution but not knowing about the exact realization of the shocks. The experiment here is that the policy-maker decides about their instruments at the end of period $t = -1$, maximizing the average utility in the union under the expectation that a set of surprise shocks might materialize in the next period. According to a known distribution—let $\mathbb{E}_{-1}$ denote the operator drawing the expectation with respect to the rational expectations measure and conditional on (i) and (ii). Specifically,

$$\mathbb{W}_{-1} := \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \gamma \sum_{i=1}^N u_{i,t}$$

$$\mathbb{W} := \mathbb{E} \gamma \sum_{i=1}^N u_{i,t}$$

where $u_{i,t}$ is the utility function of region i .⁵⁴ We approximate both criteria to the second order around the non-stochastic steady state: Assuming that the allocation in the non-stochastic steady state is symmetric, we have that

$$\mathbb{W}_{-1} = \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \gamma \sum_{i=1}^N \left[-\frac{c^{-\sigma}}{2} (\kappa y \pi_{iit}^2 + \sigma c \widehat{c}_{it}^2 + \xi_{hi} h^{1-\nu} c^\sigma \widehat{h}_{it}^2 + \varphi n^{1+\varphi} c^\sigma \widehat{n}_{it}^2 + x(1-\eta) \widehat{x}_{it}^2 + \psi y^2 \widehat{b}_{it}^2) \right. \\ \left. + n^{1+\varphi} \widehat{\xi}_{at} \widehat{n}_{it} + c^{-\sigma} \bar{s}_3 \sum_{j \neq i}^N \widehat{s}_{ijt}^2 \right] + O(3) + tips$$

where $\bar{s}_3$ consists only of model parameters, $O(3)$ are terms of the third or higher order and “t.i.p.” stands for the “terms independent of the policy,” that is, the variables that are beyond the policy-maker’s control and, thus, are irrelevant for the policy selection. Re-writing the loss function using

⁵⁴Note that it holds that $\mathbb{E} \mathbb{W}_{-1} = \mathbb{W}$.

market-clearing conditions gives:

$$\frac{\mathbb{W}_{-1}}{c_{ss}} = -c_{ss}^{\sigma} \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \sum_{i=1}^N \left[\frac{\gamma}{2} \Omega_{i,t} + \frac{1}{2} \Gamma^s \sum_{j \neq i}^N \widehat{s}_{ijt}^2 + O(3) + \text{t.i.p.} \right] \quad (\text{D.1})$$

$\Omega_{i,t}$ collects the welfare-relevant choices and is defined as

$$\Omega_{i,t} = \Gamma^{\pi} \widehat{\pi}_{ii,t}^2 + \Gamma^c \widehat{c}_{i,t}^2 + \Gamma^h \widehat{h}_{i,t}^2 + \Gamma^n \widehat{n}_{i,t}^2 + \Gamma^x \widehat{x}_{i,t}^2 + \Gamma^b \widehat{b}_{i,t}^2 - \Gamma^{\xi_a} \widehat{\xi}_{a,t} \widehat{n}_{i,t}$$

This can be easily rewritten in the form presented in the main text.